

More Information >>

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International Market



whatsapp



EN website



Youtube

Domestic Market



Sales Manager



CN website



WeChat

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GENERAL CATALOG

PRODUCTS 2025

Planetary Actuator-RMD-X V4 Series

Harmonic Actuator-RH Series

Harmonic Mini Actuator-HM Series

Cycloidal Actuator-CEM Series

Hollow Direct Drive Actuator-H Series

Frameless Motor-FL/FLO Series

Micro Servo · Product Solutions

All Support Customization



MYACTUATOR is the provider of integrated intelligent power system solutions for robots

As an innovation-driven micro-power system technology company, MYACTUATOR is committed to becoming a globally renowned provider of integrated intelligent power solutions for various robots, accelerating their application across numerous industries and bringing them into households worldwide.

Suzhou Micro Actuator Technology Co., Ltd. was established in 2020 and is a key technology-based enterprise in Suzhou. It has won honors such as "JiangSu Science-based SMEs", "Suzhou Talent Science and Technology Innovation Enterprise, Kunshan Science and Technology Talent Enterprise", and "High-tech Enterprise", etc.

MYACTUATOR currently has more than 70 technical talents of various types, more than 30 technical patents, forming more than 100 product models, with a comprehensive operation area of more than 4,000 square meters, with perfect motor design and development, manufacturing and testing equipment and advanced quality management concepts.



Corporate Culture



Product Positioning

A full-stack intelligent hardware solution provider for robots.



Career Value

Unleash human creativity and accelerate the integration of artificial intelligence products into diverse scenarios.



Service Group

Innovative technology companies/research institutes empowered by artificial intelligence.



Team Culture

Strive for excellence and truth, be honest and open.



Product Model

100⁺ tower



Technical Talent

70⁺ people



Technology Patent

30⁺ item



Operating Area

4000⁺ Square meter



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Development History

MYACTUATOR

2021

Won many honors

Suzhou City-Entrepreneurship and Innovation Talent Enterprise
Kunshan Science and Technology Talent Enterprise
Suzhou Talent Science and Technology Innovation Enterprise
Jiangsu Province Science and Technology Small and Medium Enterprises
High-tech enterprise



2020

MYACTUATOR Founded

Suzhou Micro Actuator Technology Co., Ltd. Founded



2022

Entering overseas markets



2023

Angel round investment

Received angel round investment [Dingzhi Technology (873593.BJ Jiangsu Leili (300660.SZ) led the investment
Awarded the title of "High-tech Enterprise"



2025

New product release

Integrated cycloidal module shipment
Shipment of lightweight humanoid modules
Embodied intelligent power system solution released



2024

Relocation of the new plant



Intellectual Property

Until May 2024

22
Utility Model Patent

4
Software copyright patent

2
Invention Patent

2
Design patent

Honors & Qualifications

At present, MYACTUATOR has applied for 30 patents and software copyrights, including 2 appearance patents, 2 invention patents, 4 software copyrights, and 22 utility model patents. A number of invention patents have formed independent intellectual property rights for key technologies of products, which protects the technical advantages of the company's products and gives it a head start in competition.

- High-tech Enterprise
- Suzhou Talent Science and Technology Innovation Enterprise
- Suzhou City-Entrepreneurship and Innovation Talent Enterprise
- Kunshan Science and Technology Talent Enterprise
- Jiangsu Province Science and Technology Small and Medium Enterprises



Product Guarantee

MYACTUATOR is committed to providing customers with safe and reliable motor module products. Our products have passed EU CE, RoHS certification, ISO9001 and other certifications to ensure the safety and reliability of the product throughout its life cycle.

EU CE, RoHS, ISO9001 Certification



Product Warranty Statement

Our company provides quality certificates when delivering our products. Customers can test the motors according to existing technical conditions, drawings or relevant requirements.

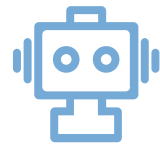
Our company's product warranty period is one year (calculated from the date of shipment, the motor is attached with SN code), and customers need to refer to the product manual to correctly and reasonably store and use the product.

If our product is damaged or cannot work properly due to quality problems within the specified service life or warranty period, our company will provide free repairs.

The following situations are not within the scope of free repair:

1. Expiration of the shelf life; (the tear-off label is torn or man-made damage is assumed to be exceeded)
2. Damage caused by improper use;
3. Man-made damage;
4. Products that have not been disassembled or repaired by our company;
5. Failures are caused by force majeure such as natural disasters.

Application



Robotic



Humanoid Robot



Healthcare



Wheeled Robot



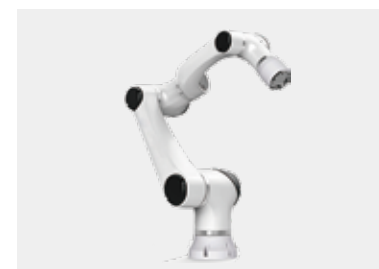
Photography



Gimbal



Industrial Automation



Robotic Arm



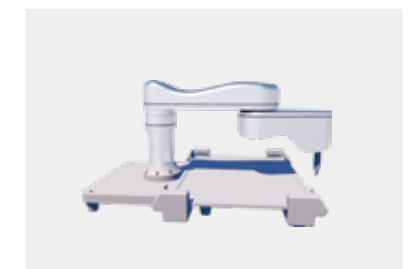
Smart warehousing



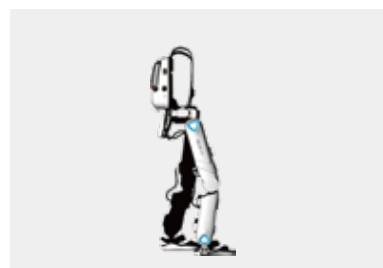
AGV Smart Car



Other Industries



Inspection



Exoskeleton Robot



Medical Equipment



PTZ



CNC Machine Tools



Warehouse Robot



Quadruped Robot



Industrial Robots



Medical Equipment



Gimbal



Automation



Smart Warehouse



Smart Agriculture

RMD-X V4 Series Planetary Actuator

Special Series For Robot

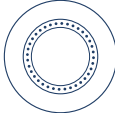


* The pictures of this series of products are subject to the actual products.

RMD-X V4 Series Product Features



CAN BUS & EtherCAT



Crossed Roller Bearings



Dual Encoder



High Torque Density

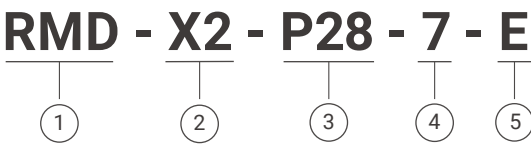


High Precision



Hollow Design

RMD-X V4 Series Naming Conventions



- 1

RMD: Brand Name R-Reducer M-Motor D-Drive
- 2

X2: X Stands For The Series Name: Integrated Planetary Actuator, 2 represent motor model number e.g:X2 X4 X6 X8 etc
- 3

P28: Planetary gear ratio e.g:P12 P28 P32 etc
- 4

7: Peak torque 7N.m
- 5

E: Communication E: CAN BUS & EtherCAT

For example

Model	RMD-X8-P20-120-E
Illustrate	RMD: Brand Name
	X8: Motor model name
	P20: Planetary gear ratio 20:1
	120: Peak torque
	E: Communication E: CAN BUS & EtherCAT

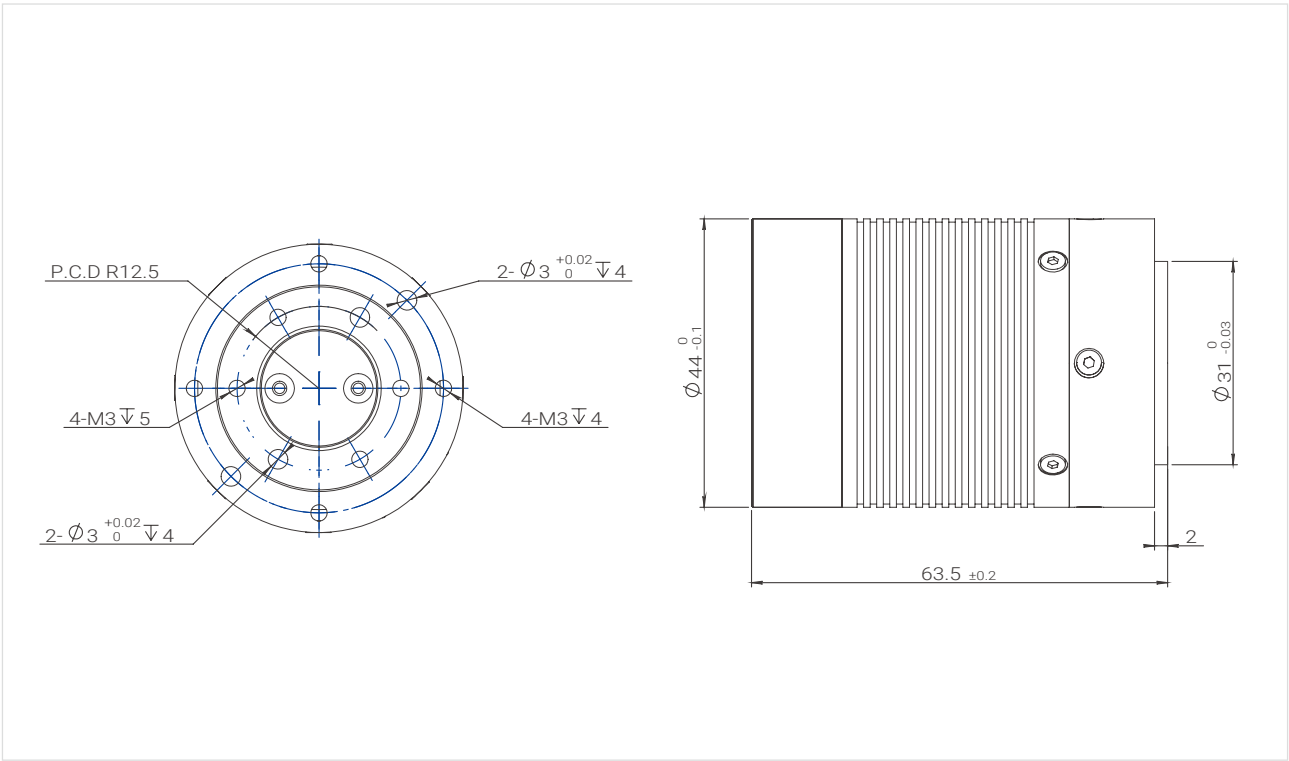
Series Name
RMD-X
Motor Simplified Name
X2-7



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X2-P28-7-E	N (without Brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters	Unit	X2-7
Gear Ratio	—	28.17
Input Voltage	V	24
No Load Speed	RPM	178
No-Load Input Current	A	1
Rated Speed	RPM	142
Rated Torque	N.m	2.5
Rated Output Power	W	37
Rated Phase Current	A(rms)	3
Peak Torque	N.m	7
Peak Phase Current	A(rms)	8.1
Efficiency	%	63
Motor Back-EMF Constant	Vdc/Krpm	4.3
Module Torque Constant	N.m/A	0.8
Motor Phase Resistance	Ω	0.61
Motor Phase Inductance	mH	0.13
Pole Pair	—	13
3 Phase Connection	—	Y
Back Drive Torque	N.m	0.4
Backlash	Arcmin	12
Output Bearing Type	—	Deep Groove Ball Bearings
Axial Load	Suffer	KN
	Stress	KN
Radial Load	KN	1
Inertia	Kg.cm ²	0.17
Encoder Type & Interface	—	Dual Encoder ABS-17BIT(Input) / 18BIT(Output)
Control Accuracy	Degree	<0.01
Communication	—	CAN BUS / EtherCAT
Weight	Kg	0.26
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X2-7	3.75	20	15	4.3
	5	48	10	5.7
	6.25	31	8	7.4
	7.5	59	5	8.6

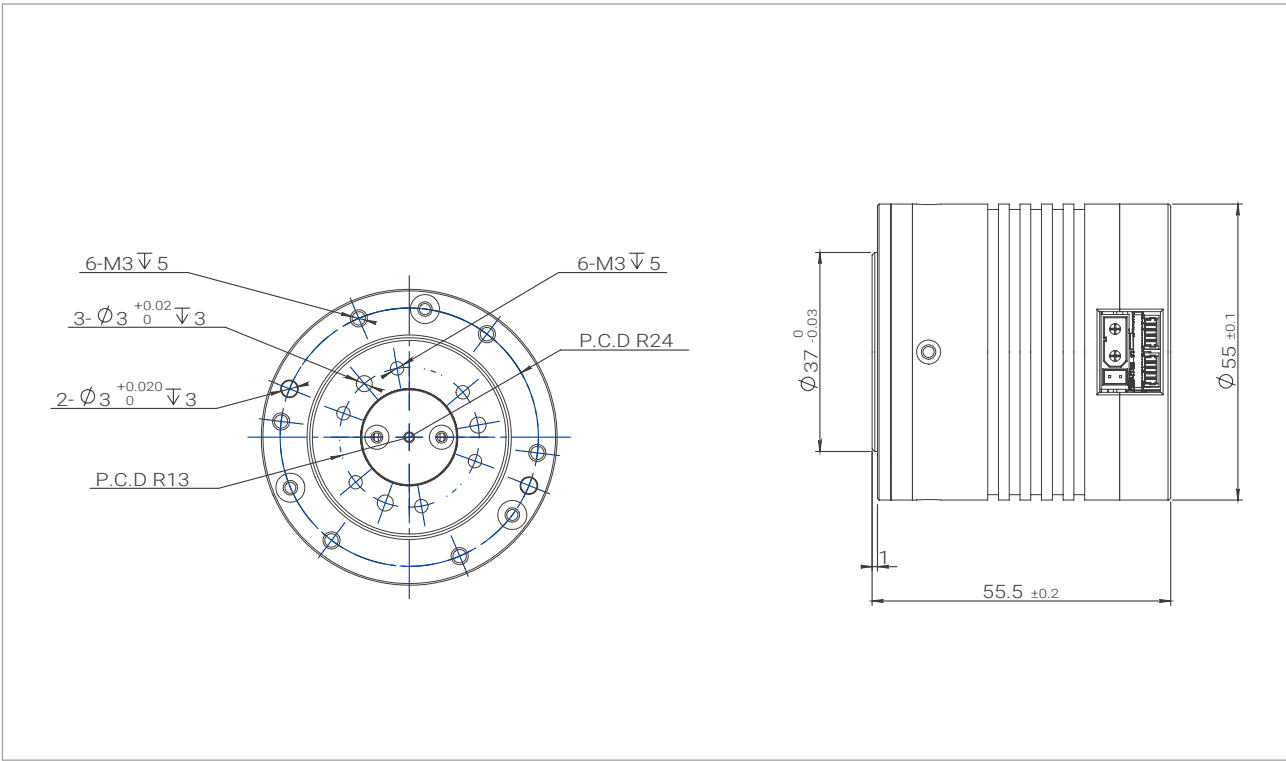
Series Name
RMD-X
Motor Simplified Name
X4-10



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X4-P12-10-E	N (without Brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters	Unit	X4-10
Gear Ratio	—	12.6
Input Voltage	V	24
No Load Speed	RPM	317
No-Load Input Current	A	1
Rated Speed	RPM	238
Rated Torque	N.m	4
Rated Output Power	W	100
Rated Phase Current	A(rms)	7.8
Peak Torque	N.m	10
Peak Phase Current	A(rms)	19.5
Efficiency	%	69.5
Motor Back-EMF Constant	Vdc/Krpm	6
Module Torque Constant	N.m/A	0.8
Motor Phase Resistance	Ω	0.32
Motor Phase Inductance	mH	0.14
Pole Pair	—	13
3 Phase Connection	—	Y
Back Drive Torque	N.m	0.8
Backlash	Arcmin	10
Output Bearing Type	—	Deep Groove Ball Bearings
Axial Load	Suffer	KN
	Stress	KN
Radial Load	KN	1.2
Inertia	Kg.cm ²	0.25
Encoder Type & Interface	—	Dual Encoder ABS-17BIT (Input) / 18BIT(Output)
Control Accuracy	Degree	<0.01
Communication	—	CAN BUS / EtherCAT
Weight	Kg	0.33
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X4-10	5.2	41	10	7.8
	6	16	8	9.1
	7.2	26	5	11
	8	30	3	12.1

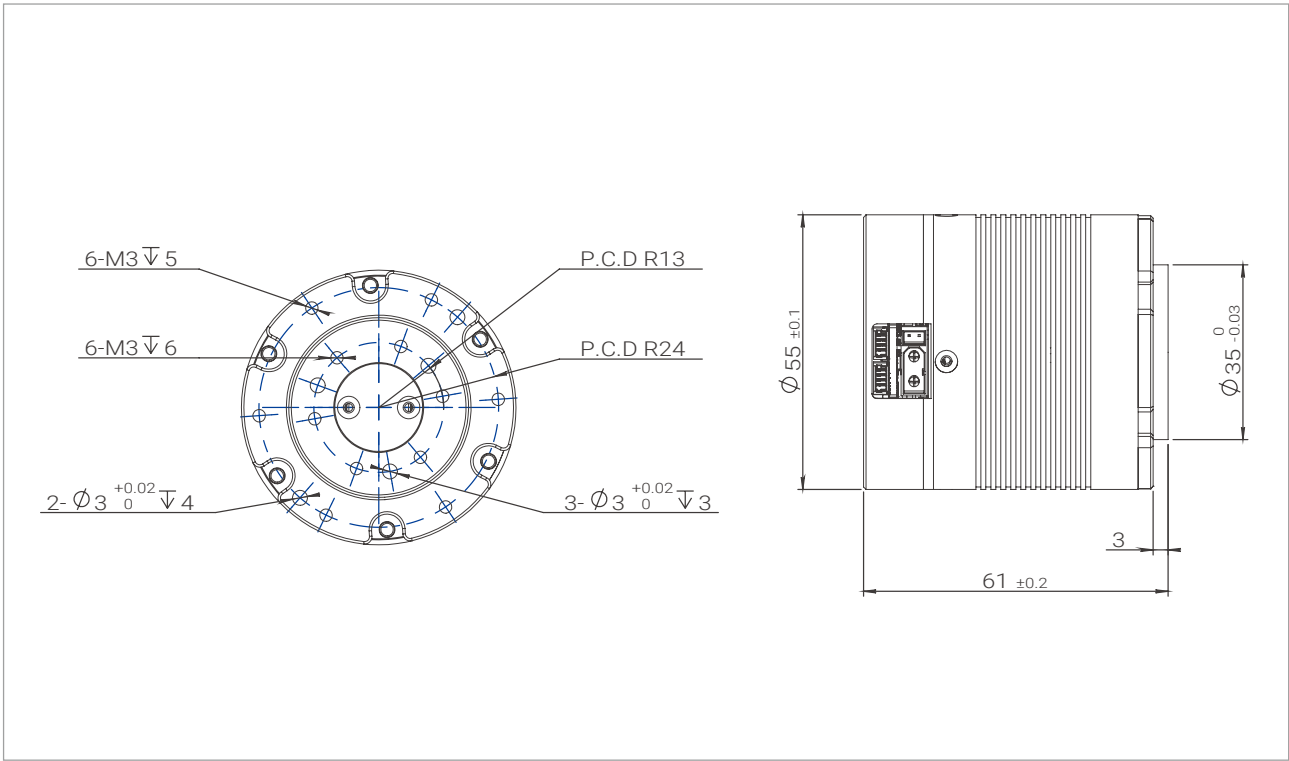
Series Name
RMD-X
Motor Simplified Name
X4-36



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X4-P36-36-E	N (without Brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters	Unit	X4-36
Gear Ratio	—	36
Input Voltage	V	24
No Load Speed	RPM	111
No-Load Input Current	A	0.9
Rated Speed	RPM	83
Rated Torque	N.m	10.5
Rated Output Power	W	100
Rated Phase Current	A(rms)	6.1
Peak Torque	N.m	34
Peak Phase Current	A(rms)	21.5
Efficiency	%	63.1
Motor Back-EMF Constant	Vdc/Krpm	6
Module Torque Constant	N.m/A	1.9
Motor Phase Resistance	Ω	0.35
Motor Phase Inductance	mH	0.17
Pole Pair	—	13
3 Phase Connection	—	Y
Back Drive Torque	N.m	1.14
Backlash	Arcmin	10
Output Bearing Type	—	Crossed Roller Bearings
Axial Load	Suffer	KN
	Stress	KN
Radial Load	KN	1.5
Inertia	Kg.cm ²	0.3
Encoder Type & Interface	—	Dual Encoder ABS-17BIT (Input) / 18BIT (Output)
Control Accuracy	Degree	<0.01
Communication	—	CAN BUS / EtherCAT
Weight	Kg	0.36
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X4-36	17.25	30	15	9.2
	23	58	10	12.7
	28.75	41	5	16.3
	34.5	50	3	21.2

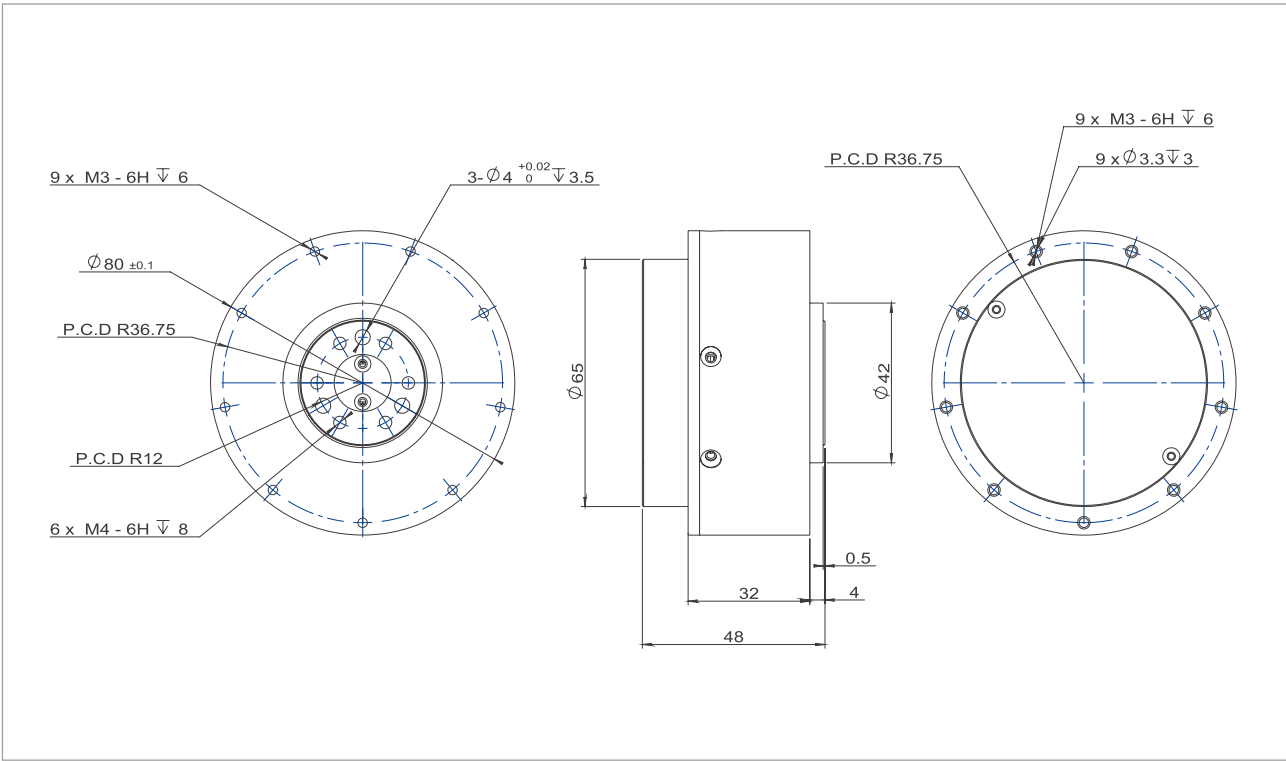
Series Name
RMD-X
Motor Simplified Name
X6-8



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X6-P8-8-C	N (without Brake)	CAN BUS

Installation Drawing



Product Parameters

Parameters	Unit	X6-8
Gear Ratio	—	8
Input Voltage	V	48
No Load Speed	RPM	387
No-Load Input Current	A	1.1
Rated Speed	RPM	310
Rated Torque	N.m	4.5
Rated Output Power	W	135
Rated Phase Current	A(rms)	3.6
Peak Torque	N.m	8
Peak Phase Current	A(rms)	7.2
Efficiency	%	78
Motor Back-EMF Constant	Vdc/Krpm	19.4
Module Torque Constant	N.m/A	1.3
Motor Phase Resistance	Ω	1.1
Motor Phase Inductance	mH	0.57
Pole Pair	—	14
3 Phase Connection	—	Y
Back Drive Torque	N.m	0.8
Backlash	Arcmin	10
Output Bearing Type	—	Deep Groove Ball Bearings
Axial Load	Suffer	KN
	Stress	KN
Radial Load	KN	1.04
Inertia	Kg.cm ²	0.61
Encoder Type & Interface	—	Dual Encoder ABS-17BIT(Input) / 18BIT(Output)
Control Accuracy	Degree	<0.01
Communication	—	CAN BUS
Weight	Kg	0.49
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X6-8	6.75	30	10	7
	9	49	8	9.5
	11.25	31	5	11.3
	13.5	19	3	12.7

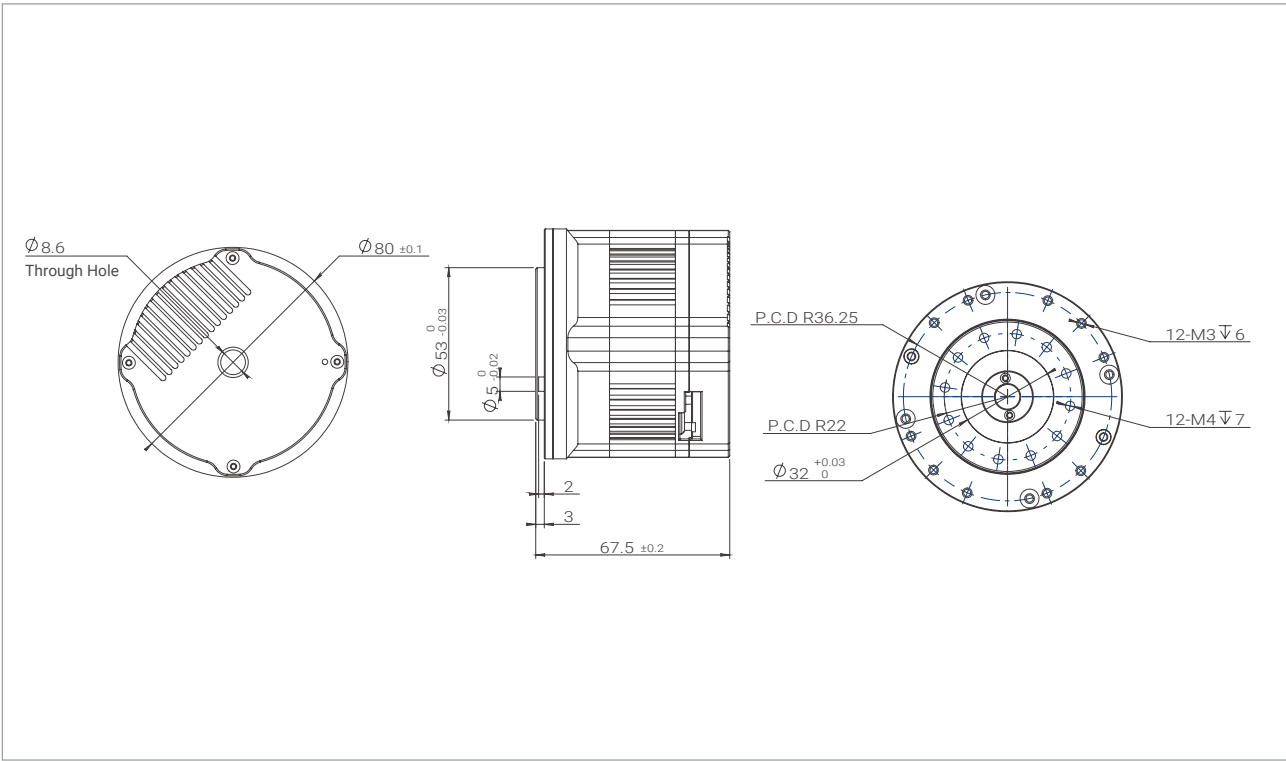
Series Name
RMD-X
Motor Simplified Name
X6-60



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X6-P20-60-E	N (without Brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters		Unit	X6-60
Gear Ratio		—	19.612
Input Voltage		V	48
No Load Speed		RPM	176
No-Load Input Current		A	0.9
Rated Speed		RPM	153
Rated Torque		N.m	20
Rated Output Power		W	320
Rated Phase Current		A(rms)	9.5
Peak Torque		N.m	60
Peak Phase Current		A(rms)	29.1
Efficiency		%	72.7
Motor Back-EMF Constant		Vdc/Krpm	16
Module Torque Constant		N.m/A	2.1
Motor Phase Resistance		Ω	0.41
Motor Phase Inductance		mH	0.51
Pole Pair		—	10
3 Phase Connection		—	Y
Back Drive Torque		N.m	1.6
Backlash		Arcmin	10
Output Bearing Type		—	Crossed Roller Bearings
Axial Load	Suffer	KN	1.8
	Stress	KN	0.8
Radial Load		KN	2
Inertia		Kg.cm²	0.66
Encoder Type & Interface		—	Dual Encoder ABS-17BIT(Input) / 17BIT(Output)
Control Accuracy		Degree	<0.01
Communication		—	CAN BUS & EtherCAT
Weight		Kg	0.82
Insulation Grade		—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X6-60	30	17	15	12.7
	40	29	10	17.7
	50	37	8	22.6
	60	24	5	28.3

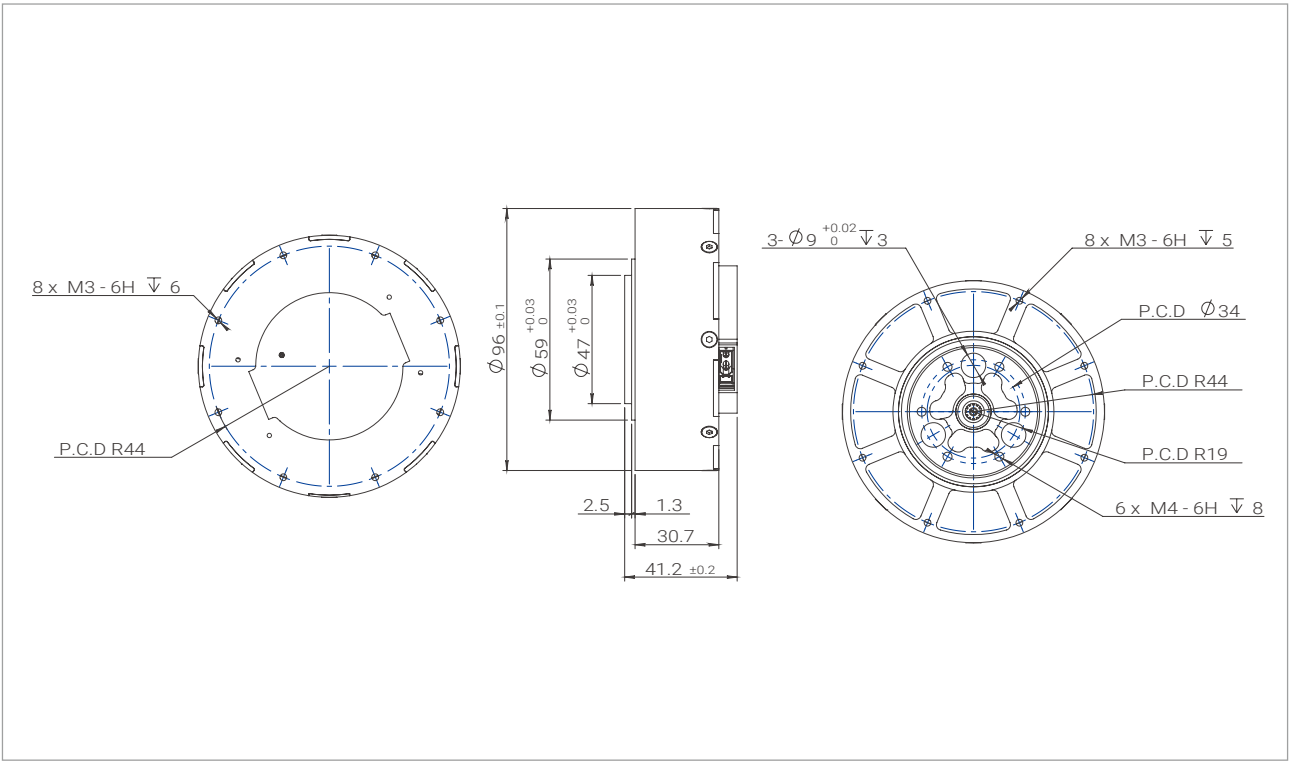
Series Name
RMD-X
Motor Simplified Name
X8-32



Single Encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X8-P9-32-R	N (without Brake)	RS485

Installation Drawing



Product Parameters

Parameters	Unit	X8-32
Gear Ratio	—	9
Input Voltage	V	24
No Load Speed	RPM	277
No-Load Input Current	A	0.9
Rated Speed	RPM	244
Rated Torque	N.m	8
Rated Output Power	W	204
Rated Phase Current	A(rms)	6.2
Peak Torque	N.m	32
Peak Phase Current	A(rms)	30
Efficiency	%	82
Motor Back-EMF Constant	Vdc/Krpm	10.9
Module Torque Constant	N.m/A	1.3
Phase Resistance	Ω	0.13
Phase Inductance	mH	0.081
Pole Pair	—	21
3 Phase Connection	—	△
Back Drive Torque	N.m	0.8
Backlash	Arcmin	8
Output Bearing Type	—	Deep Groove Ball Bearings
Axial Load	Suffer	KN
	Stress	KN
Radial Load	KN	2
Inertia	Kg.cm ²	1.43
Encoder Type & Interface	—	Single Encoder ABS-18BIT
Control Accuracy	Degree	<0.01
Communication	—	RS485
Weight	Kg	0.55
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X8-32	15.6	6	10	18.4
	18	11	8	21.2
	24	36	5	29
	30	54	3	38.2

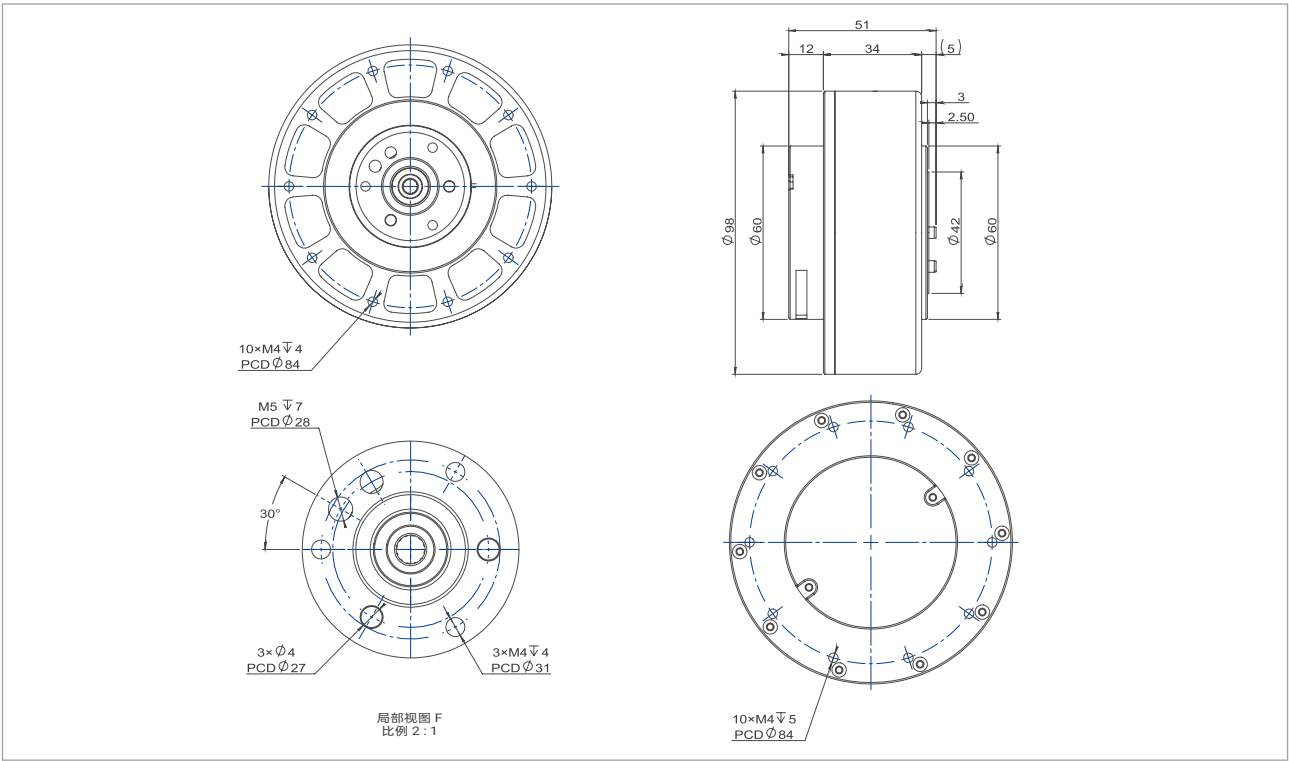
Series Name
RMD-X
Motor Simplified Name
X8-50



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X8-P9-50-C	N (without Brake)	CAN BUS
RMD-X8-P9-50-R	N (without Brake)	RS485

Installation Drawing



Product Parameters

Parameters	Unit	X8-50
Gear Ratio	—	9
Input Voltage	V	48
No Load Speed	RPM	137
No-Load Input Current	A	1.3
Rated Speed	RPM	110
Rated Torque	N.m	15
Rated Output Power	W	125
Rated Phase Current	A(rms)	3.2
Peak Torque	N.m	50
Peak Phase Current	A(rms)	16
Efficiency	%	80
Motor Back-EMF Constant	Vdc/Krpm	48.5
Module Torque Constant	N.m/A	4.7
Motor Phase Resistance	Ω	0.28
Motor Phase Inductance	mH	0.34
Pole Pair	—	21
3 Phase Connection	—	Y
Back Drive Torque	N.m	0.8
Backlash	Arcmin	10
Output Bearing Type	—	Deep groove ball bearings
Axial Load	Suffer	KN
	Stress	KN
Radial Load	KN	1.25
Inertia	Kg.cm ²	0.87
Encoder Type & Interface	—	Dual Encoder ABS-17BIT(Input) / 18BIT(Output)
Control Accuracy	Degree	-
Communication	—	RS485 / CAN BUS
Weight	Kg	0.72
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X8-50	20	8	10	6.4
	30	15	8	9.5
	40	9	5	12.7
	45	14	3	14.5

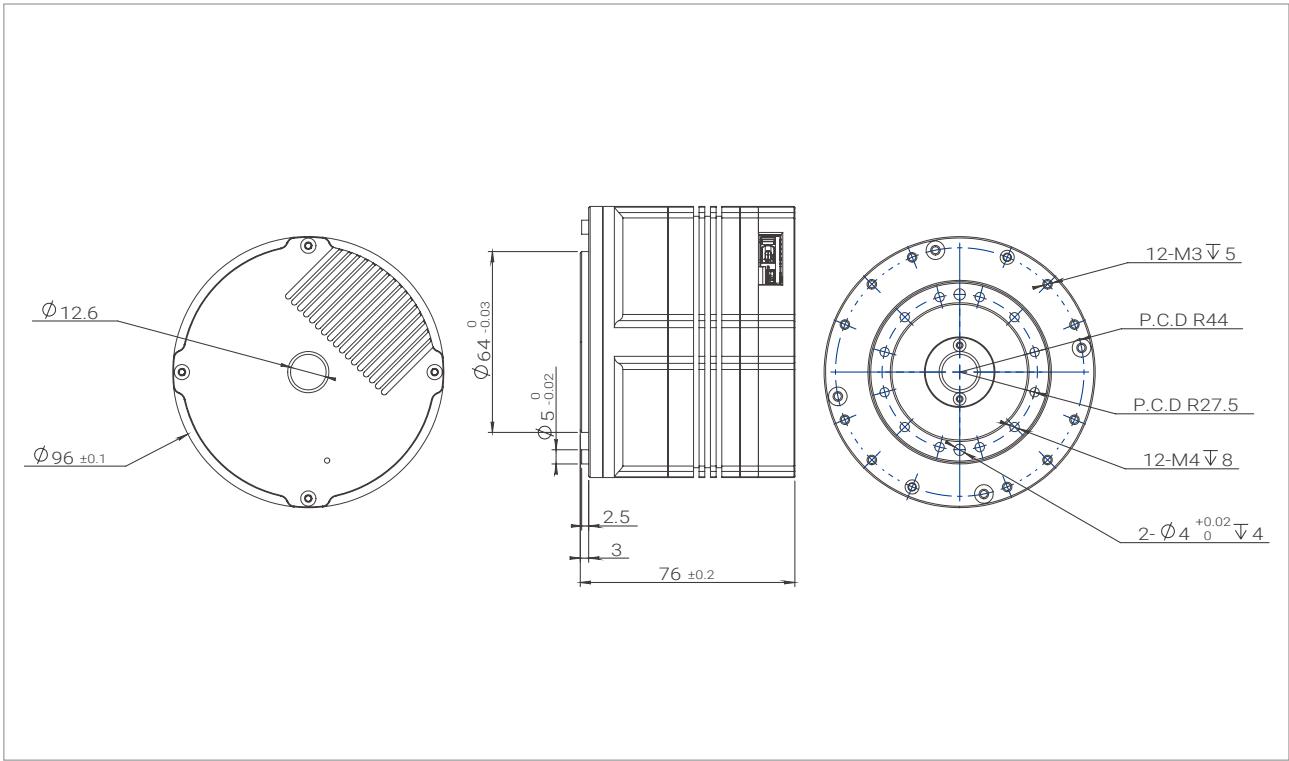
Series Name
RMD-X
Motor Simplified Name
X8-120



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X8-P20-120-E	N (without Brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters		Unit	X8-120
Gear Ratio		—	19.612
Input Voltage		V	48
No Load Speed		RPM	158
No-Load Input Current		A	1.6
Rated Speed		RPM	127
Rated Torque		N.m	43
Rated Output Power		W	574
Rated Phase Current		A(rms)	17.6
Peak Torque		N.m	120
Peak Phase Current		A(rms)	43.8
Efficiency		%	79
Motor Back-EMF Constant		Vdc/Krpm	19.2
Module Torque Constant		N.m/A	2.4
Motor Phase Resistance		Ω	0.18
Motor Phase Inductance		mH	0.31
Pole Pair		—	10
3 Phase Connection		—	Y
Back Drive Torque		N.m	3.21
Backlash		Arcmin	10
Output Bearing Type		—	Crossed Roller Bearings
Axial Load	Suffer	KN	4
	Stress	KN	1
Radial Load		KN	4.5
Inertia		Kg.cm²	1.5
Encoder Type & Interface		—	Dual Encoder ABS-17BIT (Input) / 17BIT (Output)
Control Accuracy		Degree	<0.01
Communication		—	CAN BUS / EtherCAT
Weight		Kg	1.40
Insulation Grade		—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X8-120	64.5	7	15	23.3
	86	10	10	31.1
	107.5	26	8	38.9
	129	30	5	43.8

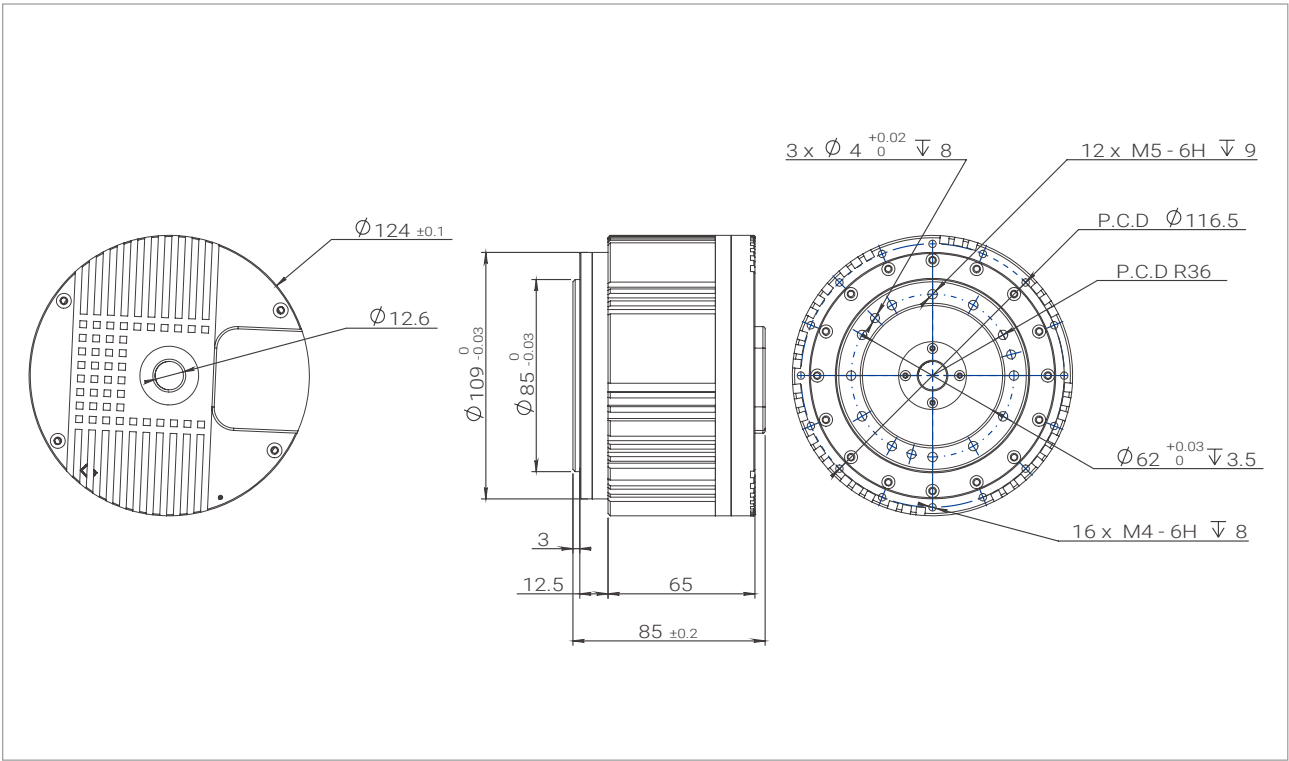
Series Name
RMD-X
Motor Simplified Name
X12-320



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X12-P20-320-E	N (without Brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters	Unit	X12-320
Gear Ratio	—	20
Input Voltage	V	48
No Load Speed	RPM	125
No-Load Input Current	A	2.7
Rated Speed	RPM	100
Rated Torque	N.m	85
Rated Output Power	W	900
Rated Phase Current	A(rms)	30
Peak Torque	N.m	320
Peak Phase Current	A(rms)	100
Efficiency	%	75
Motor Back-EMF Constant	Vdc/Krpm	17.9
Module Torque Constant	N.m/A	3.3
Motor Phase Resistance	Ω	0.12
Motor Phase Inductance	mH	0.05
Pole Pair	—	20
3 Phase Connection	—	Y
Back Drive Torque	N.m	3.8
Backlash	Arcmin	10
Output Bearing Type	—	Crossed Roller Bearings
Axial Load	Suffer	KN
	Stress	KN
Radial Load	KN	5
Inertia	Kg.cm ²	12.9
Encoder Type & Interface	—	Dual Encoder ABS-17BIT (Input) /17BIT (Output)
Control Accuracy	Degree	<0.01
Communication	—	CAN BUS / EtherCAT
Weight	Kg	2.37
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X12-320	150	13	10	37.5
	200	5	8	49.5
	250	7	7	61.5
	300	43	3	75.3

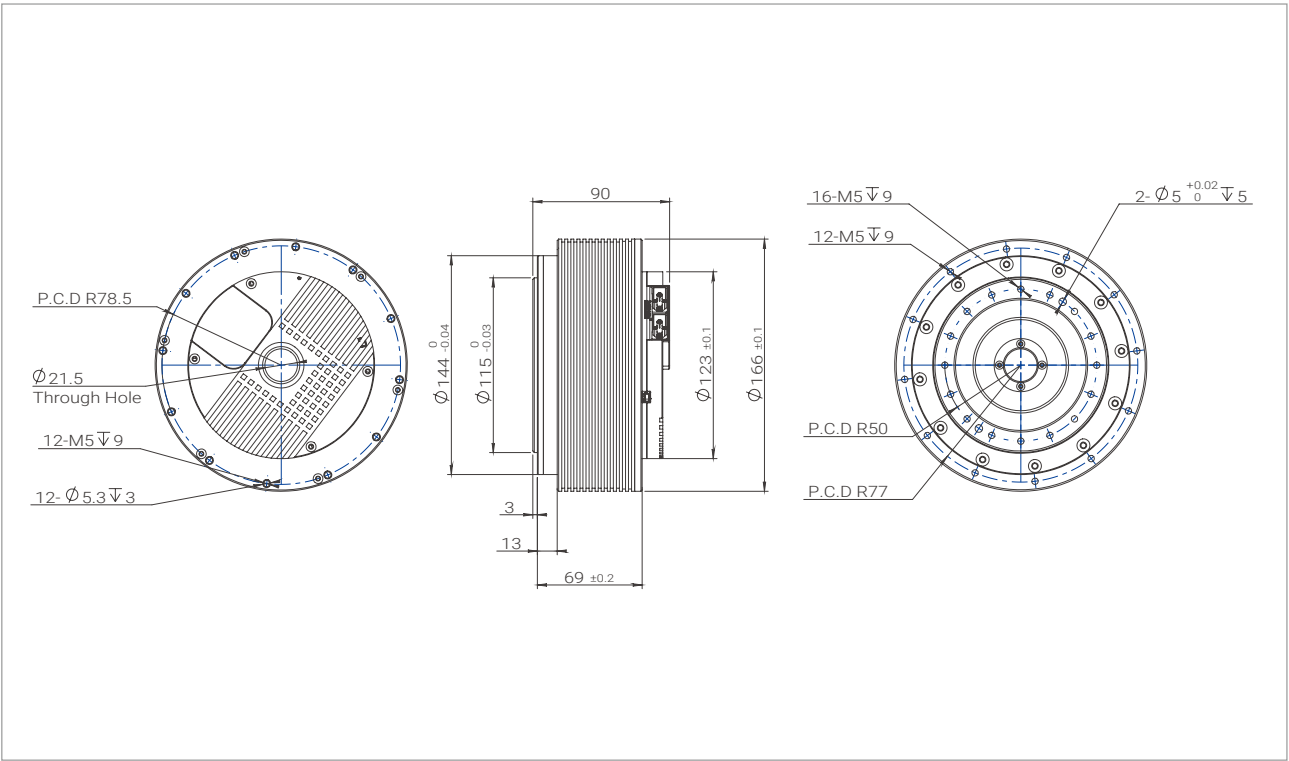
Series Name
RMD-X
Motor Simplified Name
X15-450



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-X15-P20-450-E	N (without Brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters			Unit	X15-450
Gear Ratio			—	20.25
Input Voltage			V	72
No Load Speed			RPM	108
No-Load Input Current			A	3.5
Rated Speed			RPM	98
Rated Torque			N.m	145
Rated Output Power			W	1480
Rated Phase Current			A(rms)	25
Peak Torque			N.m	450
Peak Phase Current			A(rms)	69.2
Efficiency			%	82.4
Motor Back-EMF Constant			Vdc/Krpm	29.9
Module Torque Constant			N.m/A	5.8
Motor Phase Resistance			Ω	0.08
Motor Phase Inductance			mH	0.14
Pole Pair			—	20
3 Phase Connection			—	Y
Back Drive Torque			N.m	4
Backlash			Arcmin	10
Output Bearing Type			—	Crossed Roller Bearings
Axial Load	Suffer		KN	5.4
	Stress		KN	5.4
Radial Load			KN	6
Inertia			Kg.cm²	31.6
Encoder Type & Interface			—	Dual Encoder ABS-17BIT (Input) / 17BIT (Output)
Control Accuracy			Degree	<0.01
Communication			—	CAN BUS / EtherCAT
Weight			Kg	3.50
Insulation Grade			—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
X15-450	217.5	15	15	31.1
	290	15	10	41
	362.5	20	8	51.6
	435	25	5	67.2

RH Series Harmonic Actuator

CAN BUS & EtherCAT



* The pictures of this series of products are subject to the actual products.

Product Features



High Precision



Low Noise



Hollow Design



High Efficiency

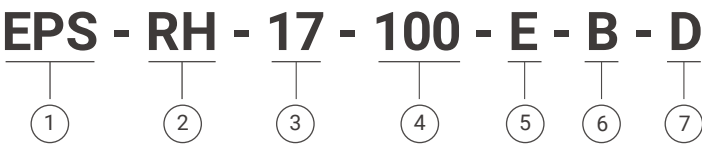


CAN BUS & EtherCAT



Dual Encoder

EPS-RH Series Naming Conventions



- 1

EPS:

Brand Name

E-Efficiency P-Precision S-Smart
- 2

RH:

RH Stands For The Series Name:

Harmonic Actuator
- 3

17:

Harmonic Model e.g:

14/17/20/25/32 etc
- 4

100:

Reduction ratio

100:1
- 5

E:

Communication

E: CAN BUS & EtherCAT
- 6

B:

N: Without Brake

B: with Brake
- 7

D:

D: Dual Encoder

S: Single Encoder

For example

Model	EPS-RH-20-100-E-N-D
Illustrate	EPS: Brand Name
	RH: Harmonic Module
	20: Harmonic Model
	100: Gear ratio 100:1
	E: Communication E: CAN BUS & EtherCAT
	N: Without Brake
	D: Dual encoder

Series Name

EPS-RH

Motor Simplified Name

RH-11



Dual encoder

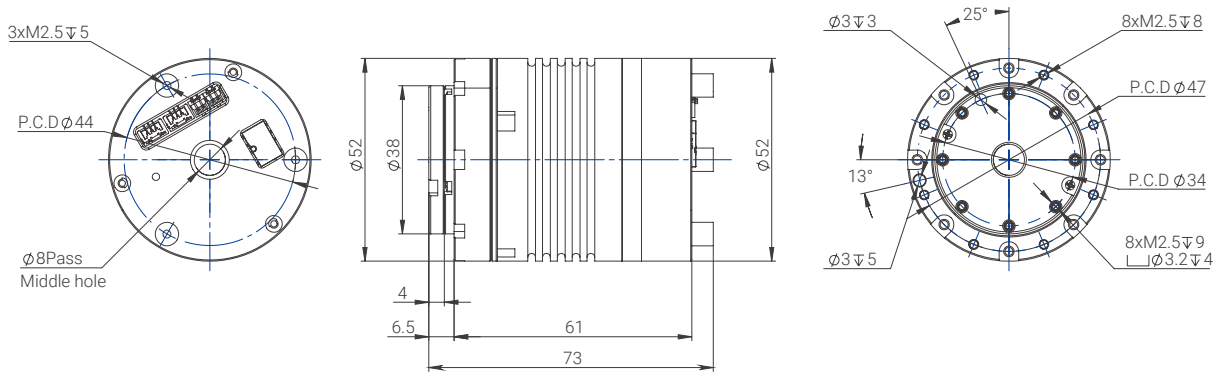
Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-RH-11-80-C-N-D	CAN BUS	N (without brake)	80

* 1. The standard model only supports without brake.

* 2. The standard model does not have waterproof function. For IP54, please contact sales before ordering.

Installation Drawing

EPS-RH-11-80-C-N-D



Product Parameters

Parameters	Unit	RH-11
Gear ratio	—	80
Input Voltage	V	24
No Load Speed	RPM	40
No Load Current	A	0.25
Rated Speed	RPM	25
Rated Torque	N.m	5
Rated Power	W	13
Rated Current	A(rms)	1.4
Peak Torque	N.m	21
Peak Current	A(rms)	5
Motor Back-EMF Constant	Vdc/Krpm	12
Module Torque Constant	N.m/A	3.65
Motor Phase Resistance	Ω	2.7
Motor Phase Inductance	mH	0.47
Pole Pair	—	13
3 Phase Connection	—	Y
Backlash	Arcsec	<40
Radial Load	Static Load	KN
	Dynamic Load	KN
Axial Load	Static Load	KN
	Dynamic Load	KN
Inertia	N	kg.m ²
	B	kg.m ²
Encoder Type	—	Dual Encoder ABS-14BIT (Input) / ABS-17BIT (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	N	Kg
	B	Kg
Insulation Grade	—	F

*** Rated torque test method:** When the ambient temperature is 24 degrees Celsius (no other heat dissipation methods), the test is performed at the rated speed. The motor torque reaches temperature balance under the condition of a temperature rise of 60 degrees Celsius, and the long-term working point is the rated torque value of the motor.

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
RH-11	7.5	11	15	2.1
	10	37	10	2.8
	15	44	8	4.1
	21	38	5	5.9

Series Name

EPS-RH

Motor Simplified Name

RH-14



Dual encoder

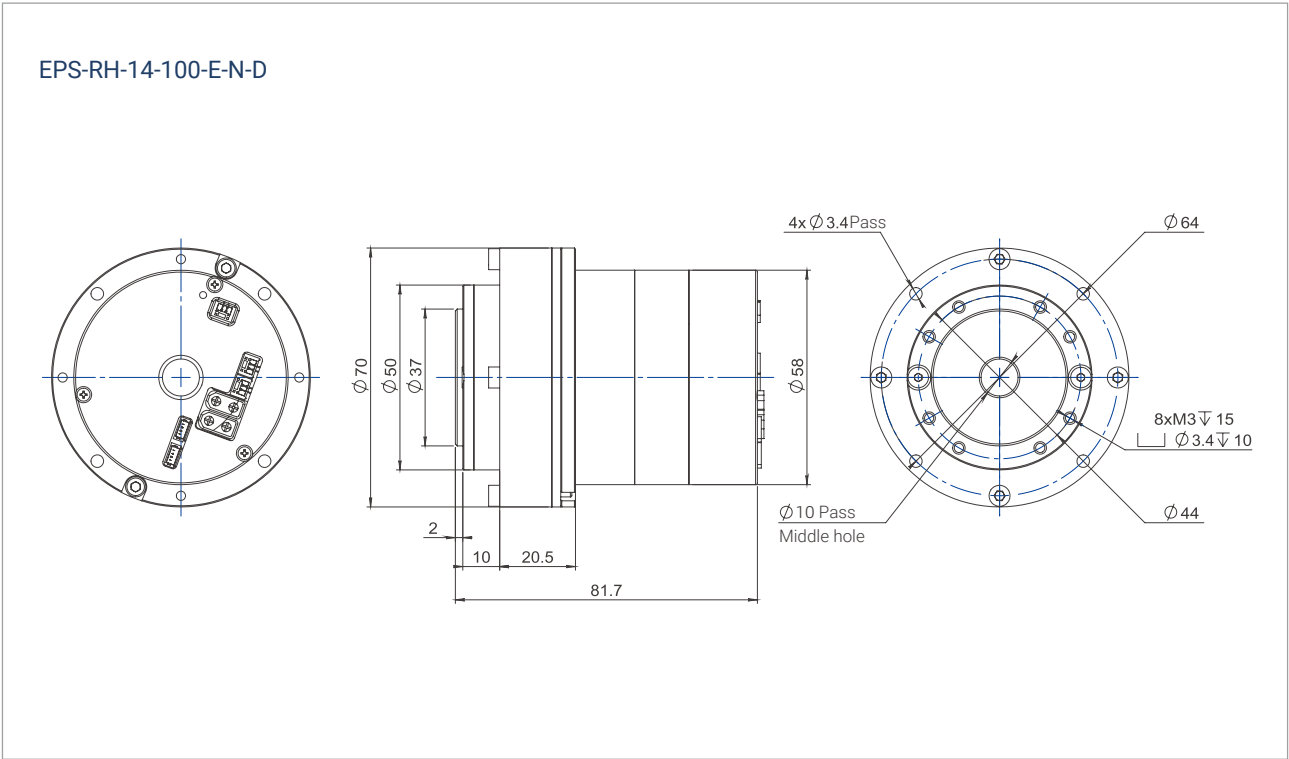
Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-RH-14-100-E-N-D	CAN BUS & EtherCAT	N (without brake)	100

* 1. The standard model only supports without brake.

* 2. The standard model does not have waterproof function. For IP54, please contact sales before ordering.

* 3. 50:1 gear ratio is available, please contact sales to confirm the lead time before ordering.

Installation Drawing



Product Parameters

Parameters		Unit	RH-14
Gear ratio		—	50 100
Input Voltage		V	48
No Load Speed		RPM	60 30
No Load Current		A	0.4 0.4
Rated Speed		RPM	50 25
Rated Torque		N.m	5.5 11
Rated Power		W	28
Rated Current		A(rms)	2.8
Peak Torque		N.m	14 28
Peak Current		A(rms)	5.7
Motor Back-EMF Constant		Vdc/Krpm	19.2
Module Torque Constant		N.m/A	2 4
Motor Phase Resistance		Ω	0.62
Motor Phase Inductance		mH	0.43
Pole Pair		—	10
3 Phase Connection		—	Y
Backlash		Arcsec	< 40
Radial Load	Static Load	KN	8.6
	Dynamic Load	KN	5.8
Axial Load	Static Load	KN	58.7
	Dynamic Load	KN	16.2
Inertia	N	kg.m ²	0.11 0.29
	B	kg.m ²	— —
Encoder Type		—	Dual Encoder ABS-17BIT (Input) ABS-17BIT (Output)
Repeat Position Accuracy		Degree	< 0.01
Communication		—	CAN BUS & EtherCAT
Weight	N	Kg	0.78
	B	Kg	/
Insulation Grade		—	F

* **Rated torque test method:** When the ambient temperature is 24 degrees Celsius (no other heat dissipation methods), the test is performed at the rated speed. The motor torque reaches temperature balance under the condition of a temperature rise of 60 degrees Celsius, and the long-term working point is the rated torque value of the motor.

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
RH-14	16.5	3	15	3.7
	22	4	10	4.8
	27.5	6	8	6.6

Series Name

EPS-RH

Motor Simplified Name

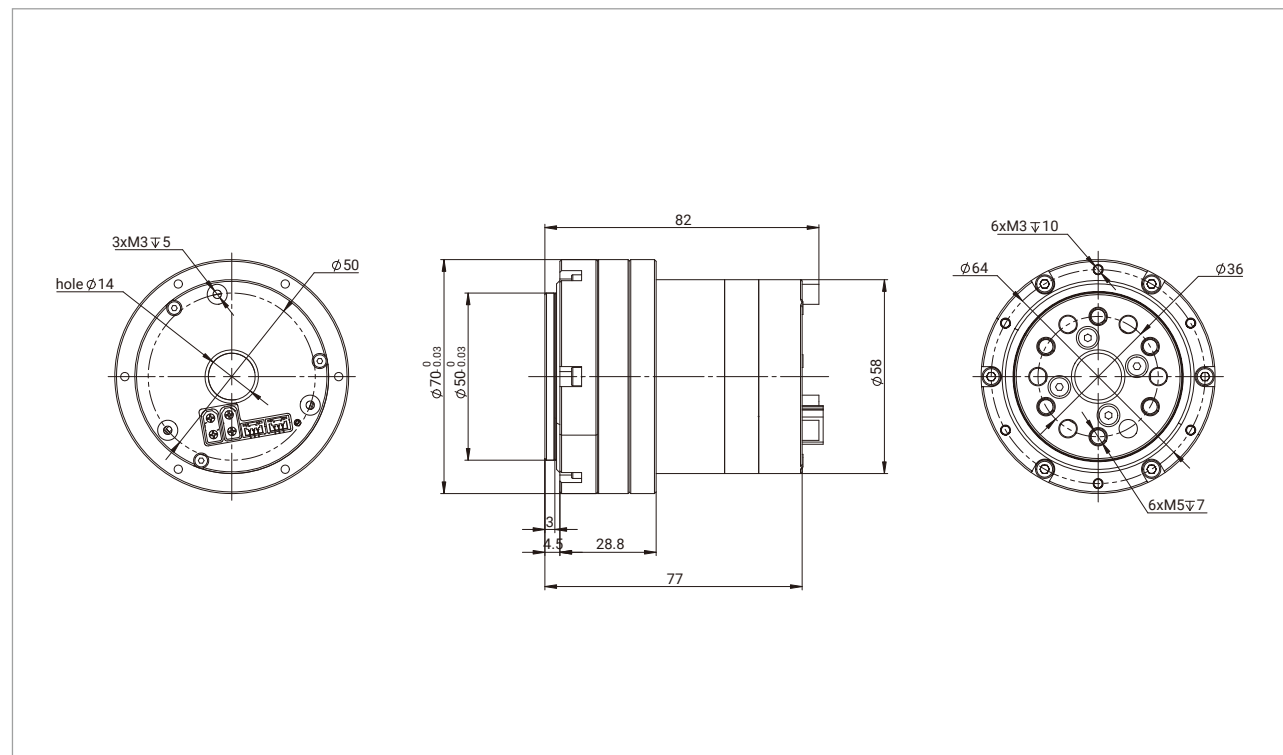
RH-17 Lite



Dual encoder

Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-RH-17-Lite-100-C-N-D	CAN BUS	N (without brake)	100

■ Installation Drawing



■ Product Parameters

Parameters	Unit	RH-17 Lite
Gear ratio	—	100
Input Voltage	V	24
No Load Speed	RPM	25
No Load Current	A	0.25
Rated Speed	RPM	20
Rated Torque	N.m	26
Rated Power	W	55
Rated Current	A(rms)	5.0
Peak Torque	N.m	65.0
Peak Current	A(rms)	12.6
Motor Back-EMF Constant	Vdc/Krpm	12.00
Module Torque Constant	N.m/A	5.17
Motor Phase Resistance	Ω	0.94
Motor Phase Inductance	mH	0.41
Pole Pair	—	10
3 Phase Connection	—	Y
Backlash	Arcsec	<40
Inertia	kg.m²	0.32
Encoder Type	—	Dual Encoder ABS-14BIT (Input) / ABS-17BIT (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	Kg	0.8
Insulation Grade	—	F

* **Rated torque test method:** When the ambient temperature is 24 degrees Celsius (no other heat dissipation methods), the test is performed at the rated speed. The motor torque reaches temperature balance under the condition of a temperature rise of 60 degrees Celsius, and the long-term working point is the rated torque value of the motor.

■ Stall Torque Data

RH-17 Lite	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
	39	26	15	7.5
	47	44	10	9.2
	52	47	8	10
	65	36	5	12.3

Series Name

EPS-RH

Motor Simplified Name

RH-17



Dual encoder

Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-RH-17-100-E-N-D	CAN BUS & EtherCAT	N (without brake)	100
EPS-RH-17-100-E-B-D	CAN BUS & EtherCAT	B (with brake)	100

* 1. The standard model supports brake or without brake.

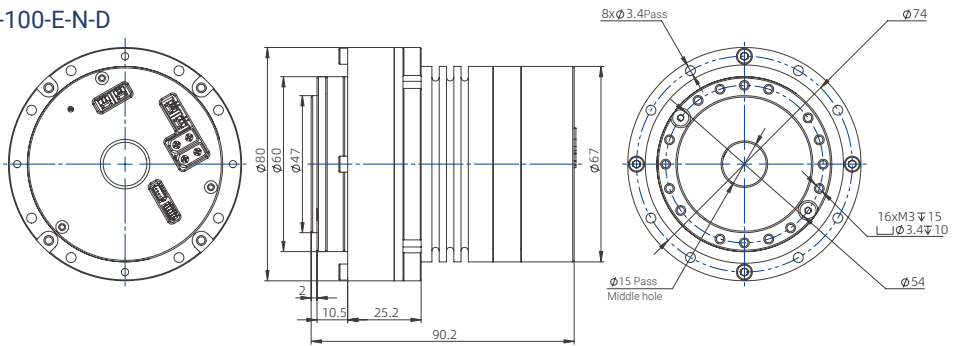
* 2. The standard model supports both CAN BUS & EtherCAT.

* 3. The standard model does not have waterproof function. For IP54, please contact sales before ordering.

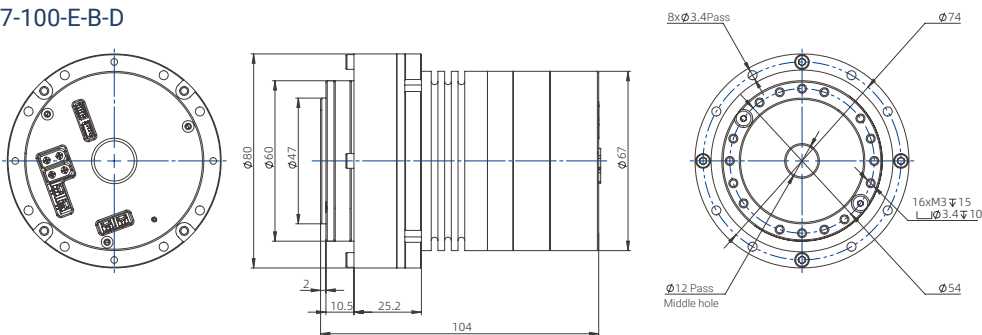
* 4. 50:1 gear ratio is available, please contact sales to confirm the lead time before ordering.

Installation Drawing

EPS-RH-17-100-E-N-D



EPS-RH-17-100-E-B-D



Product Parameters

Parameters		Unit	RH-17
Gear ratio		—	50 100
Input Voltage		V	48
No Load Speed		RPM	60 30
No Load Current		A	0.6 0.6
Rated Speed		RPM	50 25
Rated Torque		N.m	17.5 35
Rated Power		W	91
Rated Current		A(rms)	4.7
Peak Torque		N.m	27 54
Peak Current		A(rms)	7.4
Motor Back-EMF Constant		Vdc/Krpm	19.2
Module Torque Constant		N.m/A	3.7 7.4
Motor Phase Resistance		Ω	0.7
Motor Phase Inductance		mH	0.47
Pole Pair		—	10
3 Phase Connection		—	Y
Backlash		Arcsec	< 40
Radial Load	Static Load	KN	16.3
	Dynamic Load	KN	10.4
Axial Load	Static Load	KN	78.2
	Dynamic Load	KN	20.8
Inertia	N	kg.m ²	0.16 0.52
	B	kg.m ²	0.18 0.56
Encoder Type		—	Dual Encoder ABS-17BIT (Input) ABS-17BIT (Output)
Repeat Position Accuracy		Degree	< 0.01
Communication		—	CAN BUS & EtherCAT
Weight	N	Kg	1.11
	B	Kg	1.28
Insulation Grade		—	F

*** Rated torque test method:** When the ambient temperature is 24 degrees Celsius (no other heat dissipation methods), the test is performed at the rated speed. The motor torque reaches temperature balance under the condition of a temperature rise of 60 degrees Celsius, and the long-term working point is the rated torque value of the motor.

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
RH-17	52.5	15	15	6.9
	70	7	10	9.8
	87.5	35	8	11.9

Series Name

EPS-RH

Motor Simplified Name

RH-20



Dual encoder

Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-RH-20-100-E-N-D	CAN BUS & EtherCAT	N (without brake)	100
EPS-RH-20-100-E-B-D	CAN BUS & EtherCAT	B (with brake)	100

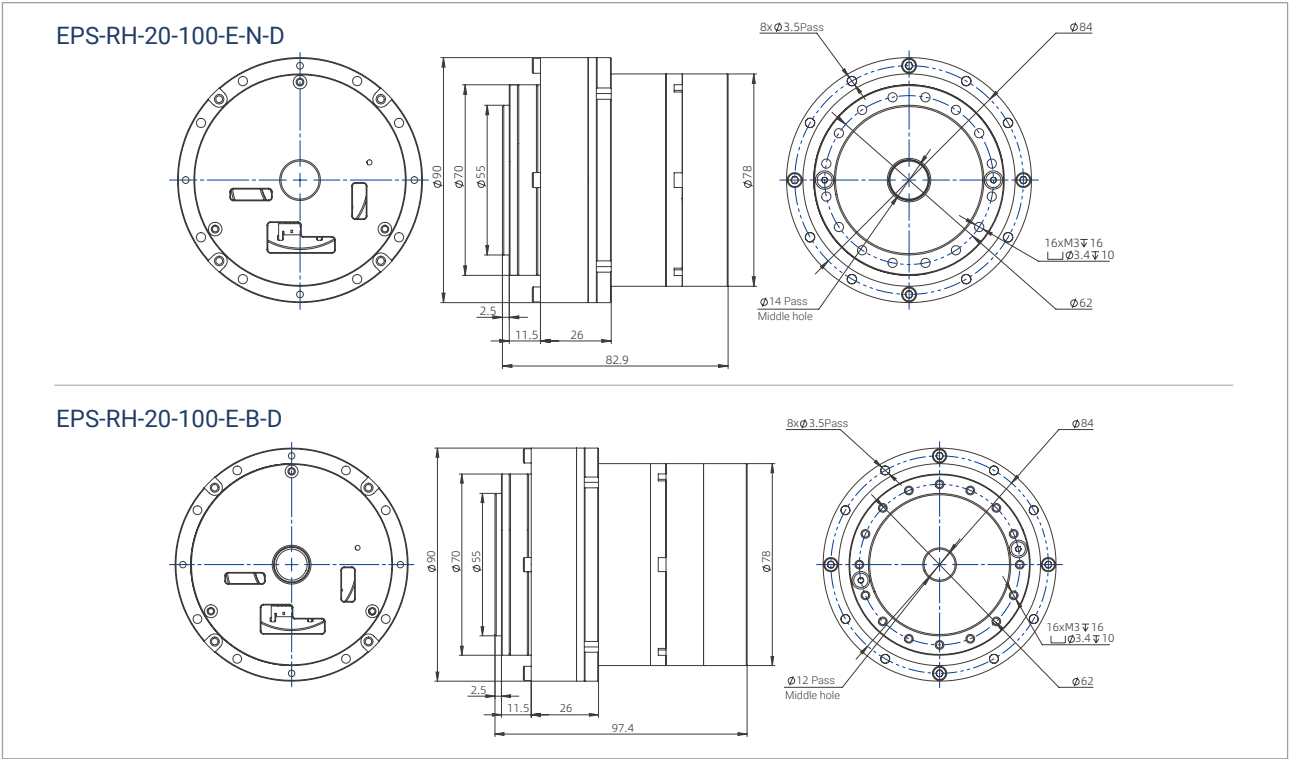
* 1. The standard model supports brake or without brake.

* 2. The standard model supports both CAN BUS & EtherCAT.

* 3. The standard model does not have waterproof function. For IP54, please contact sales before ordering.

* 4. 50:1 gear ratio is available, please contact sales to confirm the lead time before ordering.

Installation Drawing



Product Parameters

Parameters	Unit	RH-20
Gear ratio	—	50 100
Input Voltage	V	48
No Load Speed	RPM	60 30
No Load Current	A	1.1 1.2
Rated Speed	RPM	50 25
Rated Torque	N.m	25 50
Rated Power	W	130
Rated Current	A(rms)	6.8
Peak Torque	N.m	40 80
Peak Current	A(rms)	10.6
Motor Back-EMF Constant	Vdc/Krpm	19.2
Module Torque Constant	N.m/A	3.65 7.3
Motor Phase Resistance	Ω	0.36
Motor Phase Inductance	mH	0.51
Pole Pair	—	10
3 Phase Connection	—	Y
Backlash	Arcsec	< 40
Radial Load	Static Load	KN
	Dynamic Load	KN
Axial Load	Static Load	KN
	Dynamic Load	KN
Inertia	N	kg.m ²
	B	kg.m ²
Encoder Type	—	Dual Encoder ABS-17BIT (Input) ABS-17BIT (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS & EtherCAT
Weight	N	Kg
	B	Kg
Insulation Grade	—	F

*** Rated torque test method:** When the ambient temperature is 24 degrees Celsius (no other heat dissipation methods), the test is performed at the rated speed. The motor torque reaches temperature balance under the condition of a temperature rise of 60 degrees Celsius, and the long-term working point is the rated torque value of the motor.

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
RH-20	75	12	15	10.5
	100	8	10	13.5
	125	15	8	18

Series Name
EPS-RH

Motor Simplified Name

RH-25



Dual encoder

Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-RH-25-100-E-N-D	CAN BUS & EtherCAT	N (without brake)	100
EPS-RH-25-100-E-B-D	CAN BUS & EtherCAT	B (with brake)	100

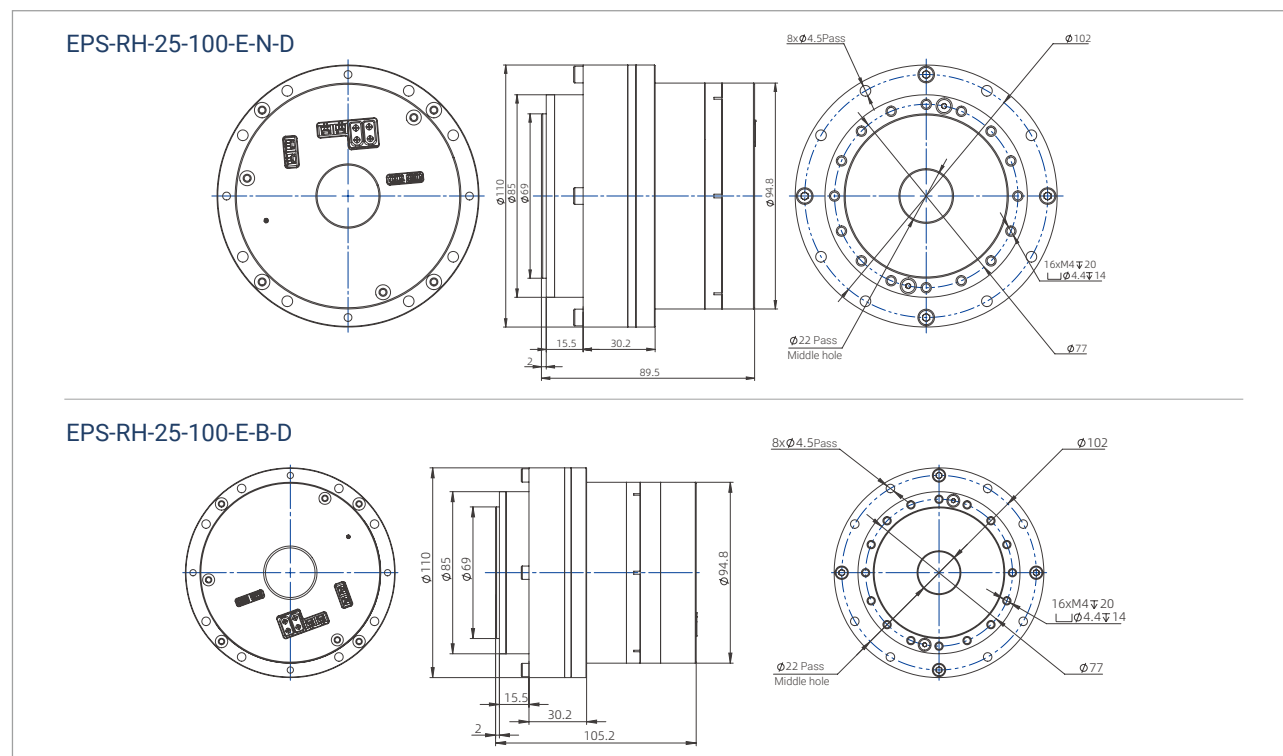
* 1. The standard model supports brake or without brake.

* 2. The standard model supports both CAN BUS & EtherCAT.

* 3. The standard model does not have waterproof function. For IP54, please contact sales before ordering.

* 4. 50:1 gear ratio is available, please contact sales to confirm the lead time before ordering.

■ Installation Drawing



■ Product Parameters

Parameters		Unit	RH-25
Gear ratio		—	50 100
Input Voltage		V	48
No Load Speed		RPM	60 30
No Load Current		A	1.8 2
Rated Speed		RPM	50 25
Rated Torque		N.m	54 108
Rated Power		W	282
Rated Current		A(rms)	10.6
Peak Torque		N.m	78.5 157
Peak Current		A(rms)	14.1
Motor Back-EMF Constant		Vdc/Krpm	19.2
Module Torque Constant		N.m/A	5.05 10.1
Motor Phase Resistance		Ω	0.16
Motor Phase Inductance		mH	0.33
Pole Pair		—	10
3 Phase Connection		—	Y
Backlash		Arcsec	< 40
Radial Load	Static Load	KN	35.8
	Dynamic Load	KN	21.8
Axial Load	Static Load	KN	198.2
	Dynamic Load	KN	49.1
Inertia	N	kg.m²	0.51 1.94
	B	kg.m²	0.59 2.32
Encoder Type		—	Dual Encoder ABS-17BIT (Input) ABS-17BIT (Output)
Repeat Position Accuracy		Degree	< 0.01
Communication		—	CAN BUS & EtherCAT
Weight	N	Kg	2.42
	B	Kg	2.74
Insulation Grade		—	F

* **Rated torque test method:** When the ambient temperature is 24 degrees Celsius (no other heat dissipation methods), the test is performed at the rated speed. The motor torque reaches temperature balance under the condition of a temperature rise of 60 degrees Celsius, and the long-term working point is the rated torque value of the motor.

■ Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
RH-25	162	3	15	18
	216	9	10	22.1
	256	39	5	24.6

Series Name

EPS-RH

Motor Simplified Name

RH-32



Dual encoder

Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-RH-32-100-E-N-D	CAN BUS & EtherCAT	N (without brake)	100
EPS-RH-32-100-E-B-D	CAN BUS & EtherCAT	B (with brake)	100

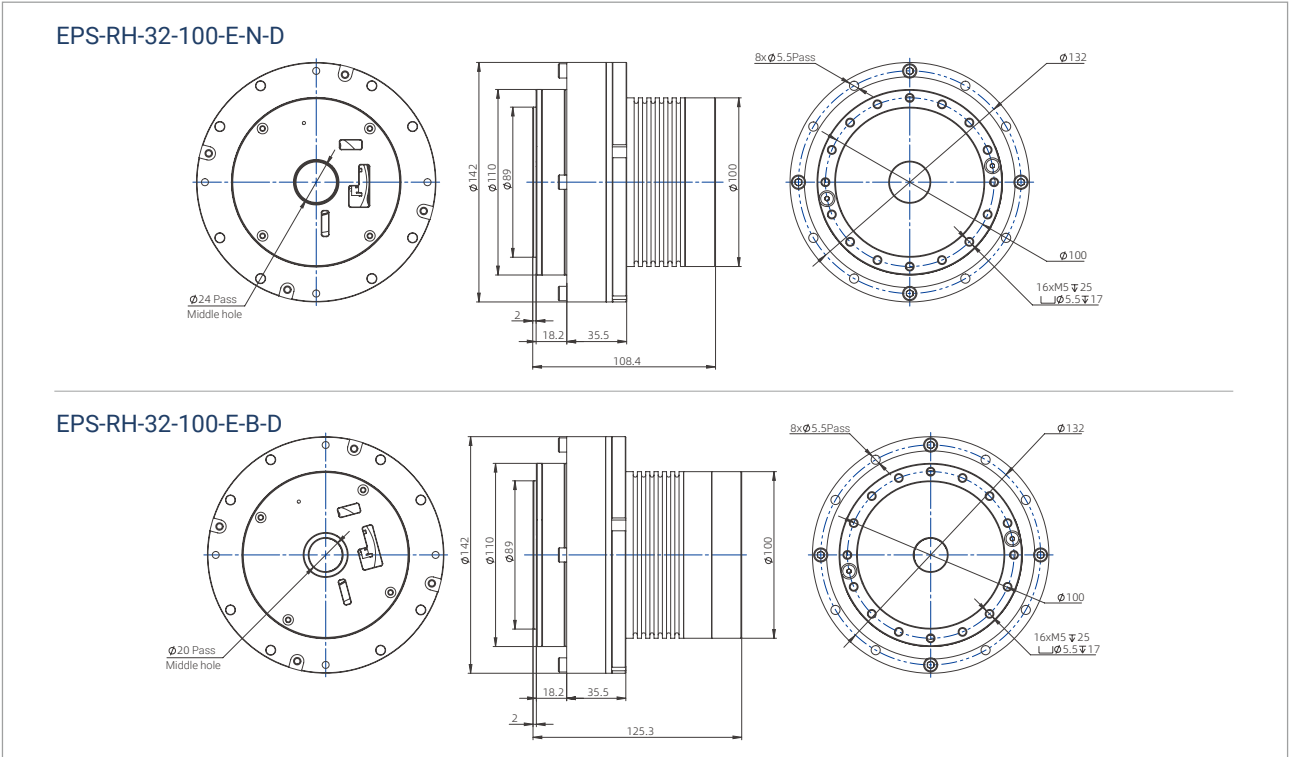
* 1. The standard model supports brake or without brake.

* 2. The standard model supports both CAN BUS & EtherCAT.

* 3. The standard model does not have waterproof function. For IP54, please contact sales before ordering.

* 4. 50:1 gear ratio is available, please contact sales to confirm the lead time before ordering.

Installation Drawing



Product Parameters

Parameters	Unit	RH-32
Gear ratio	—	50 100
Input Voltage	V	48
No Load Speed	RPM	40 20
No Load Current	A	2.6 2.8
Rated Speed	RPM	36 18
Rated Torque	N.m	75 150
Rated Power	W	282
Rated Current	A(rms)	21.8
Peak Torque	N.m	114.5 229
Peak Current	A(rms)	32.1
Motor Back-EMF Constant	Vdc/Krpm	26.7
Module Torque Constant	N.m/A	3.45 6.9
Motor Phase Resistance	Ω	0.08
Motor Phase Inductance	mH	0.18
Pole Pair	—	10
3 Phase Connection	—	Y
Backlash	Arcsec	< 40
Radial Load	Static Load	KN
	Dynamic Load	KN
Axial Load	Static Load	KN
	Dynamic Load	KN
Inertia	N	kg.m ²
	B	kg.m ²
Encoder Type	—	Dual Encoder ABS-17BIT (Input) ABS-17BIT (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS & EtherCAT
Weight	N	Kg
	B	Kg
Insulation Grade	—	F

*** Rated torque test method:** When the ambient temperature is 24 degrees Celsius (no other heat dissipation methods), the test is performed at the rated speed. The motor torque reaches temperature balance under the condition of a temperature rise of 60 degrees Celsius, and the long-term working point is the rated torque value of the motor.

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
RH-32	195	24	10	24
	225	40	8	28
	270	49	5	33.5

HM Series Harmonic Mini Actuator

CAN BUS



* The pictures of this series of products are subject to the actual products.

Product Features



Small size



LightWeight



Low Noise



High Efficiency



CAN BUS



Dual Encoder

EPS-HM Series Naming Conventions

EPS - HM - 70 - 26 - 100 - C

1

2

3

4

5

6

- 1

EPS:

Brand Name E-Efficiency P-Precision S-Smart
- 2

HM:

HM Stands For The Series Name: Harmonic Mini Actuator
- 3

70:

Module outer diameter 70mm
- 4

26:

Rated Torque 26 N.m
- 5

100:

Gear ratio 100:1
- 6

C:

Communication CAN BUS

For example

Model	EPS-HM-50-05-80-C
Illustrate	EPS: Brand Name
	HM: Harmonic Mini Actuator
	50: Module outer diameter
	05: Rated Torque
	80: Gear ratio
	C: Communication CAN BUS

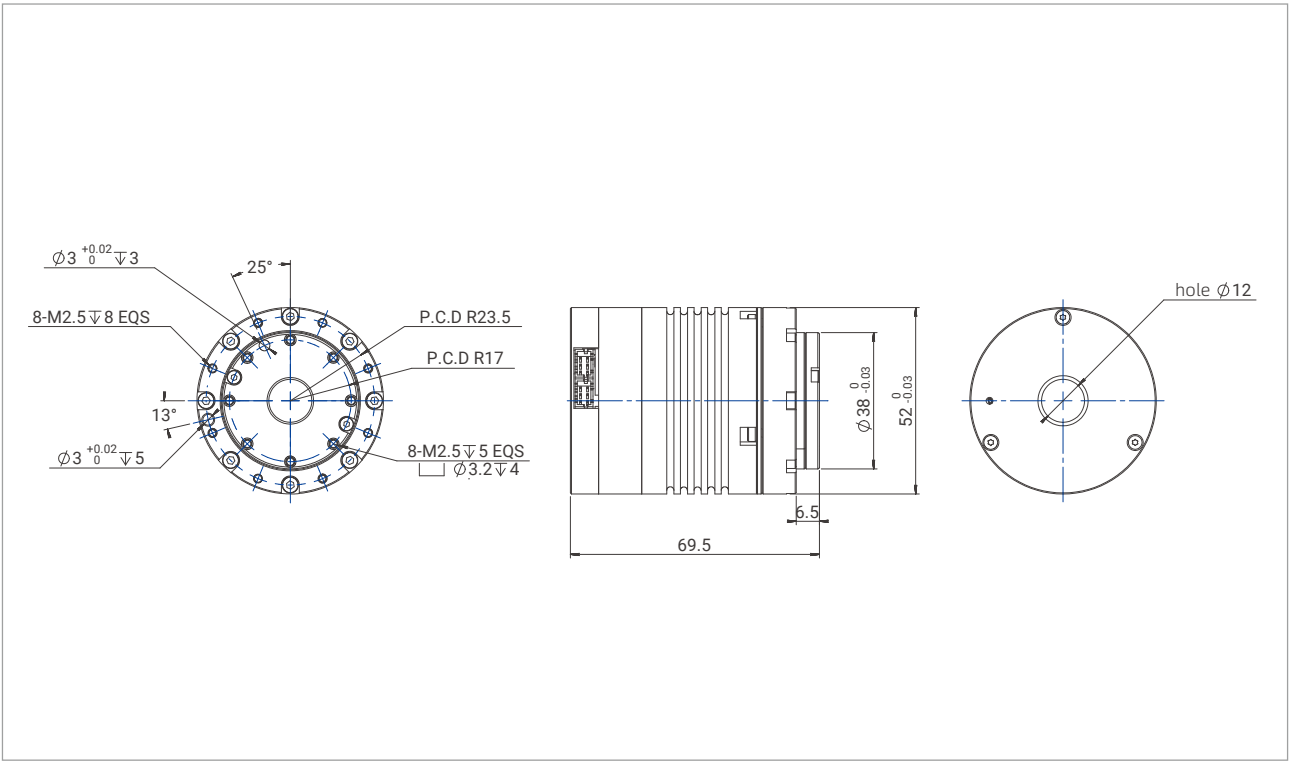
Series Name
EPS-HM
Motor Simplified Name
HM-50



Dual encoder

Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-HM-50-05-80-C	CAN BUS	N (without brake)	80

Installation Drawing



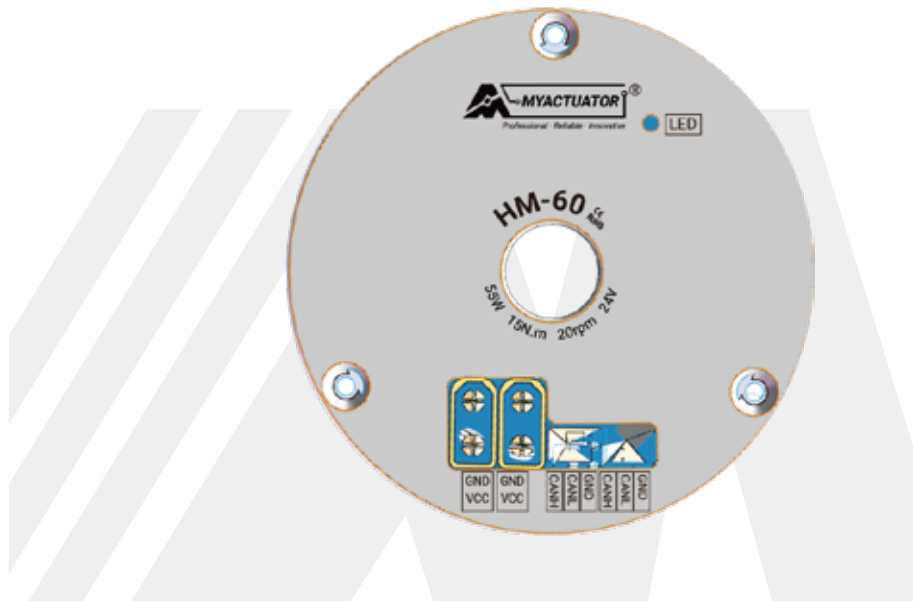
Product Parameters

Parameters	Unit	HM-50
Gear ratio	—	80
Input Voltage	V	24
No Load Speed	RPM	40
No Load Current	A	0.25
Rated Speed	RPM	25
Rated Torque	N.m	5
Rated Power	W	13
Rated Current	A(rms)	1.4
Peak Torque	N.m	21
Peak Current	A(rms)	5
Motor Back-EMF Constant	Vdc/Krpm	12
Module Torque Constant	N.m/A	3.65
Motor Phase Resistance	Ω	2.7
Motor Phase Inductance	mH	0.47
Pole Pair	—	13
3 Phase Connection	—	Y
Backlash	Arcsec	<40
Inertia	kg.m ²	0.29
Encoder Type	—	Dual Encoder ABS-14BIT (Input) / ABS-17BIT (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	Kg	0.38
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
HM-50	7.5	11	15	2.1
	10	37	10	2.8
	15	44	8	4.1
	21	38	5	5.9

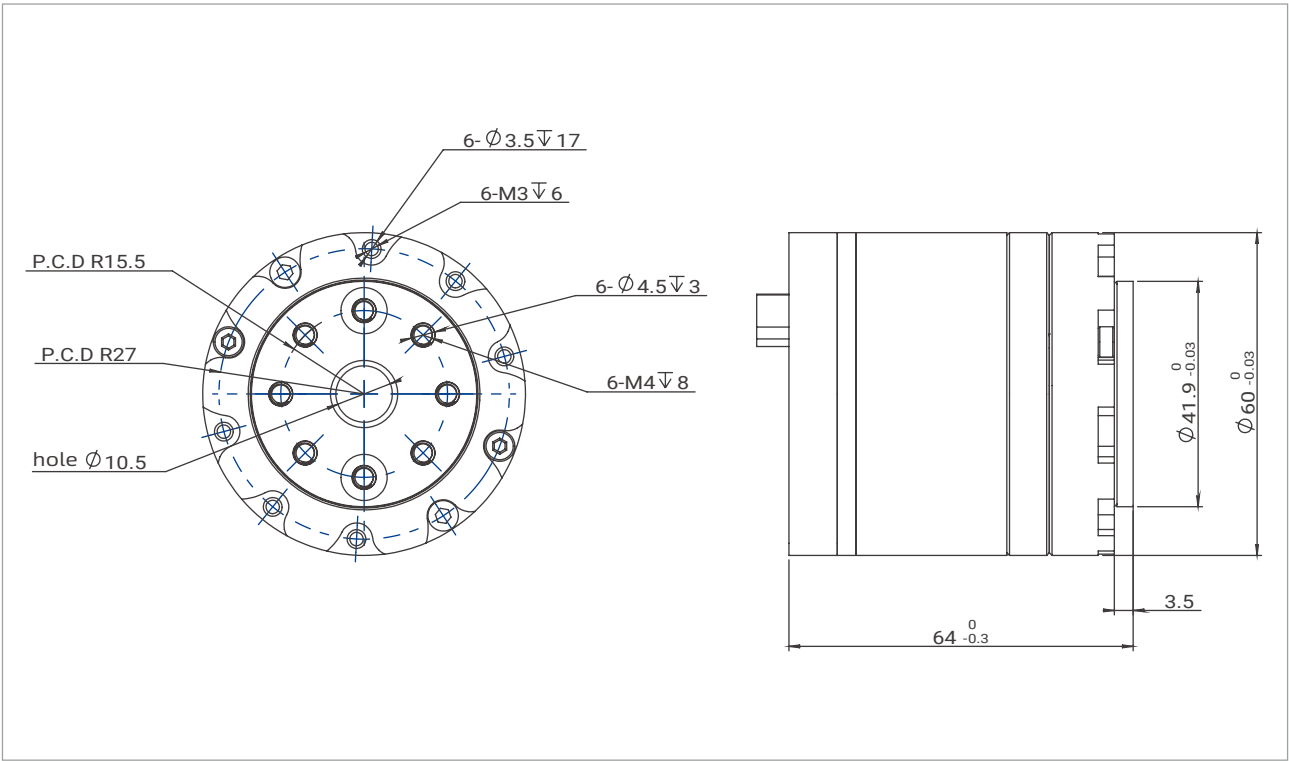
Series Name
EPS-HM
Motor Simplified Name
HM-60



Dual encoder

Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-HM-60-15-100-C	CAN BUS	N (without brake)	100

Installation Drawing



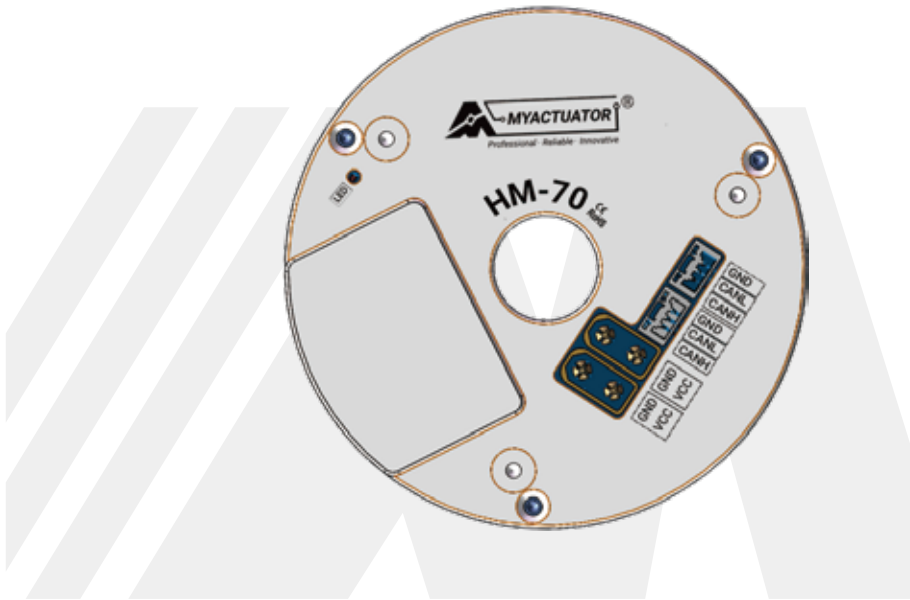
Product Parameters

Parameters	Unit	HM-60
Gear ratio	—	100
Input Voltage	V	24
No Load Speed	RPM	25
No Load Current	A	0.23
Rated Speed	RPM	20
Rated Torque	N.m	15
Rated Power	W	55
Rated Current	A(rms)	5
Peak Torque	N.m	37
Peak Current	A(rms)	12.1
Motor Back-EMF Constant	Vdc/Krpm	12
Module Torque Constant	N.m/A	3
Motor Phase Resistance	Ω	0.94
Motor Phase Inductance	mH	0.41
Pole Pair	—	10
3 Phase Connection	—	Y
Backlash	Arcsec	<40
Inertia	kg.m²	0.1
Encoder Type	—	Dual Encoder ABS-14BIT (Input) / ABS-17BIT (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	Kg	0.65
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
HM-60	15	18	15	5.2
	22	42	10	7.6
	30	45	8	9.8
	37	38	5	11.7

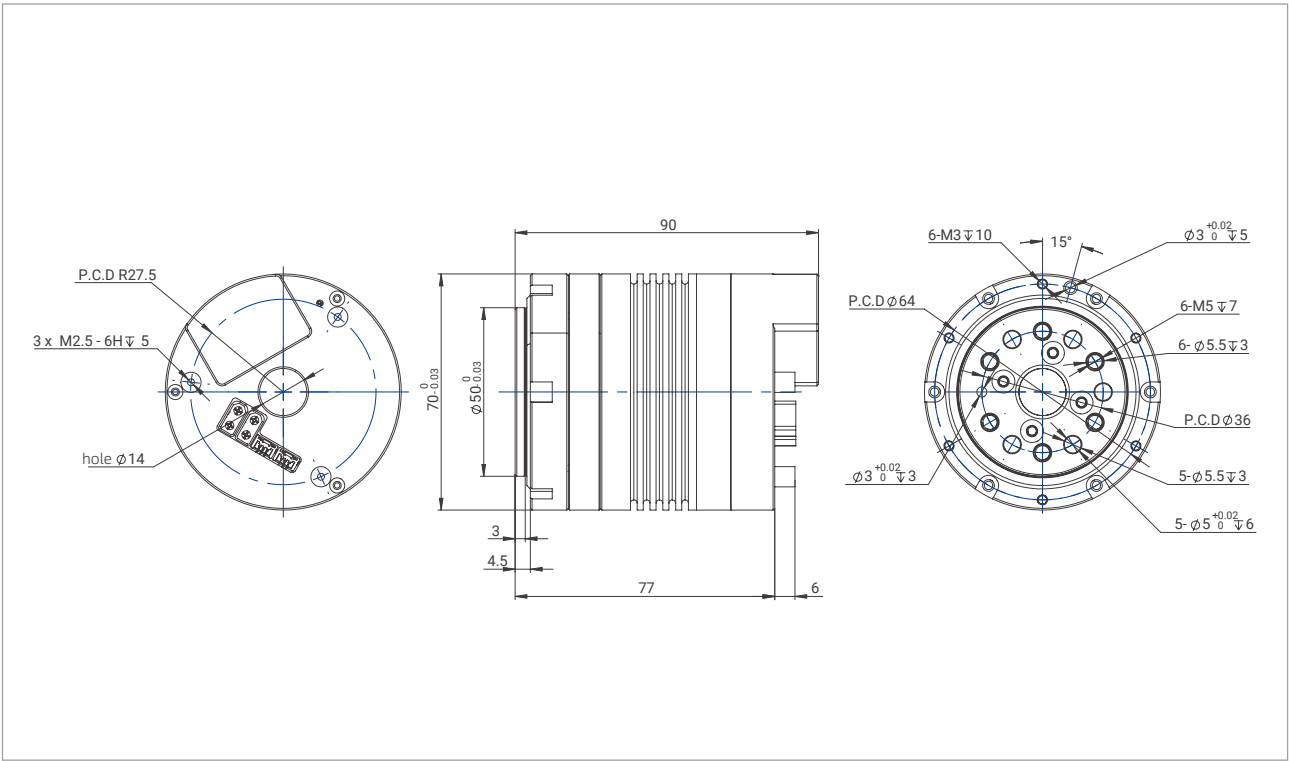
Series Name
EPS-HM
Motor Simplified Name
HM-70



Dual encoder

Actuator Full Name	Communication	With Brake/Without Brake	Gear ratio
EPS-HM-70-26-100-C	CAN BUS	N (without brake)	100

Installation Drawing



Product Parameters

Parameters	Unit	HM-70
Gear ratio	—	100
Input Voltage	V	24
No Load Speed	RPM	25
No Load Current	A	0.25
Rated Speed	RPM	20
Rated Torque	N.m	26
Rated Power	W	55
Rated Current	A(rms)	5.0
Peak Torque	N.m	65.0
Peak Current	A(rms)	12.6
Motor Back-EMF Constant	Vdc/Krpm	12.00
Module Torque Constant	N.m/A	5.17
Motor Phase Resistance	Ω	0.94
Motor Phase Inductance	mH	0.41
Pole Pair	—	10
3 Phase Connection	—	Y
Backlash	Arcsec	<40
Inertia	kg.m²	0.32
Encoder Type	—	Dual Encoder ABS-14BIT (Input) / ABS-17BIT (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	Kg	0.8
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
HM-70	39	26	15	7.5
	47	44	10	9.2
	52	47	8	10
	65	36	5	12.3

CEM Series Cycloid Actuator

CAN BUS & EtherCAT



* The pictures of this series of products are subject to the actual products.

CEM Series Product Features



Shock Resistance



Small Backlash



Compact Structure



High Precision

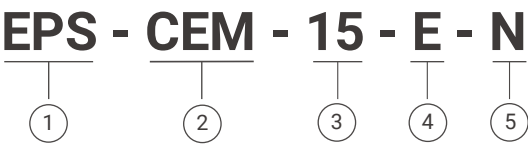


CAN BUS & EtherCAT



Dual Encoder

EPS-CEM Series Naming Conventions



- ① **EPS:** Brand Name E-Efficiency P-Precision S-Smart
- ② **CEM:** CEM Stands For The Series Name: Cycloid Actuator
- ③ **15:** Peak Torque 15 N.m
- ④ **E:** Communication C/E: CAN BUS & EtherCAT
- ⑤ **N:** N: Without Brake B: With Brake

For example

Model	EPS-CEM-25-E-N
Illustrate	EPS: Brand Name
	CEM: Cycloid Actuator
	25: Peak Torque
	E: Communication C: CAN BUS & EtherCAT
	N: Without Brake

Series Name

EPS-CEM

Motor Simplified Name

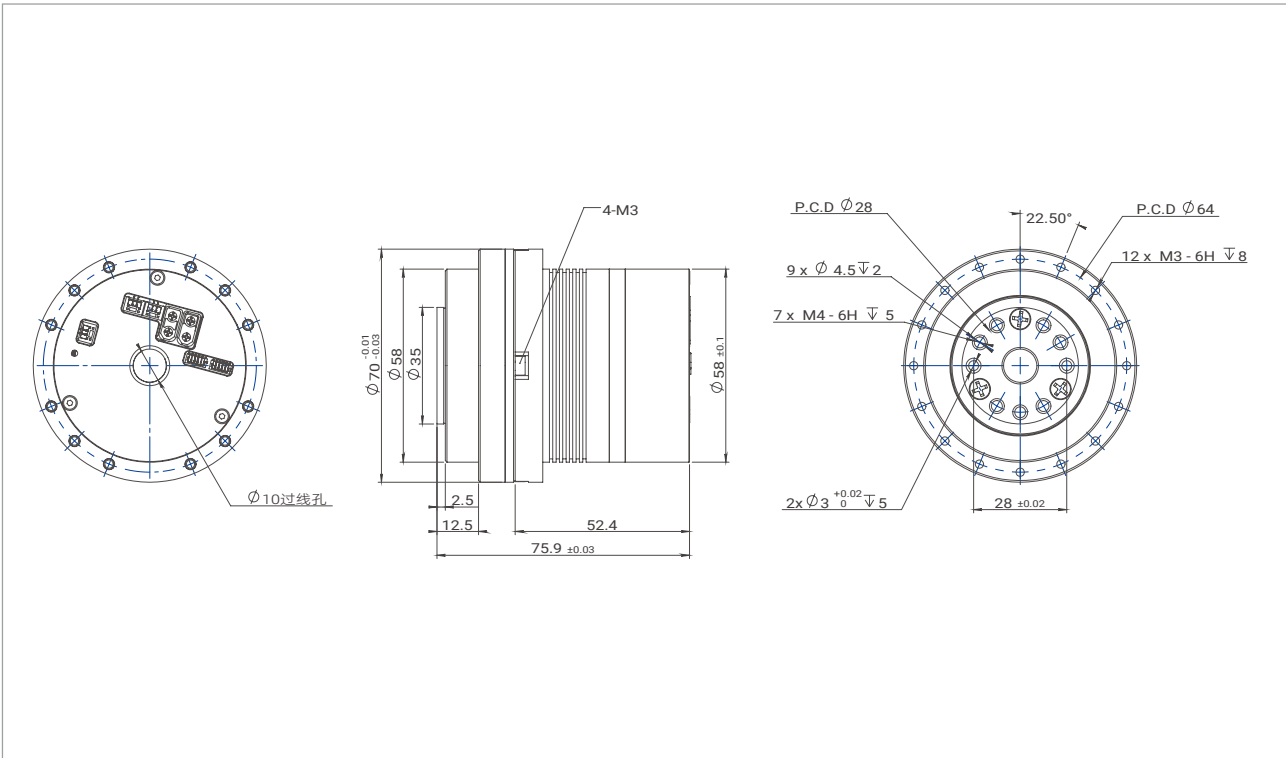
CEM-15



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
EPS-CEM-15-E-N	N (without brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters	Unit	CEM-15
Gear Ratio	—	30
Input Voltage	V	48
No Load Speed	RPM	170
Rated Speed	RPM	140
Rated Torque	N.m	9
Rated Output Power	W	132
Rated Phase Current	A(rms)	2.3
Peak Torque	N.m	15
Peak Phase Current	A(rms)	6.5
Motor Back-EMF Constant	Vdc/Krpm	5.58
Module Torque Constant	N.m/A	2.4
Motor Phase Resistance	Ω	0.62
Motor Phase Inductance	mH	0.43
Pole Pair	—	10
3 Phase Connection	—	Y
Backlash	Arcmin	<5
Max axial load	KN	1
Max radial load	KN	2
Inertia	Kg.cm ²	0.7
Encoder Type	—	ABS-17bit (Input) /17bit (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS & EtherCAT
Weight	Kg	0.81
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
CEM-15	9	13	10	7.6
	11.7	33	8	9.8
	13.5	28	5	11
	16.2	33	3	13.4

Series Name

EPS-CEM

Motor Simplified Name

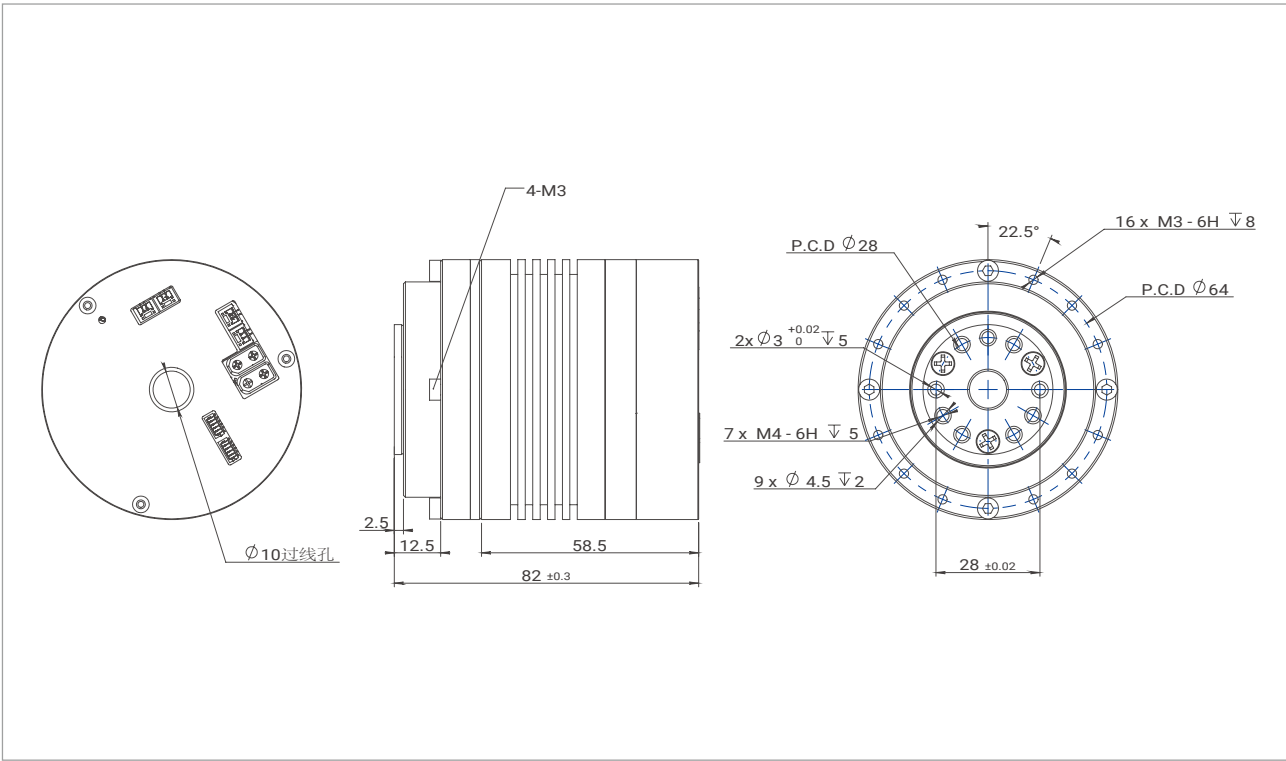
CEM-25



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
EPS-CEM-25-E-N	N (without brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters	Unit	CEM-25
Gear Ratio	—	30
Input Voltage	V	48
No Load Speed	RPM	130
Rated Speed	RPM	100
Rated Torque	N.m	15
Rated Output Power	W	157
Rated Phase Current	A(rms)	2.6
Peak Torque	N.m	25
Peak Phase Current	A(rms)	7.2
Motor Back-EMF Constant	Vdc/Krpm	8.37
Module Torque Constant	N.m/A	3.6
Motor Phase Resistance	Ω	0.7
Motor Phase Inductance	mH	0.47
Pole Pair	—	10
3 Phase Connection	—	Y
Backlash	Arcmin	<5
Max axial load	KN	1
Max radial load	KN	2
Inertia	Kg.cm ²	0.7
Encoder Type	—	ABS-17bit (Input) /17bit (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS & EtherCAT
Weight	Kg	0.94
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
CEM-25	15	31	10	9.2
	18	25	8	10.6
	22	30	5	13.1
	25	20	3	14.9

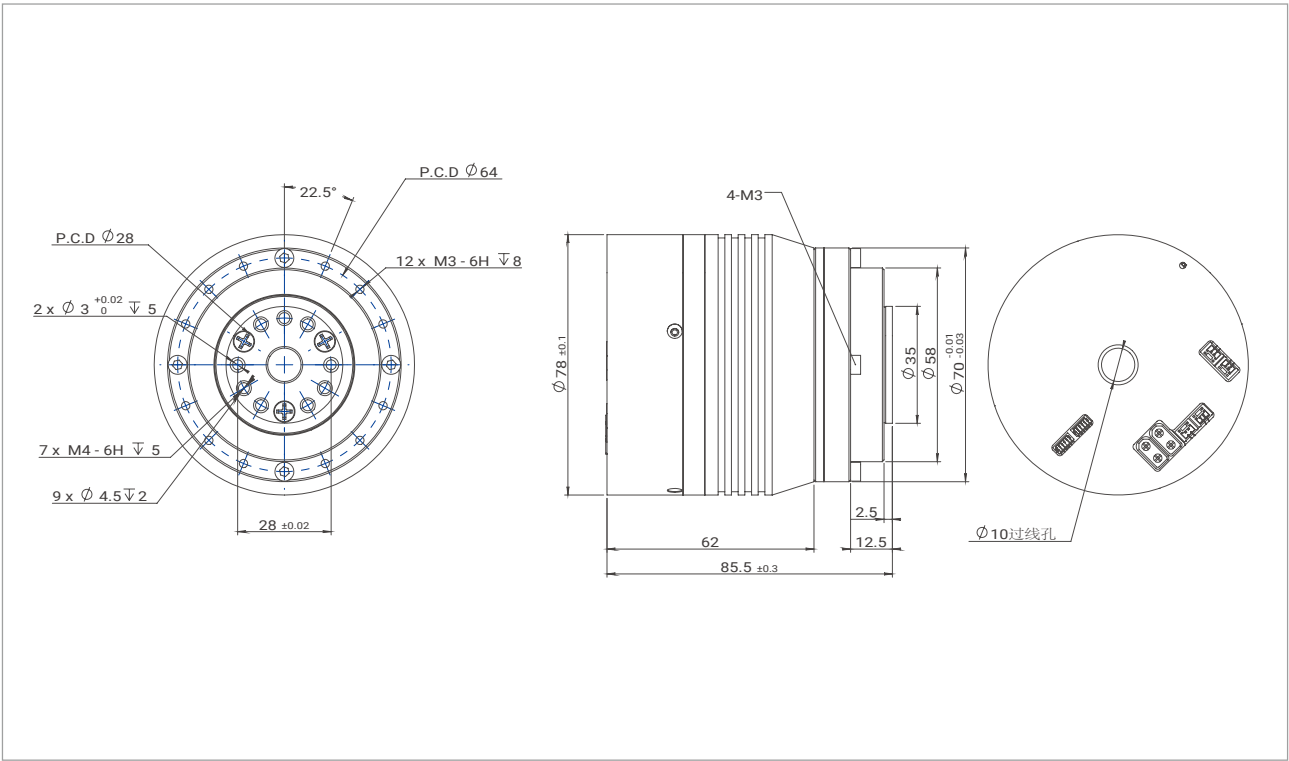
Series Name
EPS-CEM
Motor Simplified Name
CEM-45



Dual encoder

Actuator Full Name	With Brake/Without Brake	Communication
EPS-CEM-45-E-N	N (without brake)	CAN BUS & EtherCAT

Installation Drawing



Product Parameters

Parameters	Unit	CEM-45
Gear Ratio	—	30
Input Voltage	V	48
No Load Speed	RPM	100
Rated Speed	RPM	80
Rated Torque	N.m	25
Rated Output Power	W	210
Rated Phase Current	A(rms)	5.7
Peak Torque	N.m	45
Peak Phase Current	A(rms)	16.8
Motor Back-EMF Constant	Vdc/Krpm	9.77
Module Torque Constant	N.m/A	4.2
Motor Phase Resistance	Ω	0.25
Motor Phase Inductance	mH	0.38
Pole Pair	—	10
3 Phase Connection	—	Y
Backlash	Arcmin	<5
Max axial load	KN	1
Max radial load	KN	2
Inertia	Kg.cm ²	0.7
Encoder Type	—	ABS-17bit (Input) /17bit (Output)
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS & EtherCAT
Weight	Kg	1.09
Insulation Grade	—	F

Stall Torque Data

	Torque (N.m)	Temperature Rise (°C)	Stall Time (s)	Phase Current (Arms)
CEM-45	25	6	10	13.4
	35	15	8	19.1
	45	32	5	24.8
	50	50	3	27.6

H Series Hollow Direct Drive Actuator

Outer Rotor



* The pictures of this series of products are subject to the actual products.

Product Features



Hollow Design



Driver Integration



Low Speed Steady



High Precision

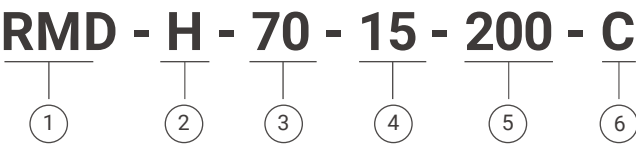


CAN BUS



Single Encoder

RMD-H Series Naming Conventions



- 1

RMD: Brand Name R-Reducer M-Motor D-Drive
- 2

H: H Stands For The Series Name: Integrated Hollow Direct Drive Actuator
- 3

70: 70 Represent Motor Model Number e.g: 30/40/50/70/90 etc
- 4

15: Stator stack height 15mm
- 5

200: Motor power 200W
- 6

C: Communication C: CAN BUS

For example

Model	RMD-H-90-15-400-C
Illustrate	RMD: Brand Name
	H: Hollow Direct Drive Actuator
	90: Motor Model Number
	15: Stator Stack Height
	400: Motor Power
	C: Communication C: CAN BUS

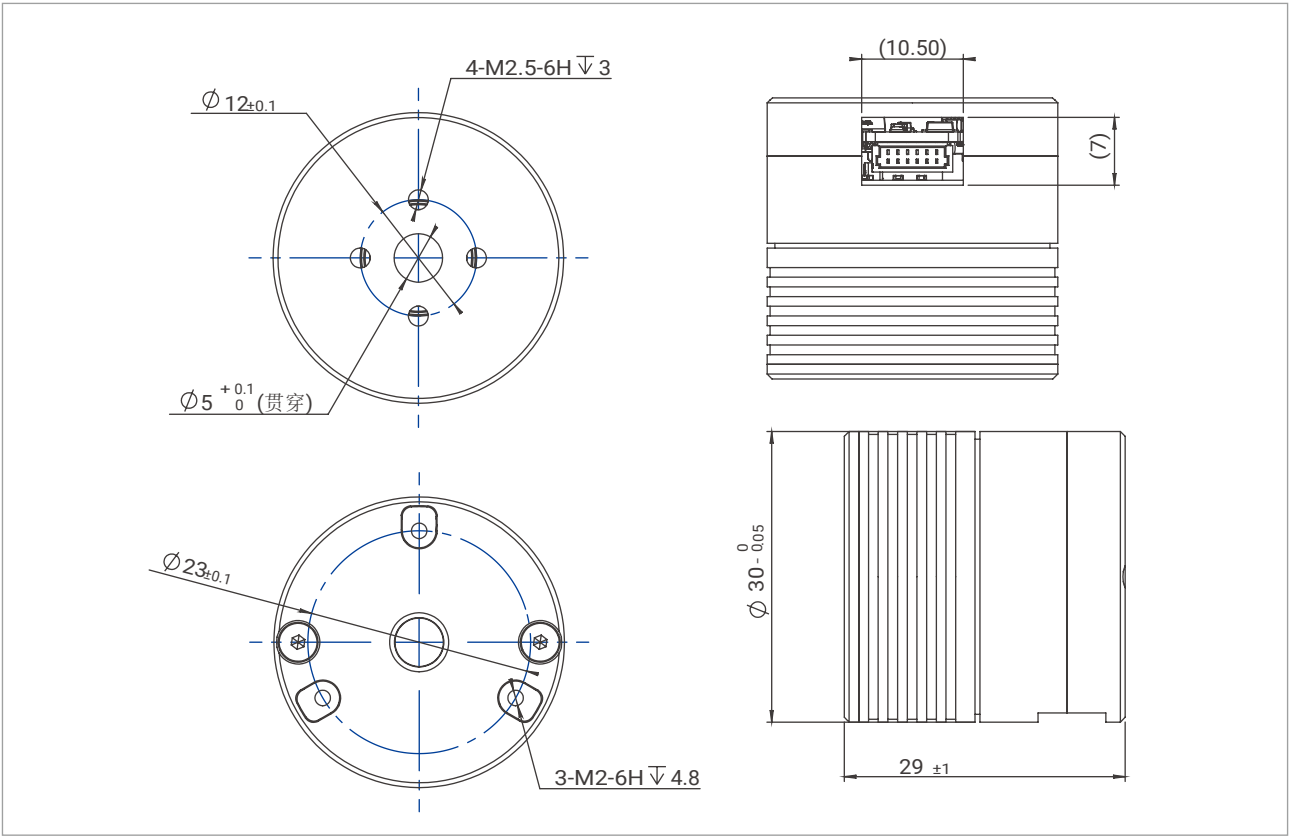
Series Name
RMD-H
Motor Simplified Name
H-30-08



Single Encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-H-30-08-15-C	N (without brake)	CAN BUS

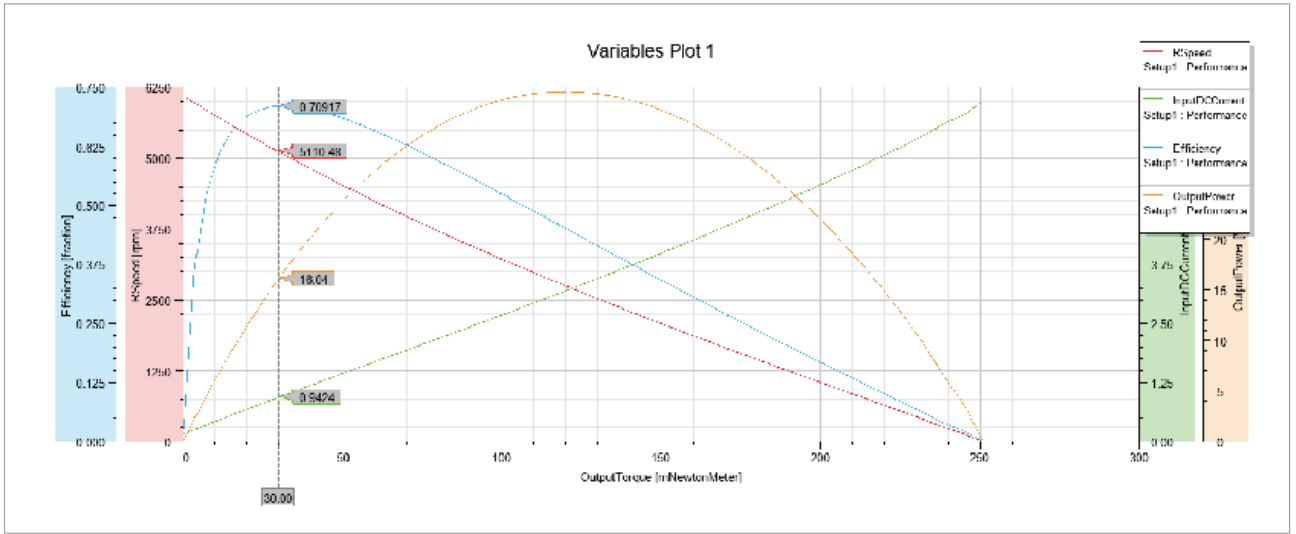
Installation Drawing



Product Parameters

Parameters	Unit	H-30-08
Input Voltage	V	24
No-Load speed	RPM	6000
No-Load Input Current	A	0.42
Rated Speed	RPM	5000
Rated Torque	N.m	0.03
Rated Output Power	W	15
Rated Phase Current	A	1.2
Peak Torque	N.m	0.1
Peak Phase Current	A	4
Efficiency	%	>60
Motor Back-EMF Constant	Vdc/Krpm	4.8
Module Torque Constant	N.m/A	0.03
Motor Phase Resistance	Ω	—
Motor Phase Inductance	mH	—
Pole Pair	—	7
Cogging Torque	mN.m	0.2
3 Phase Connection	—	Y
Max Axial Load	KN	0.18
Max Radial Load	KN	0.05
Inertia	Kg.cm ²	0.15
Encoder Type	—	ABS-17bit
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	Kg	—
Insulation Grade	—	F

Motor Characteristic Curve



Series Name
RMD-H

Motor Simplified Name

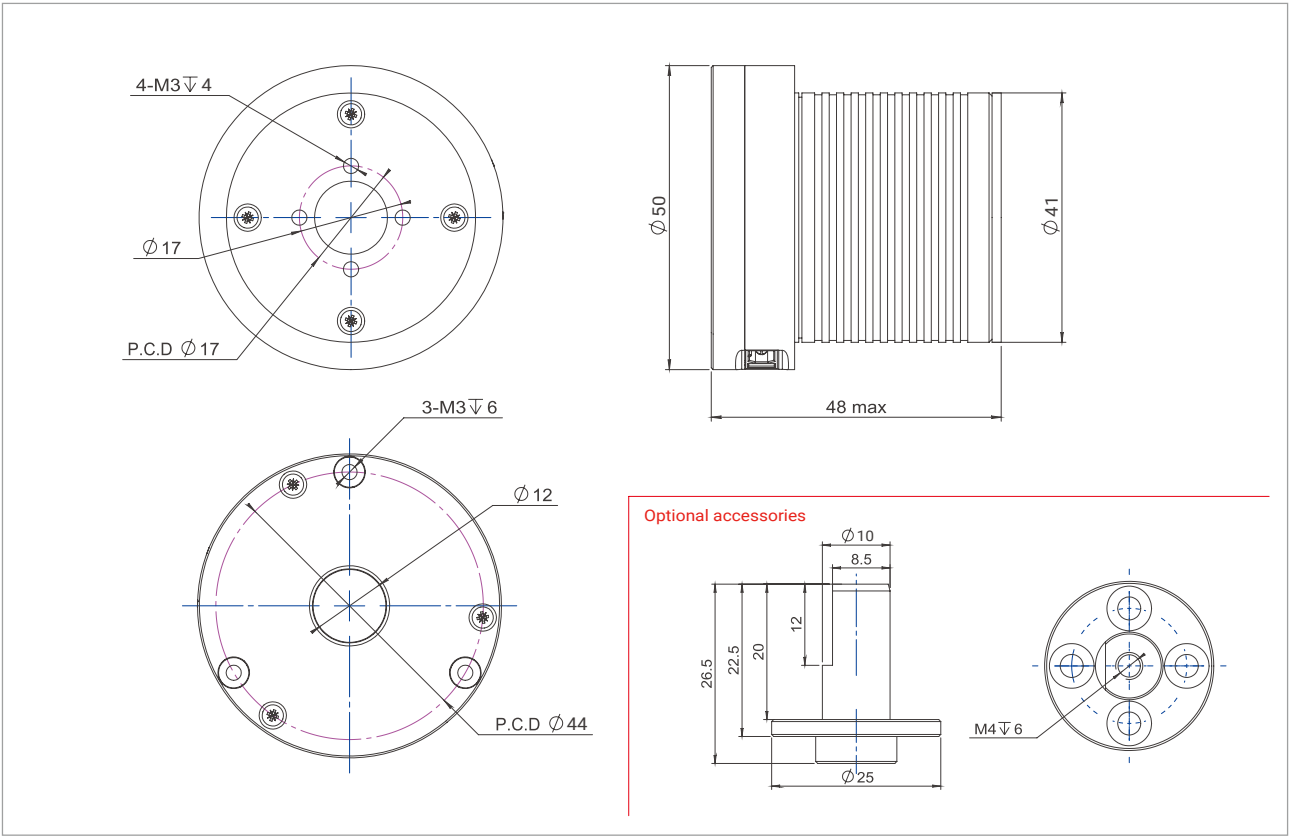
H-40-15



Single Encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-H-40-15-50-C	N (without brake)	CAN BUS

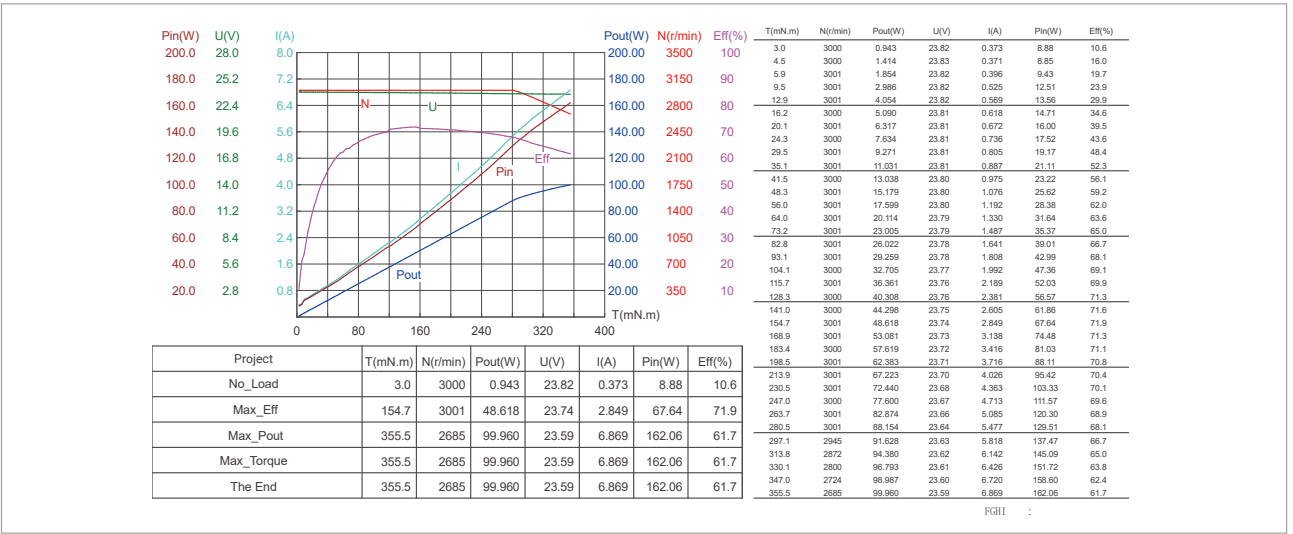
Installation Drawing



Product Parameters

Parameters	Unit	H-40-15
Input Voltage	V	24
No-Load speed	RPM	3700
No-Load Input Current	A	0.44
Rated Speed	RPM	3000
Rated Torque	N.m	0.15
Rated Output Power	W	47
Rated Phase Current	A	5.1
Peak Torque	N.m	0.3
Peak Phase Current	A	12.8
Efficiency	%	>64
Motor Back-EMF Constant	Vdc/Krpm	8
Module Torque Constant	N.m/A	0.03
Motor Phase Resistance	Ω	3.37
Motor Phase Inductance	mH	0.176
Pole Pair	—	13
Cogging Torque	mN.m	10
3 Phase Connection	—	Y
Max Axial Load	KN	0.35
Max Radial Load	KN	0.08
Inertia	Kg.cm ²	0.29
Encoder Type	—	ABS-17bit
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	Kg	0.2
Insulation Grade	—	F

Motor Characteristic Curve



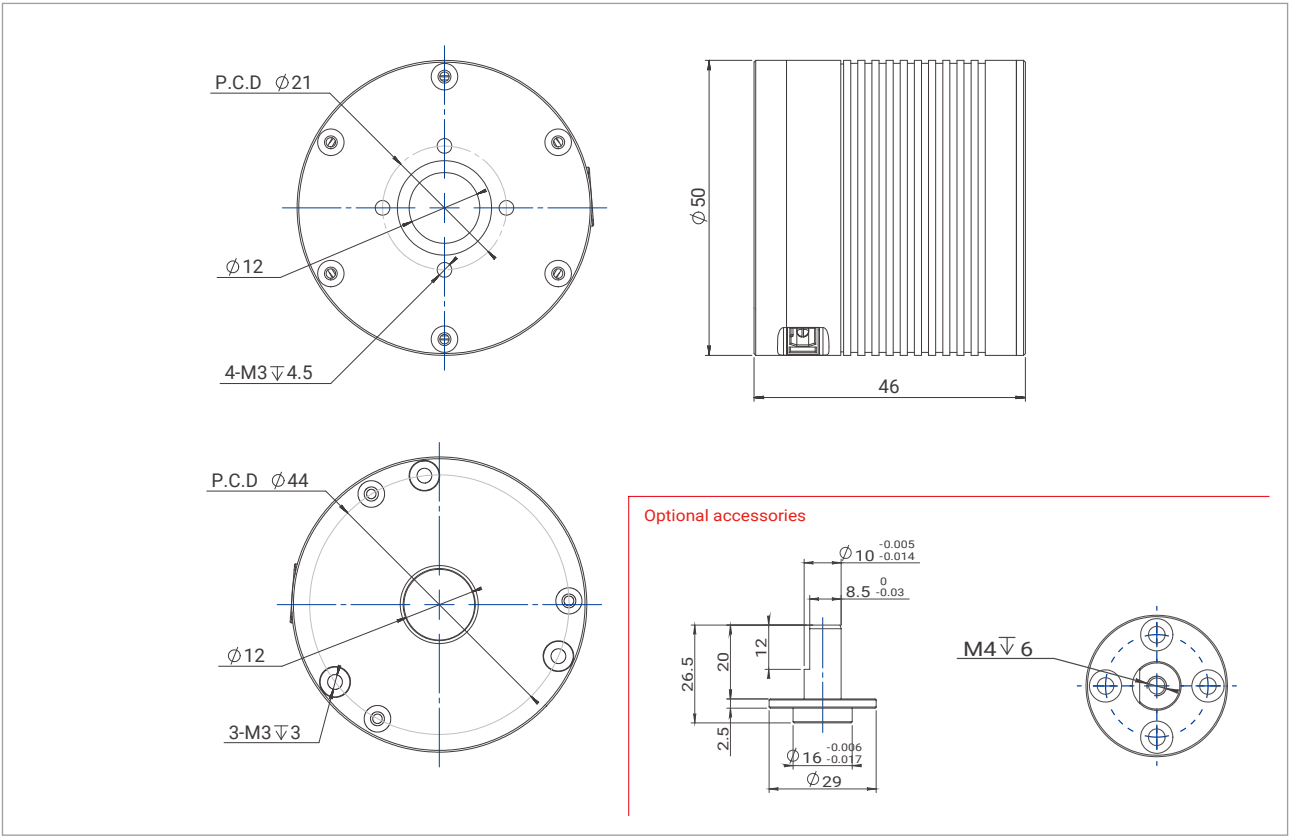
Series Name
RMD-H
Motor Simplified Name
H-50-15



Single Encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-H-50-15-100-C	N (without brake)	CAN BUS

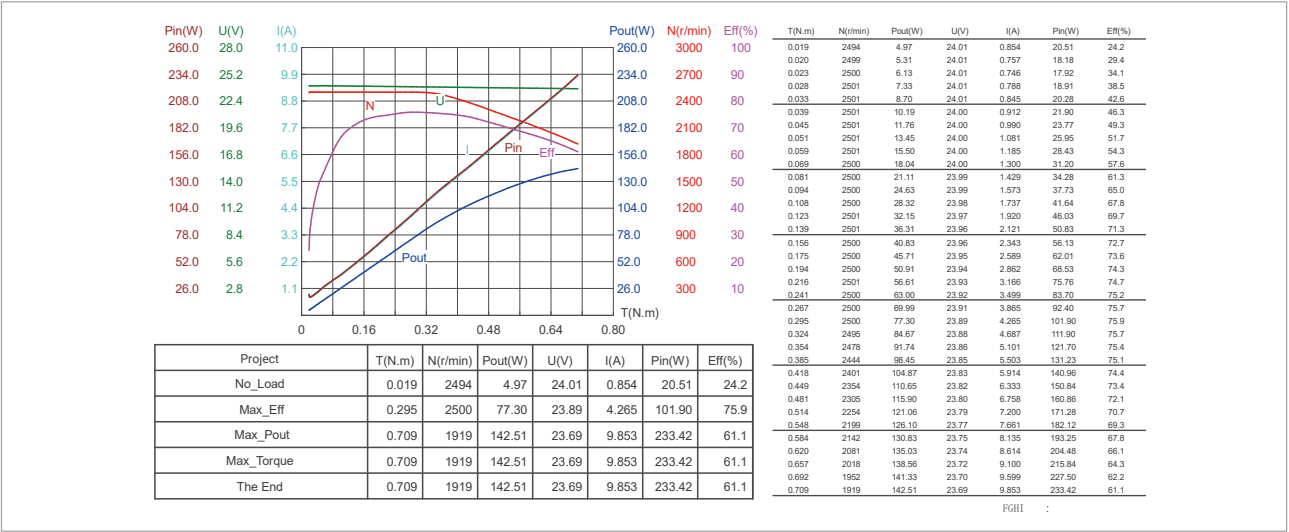
Installation Drawing



Product Parameters

Parameters	Unit	H-50-15
Input Voltage	V	24
No-Load speed	RPM	3200
No-Load Input Current	A	0.4
Rated Speed	RPM	2500
Rated Torque	N.m	0.35
Rated Output Power	W	91.6
Rated Phase Current	A	4.9
Peak Torque	N.m	0.9
Peak Phase Current	A	12.4
Efficiency	%	>83
Motor Back-EMF Constant	Vdc/Krpm	7.93
Module Torque Constant	N.m/A	0.07
Motor Phase Resistance	Ω	0.4
Motor Phase Inductance	mH	0.19
Pole Pair	—	13
Cogging Torque	mN.m	12
3 Phase Connection	—	Y
Max Axial Load	KN	0.489
Max Radial Load	KN	0.1
Inertia	Kg.cm ²	0.57
Encoder Type	—	ABS-17bit
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	Kg	0.24
Insulation Grade	—	F

Motor Characteristic Curve



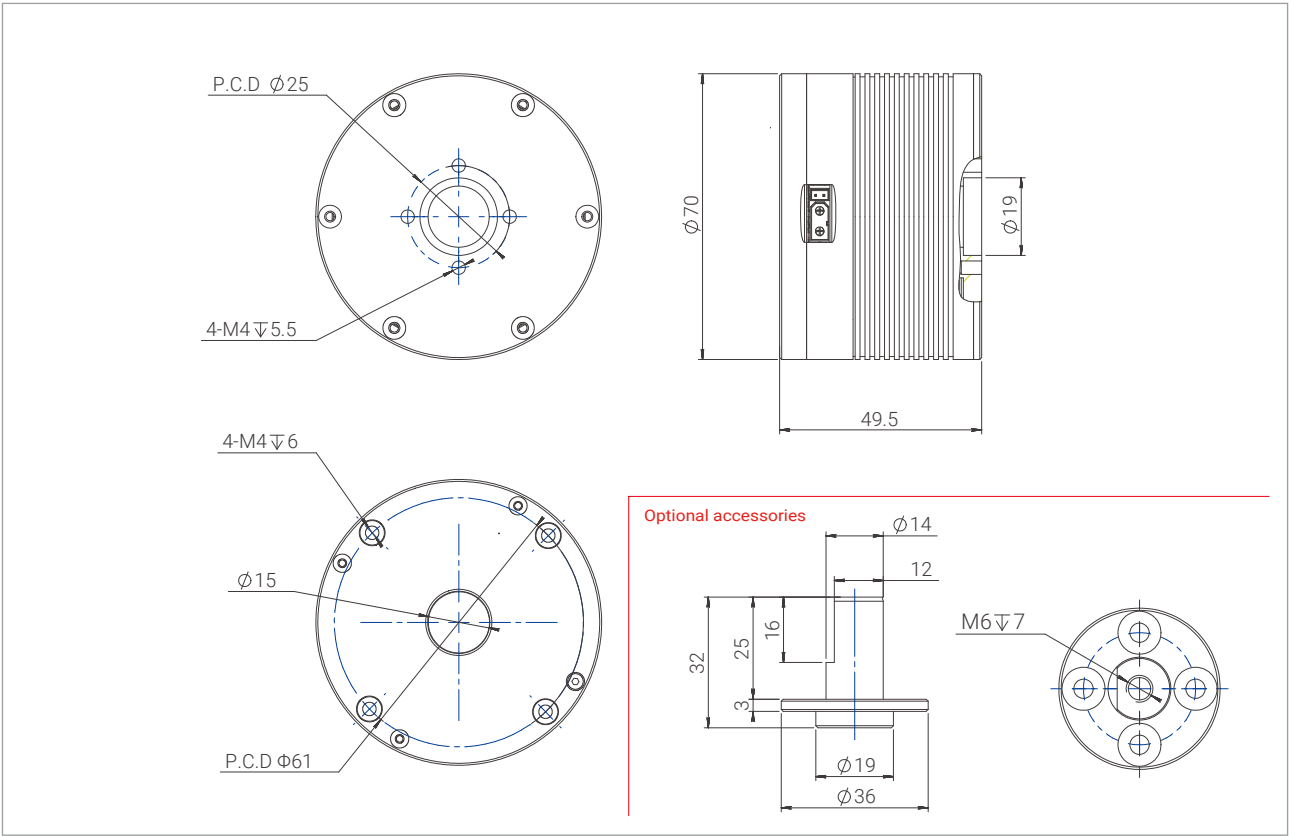
Series Name
RMD-H
Motor Simplified Name
H-70-15



Single Encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-H-70-15-200-C	N (without brake)	CAN BUS

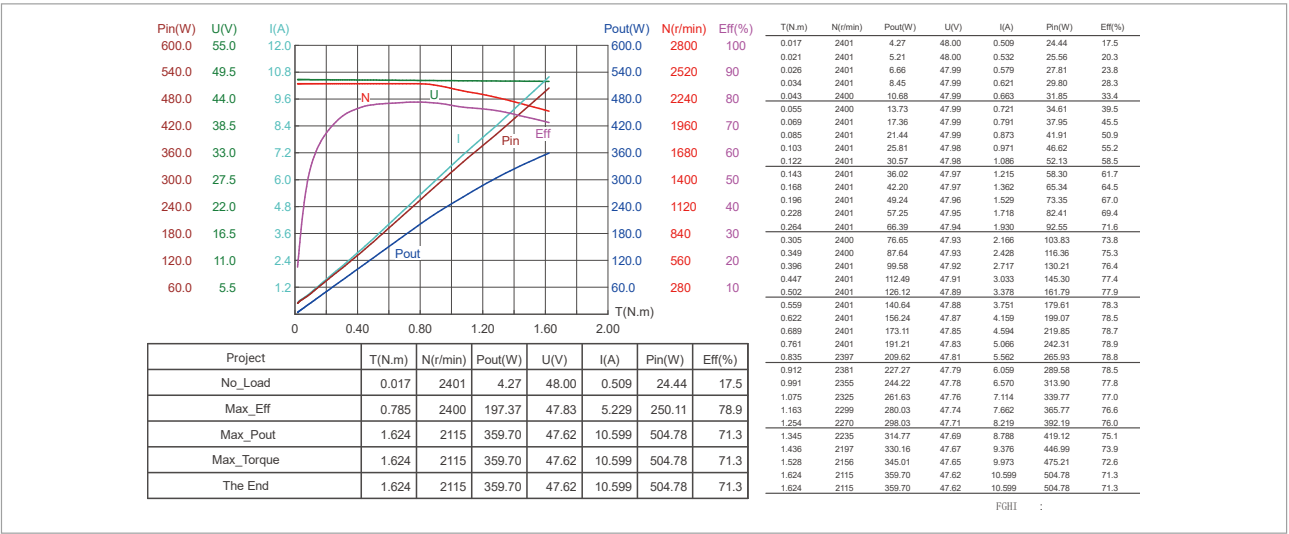
Installation Drawing



Product Parameters

Parameters	Unit	H-70-15
Input Voltage	V	48
No-Load speed	RPM	2800
No-Load Input Current	A	0.18
Rated Speed	RPM	2400
Rated Torque	N.m	0.8
Rated Output Power	W	200
Rated Phase Current	A	4.4
Peak Torque	N.m	2
Peak Phase Current	A	13.2
Efficiency	%	>84
Motor Back-EMF Constant	Vdc/Krpm	17
Module Torque Constant	N.m/A	0.18
Motor Phase Resistance	Ω	0.5
Motor Phase Inductance	mH	0.55
Pole Pair	—	13
Cogging Torque	mN.m	31
3 Phase Connection	—	Y
Max Axial Load	KN	1.87
Max Radial Load	KN	0.44
Inertia	Kg.cm ²	1.8
Encoder Type	—	ABS-17bit
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	Kg	0.47
Insulation Grade	—	F

Motor Characteristic Curve



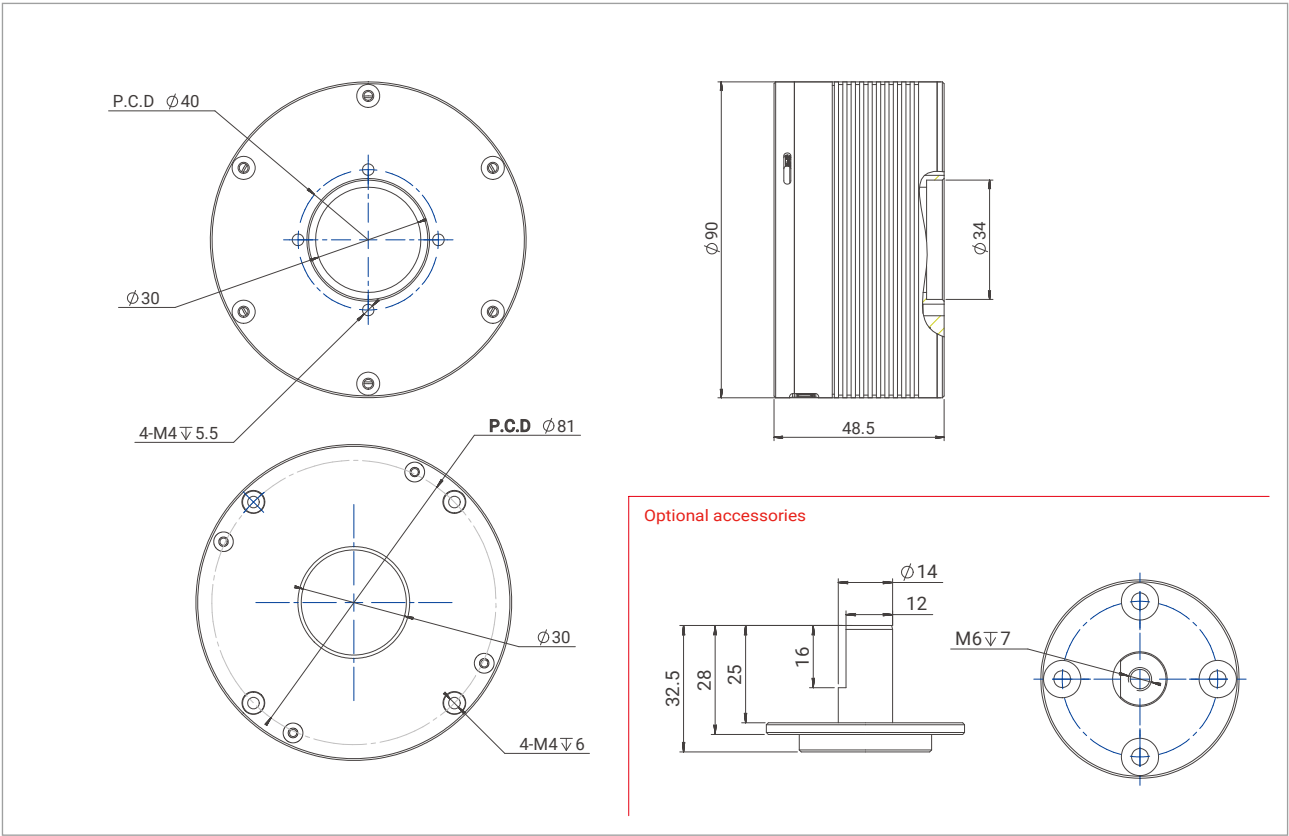
Series Name
RMD-H
Motor Simplified Name
H-90-15



Single Encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-H-90-15-400-C	N (without brake)	CAN BUS

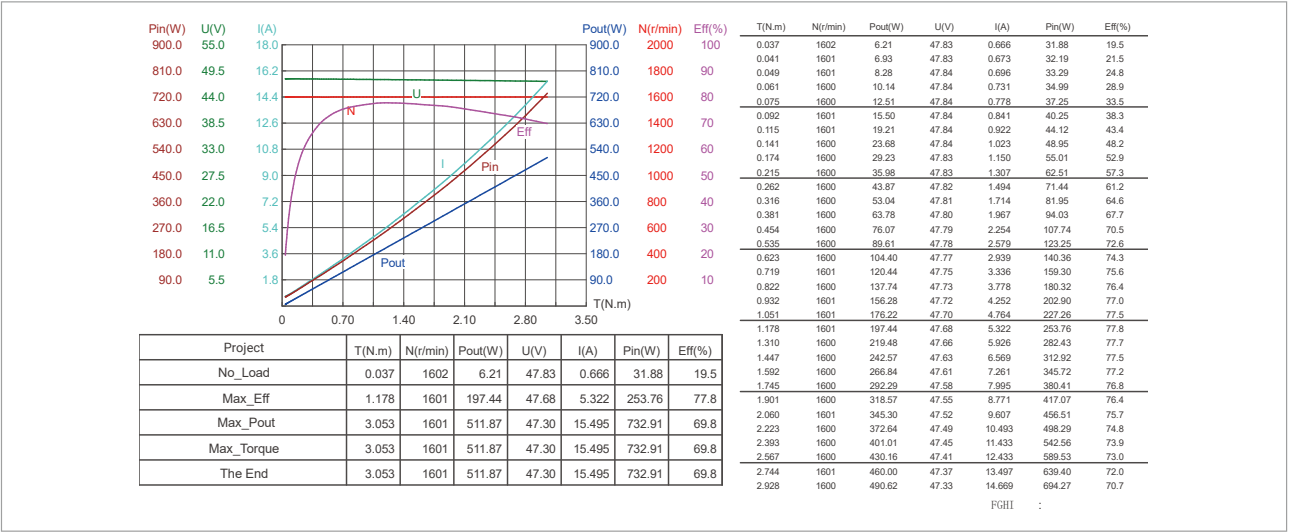
Installation Drawing



Product Parameters

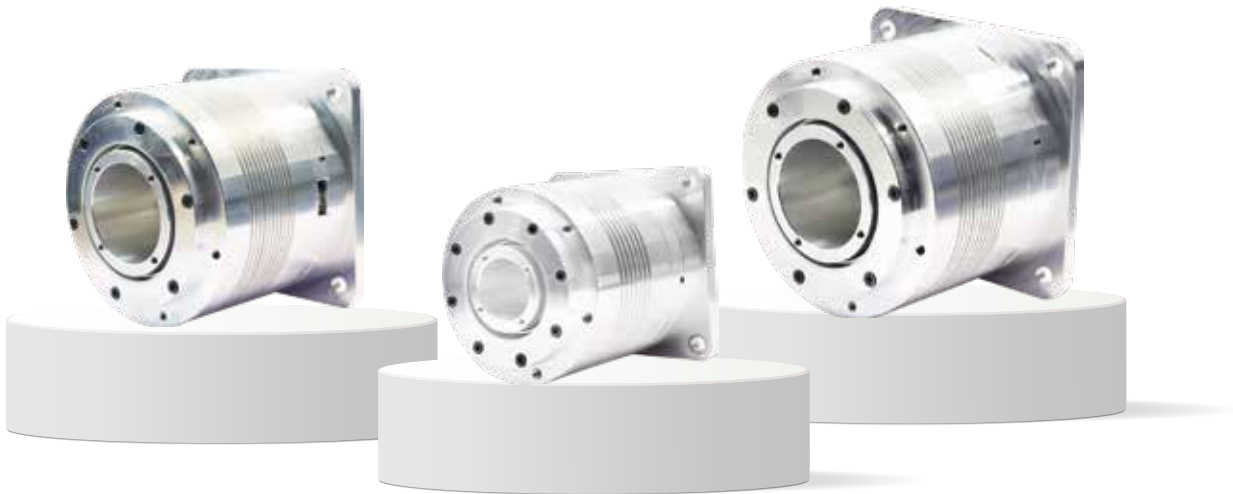
Parameters	Unit	H-90-15
Input Voltage	V	48
No-Load speed	RPM	2000
No-Load Input Current	A	0.35
Rated Speed	RPM	1600
Rated Torque	N.m	1.5
Rated Output Power	W	252
Rated Phase Current	A	6.6
Peak Torque	N.m	3.75
Peak Phase Current	A	19.8
Efficiency	%	>83
Motor Back-EMF Constant	Vdc/Krpm	21.1
Module Torque Constant	N.m/A	0.23
Motor Phase Resistance	Ω	0.35
Motor Phase Inductance	mH	0.28
Pole Pair	—	16
Cogging Torque	mN.m	64
3 Phase Connection	—	Y
Max Axial Load	KN	2.35
Max Radial Load	KN	0.77
Inertia	Kg.cm ²	4.2
Encoder Type	—	ABS-17bit
Repeat Position Accuracy	Degree	< 0.01
Communication	—	CAN BUS
Weight	Kg	0.71
Insulation Grade	—	F

Motor Characteristic Curve



H Series Hollow Direct Drive Actuator

Inner Rotor



* The pictures of this series of products are subject to the actual products.

Product Features



High Precision



Driver Integration



Low Speed Steady



Hollow Design

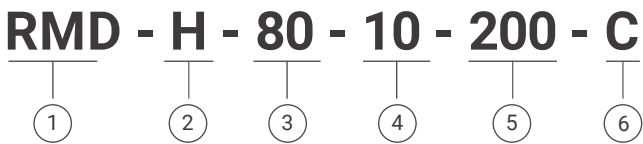


CAN BUS



Single Encoder

RMD-H Series Naming Conventions



- 1

RMD:

Brand Name R-Reducer M-Motor D-Drive
- 2

H:

H Stands For The Series Name: Integrated Hollow Direct Drive Actuator
- 3

80:

80 Represent Motor Model Number e.g: 60/80/95 etc
- 4

10:

Stator stack height 10mm
- 5

200:

Motor power 200W
- 6

C:

Communication C: CAN BUS

For example

Model	RMD-H-60-15-200-C
Illustrate	RMD: Brand Name
	H: Hollow Direct Drive Series
	60: Motor Model Number
	15: Stator Stack Height
	200: Motor Power
	C: Communication C: CAN BUS

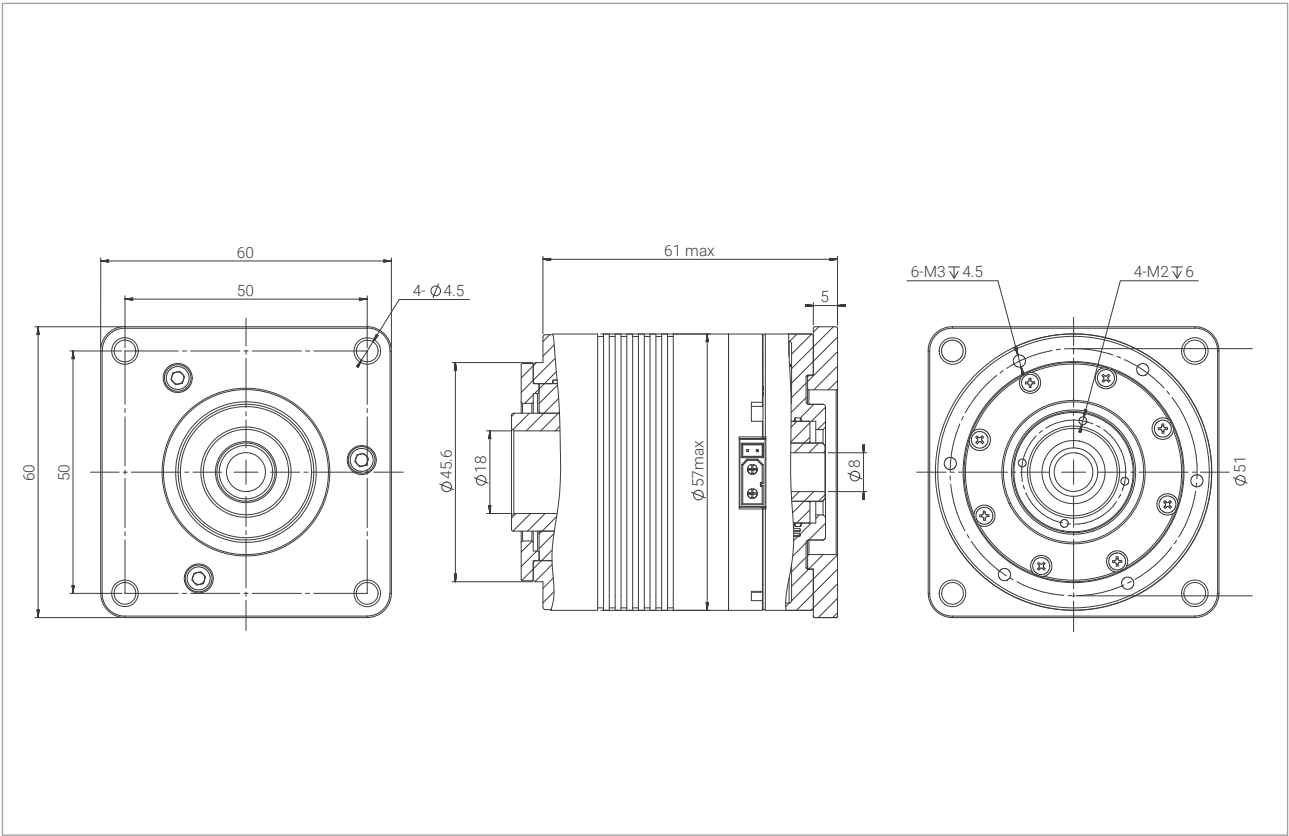
Series Name
RMD-H
Motor Simplified Name
H-60-15



Single Encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-H-60-15-200-C	N (without brake)	CAN BUS

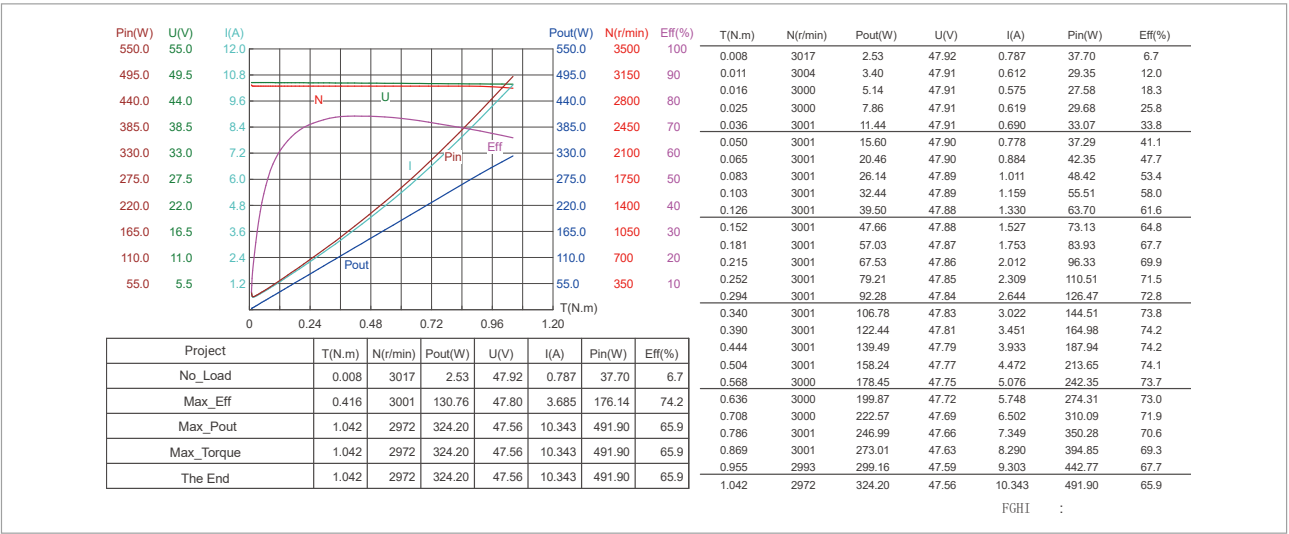
Installation Drawing



Product Parameters

Parameters	Unit	H-60-15
Input Voltage	V	48
No-Load speed	RPM	4200
No-Load Input Current	A	0.53
Rated Speed	RPM	3700
Rated Torque	N.m	0.51
Rated Output Power	W	197
Rated Phase Current	A	5
Peak Torque	N.m	1.53
Peak Phase Current	A	13.7
Efficiency	%	>80
Motor Back-EMF Constant	Vdc/Krpm	12.97
Module Torque Constant	N.m/A	0.1
Motor Phase Resistance	Ω	0.68
Motor Phase Inductance	mH	0.44
Pole Pair	—	10
Cogging Torque	mN.m	23
3 Phase Connection	—	Y
Max Axial Load	KN	1.25
Max Radial Load	KN	0.31
Inertia	Kg.cm ²	1.24
Encoder Type	—	ABS-17bit
Repeat Position Accuracy	Degree	< 0.01 Degree
Communication	—	CAN BUS
Weight	Kg	0.43
Insulation Grade	—	F

Motor Characteristic Curve



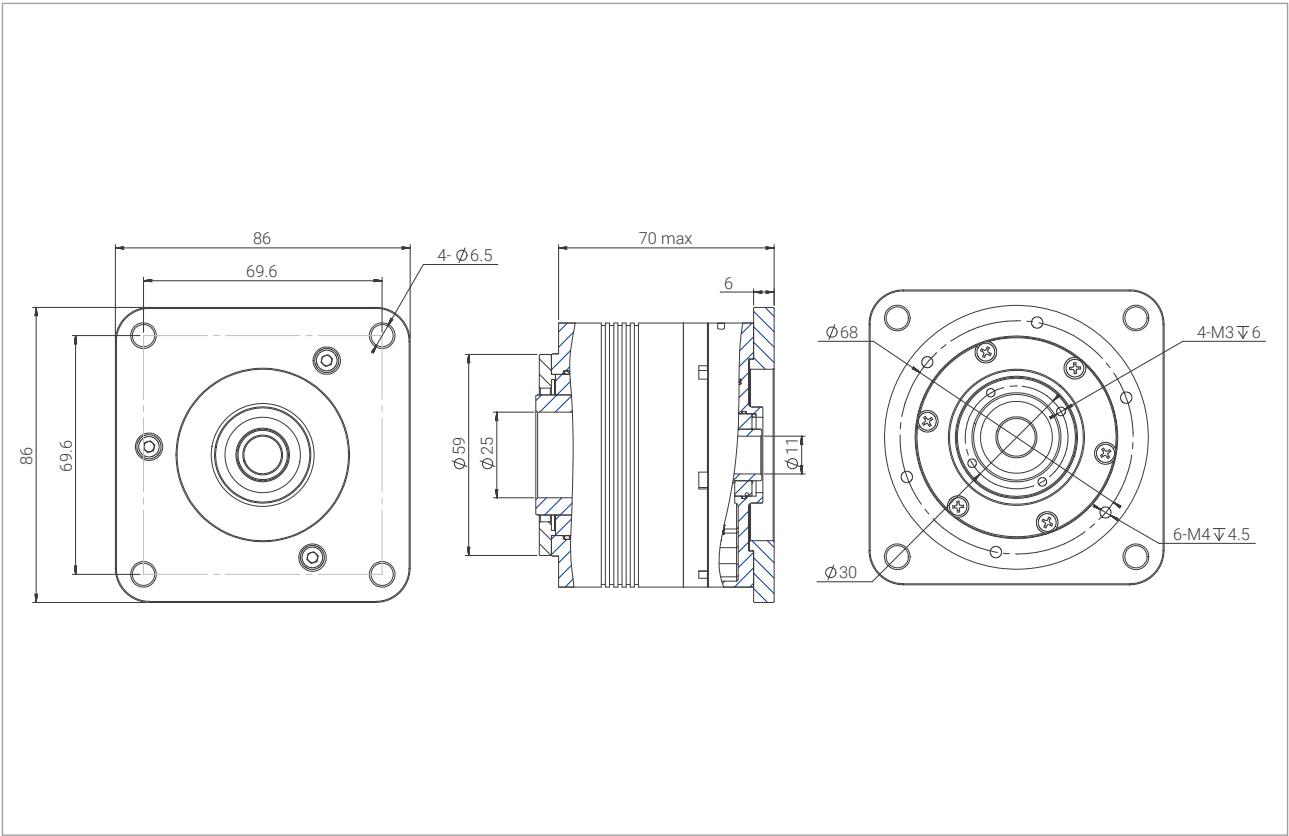
Series Name
RMD-H
Motor Simplified Name
H-80-10



Single Encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-H-80-10-200-C	N (without brake)	CAN BUS

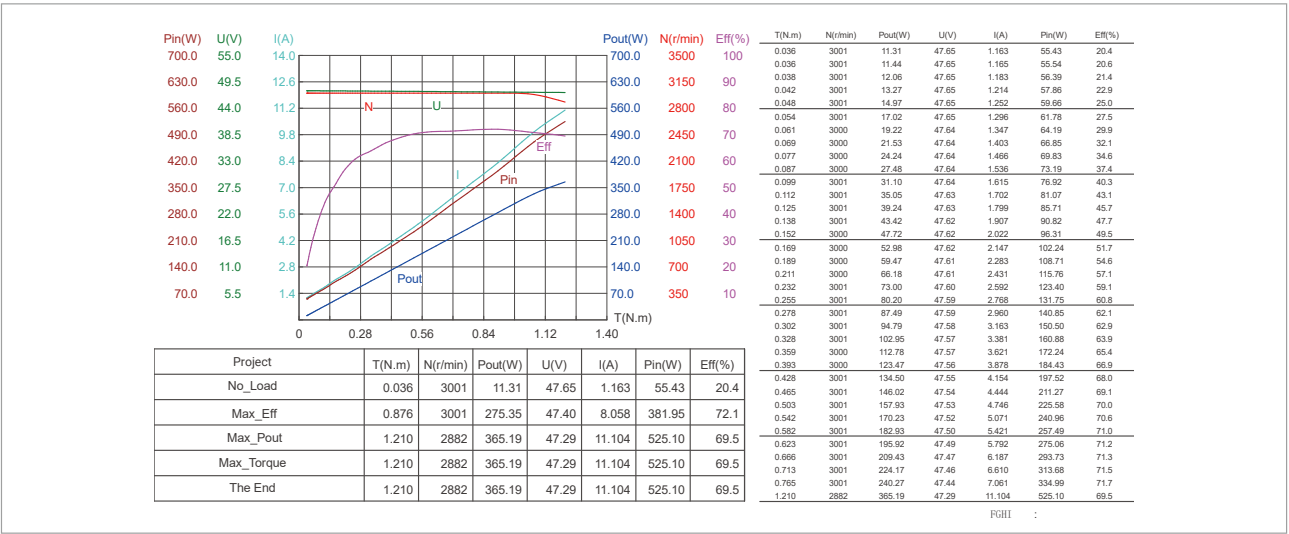
Installation Drawing



Product Parameters

Parameters	Unit	H-80-10
Input Voltage	V	48
No-Load speed	RPM	3550
No-Load Input Current	A	1.35
Rated Speed	RPM	3000
Rated Torque	N.m	0.6
Rated Output Power	W	188
Rated Phase Current	A	6
Peak Torque	N.m	1.6
Peak Phase Current	A	13.1
Efficiency	%	>67
Motor Back-EMF Constant	Vdc/Krpm	16
Module Torque Constant	N.m/A	0.1
Motor Phase Resistance	Ω	0.4
Motor Phase Inductance	mH	0.5
Pole Pair	—	10
Cogging Torque	mN.m	55
3 Phase Connection	—	Y
Max Axial Load	KN	2.03
Max Radial Load	KN	0.51
Inertia	Kg.cm ²	3.5
Encoder Type	—	ABS-17bit
Repeat Position Accuracy	Degree	< 0.01 Degree
Communication	—	CAN BUS
Weight	Kg	0.73
Insulation Grade	—	F

Motor Characteristic Curve



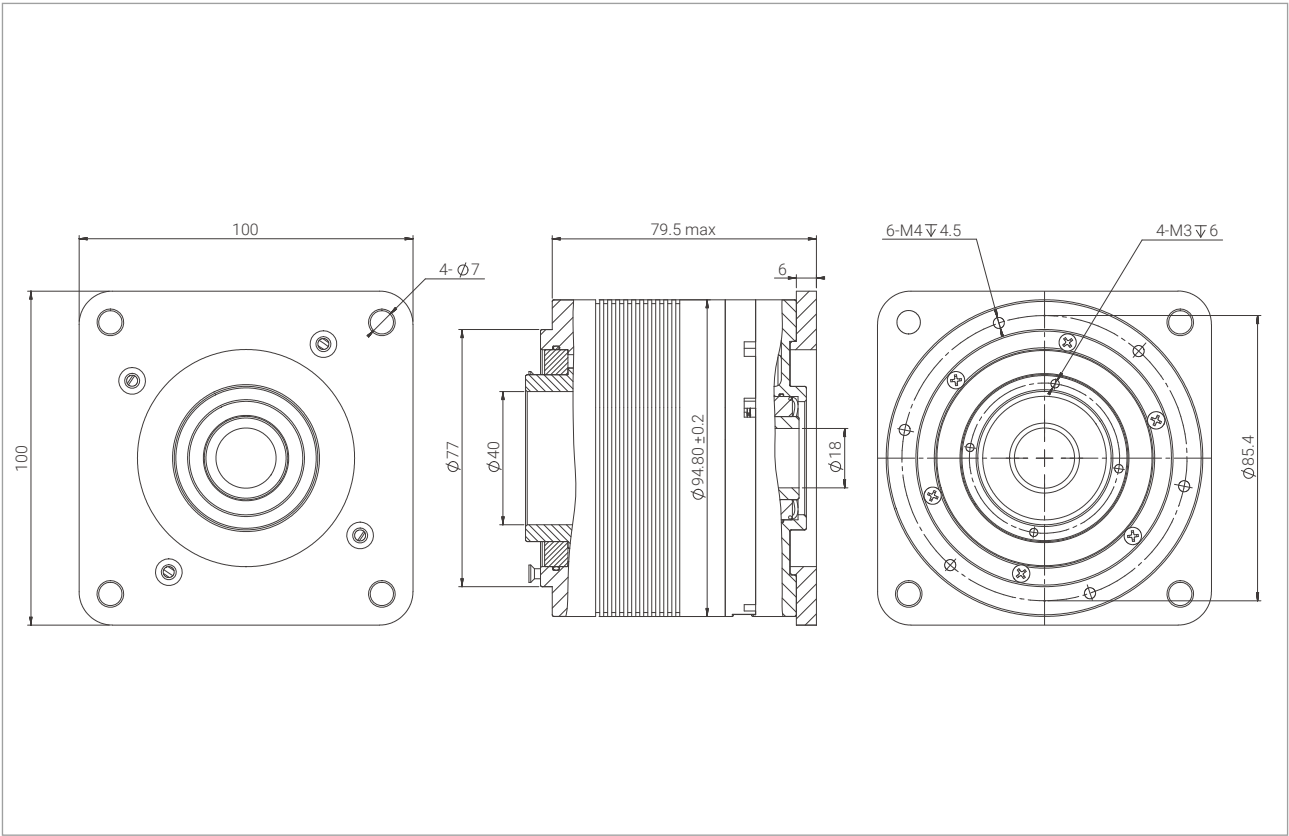
Series Name
RMD-H
Motor Simplified Name
H-95-23



Single Encoder

Actuator Full Name	With Brake/Without Brake	Communication
RMD-H-95-23-600-C	N (without brake)	CAN BUS

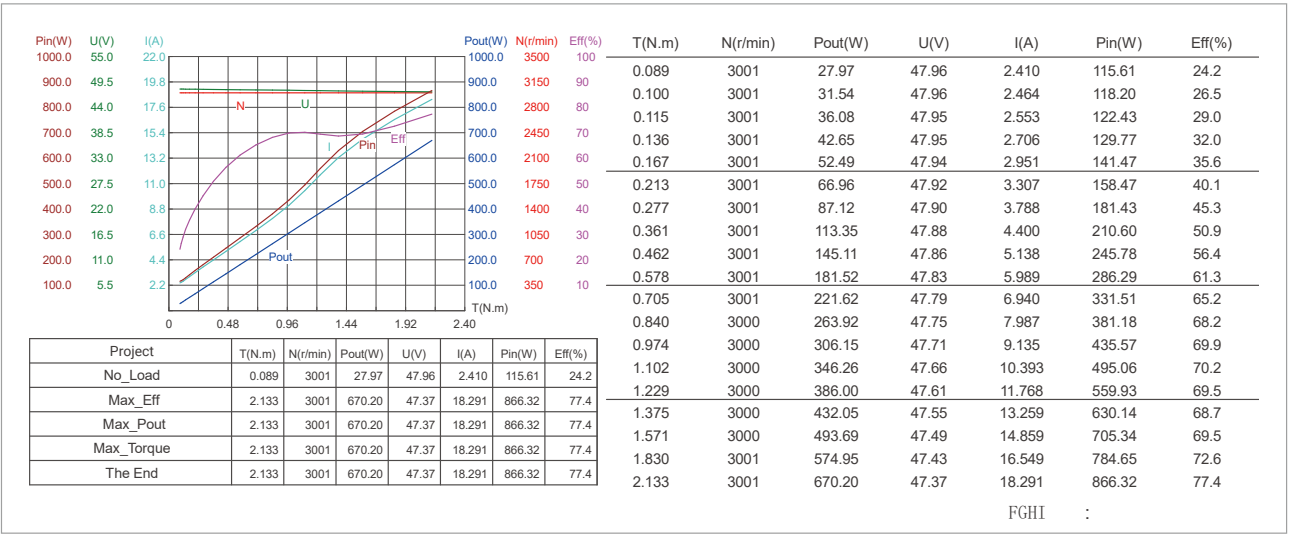
Installation Drawing



Product Parameters

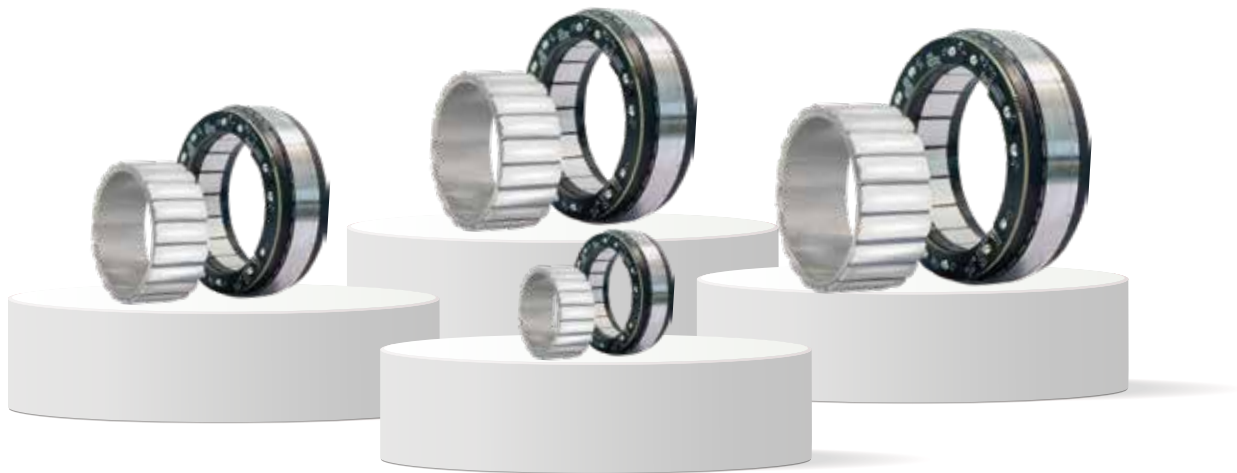
Parameters	Unit	H-95-23
Input Voltage	V	48
No-Load speed	RPM	4000
No-Load Input Current	A	2.3
Rated Speed	RPM	3000
Rated Torque	N.m	2
Rated Output Power	W	628
Rated Phase Current	A	16.2
Peak Torque	N.m	5
Peak Phase Current	A	40.5
Efficiency	%	>77
Motor Back-EMF Constant	Vdc/Krpm	16
Module Torque Constant	N.m/A	0.12
Motor Phase Resistance	Ω	0.8
Motor Phase Inductance	mH	0.12
Pole Pair	—	10
Cogging Torque	mN.m	72
3 Phase Connection	—	Y
Max Axial Load	KN	2.56
Max Radial Load	KN	0.64
Inertia	Kg.cm ²	4.6
Encoder Type	—	ABS-17bit
Repeat Position Accuracy	Degree	< 0.01 Degree
Communication	—	CAN BUS
Weight	Kg	1.5
Insulation Grade	—	F

Motor Characteristic Curve



FL Series Frameless Torque Motor

Inner Rotor



* The pictures of this series of products are subject to the actual products.

Product Features



Flexible integration



High Power Density



Good heat dissipation



High Precision

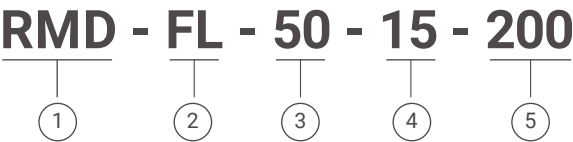


Hollow Design



Hall+INC

RMD-FL Series Naming Conventions



- 1

RMD: Brand Name R-Reducer M-Motor D-Drive
- 2

FL: FL Stands For The Series Name: Inner rotor frameless torque motor
- 3

50: Stator diameter 50mm
- 4

15: Stator height 15mm
- 5

200: Rated power 200W

For example

Model	RMD-FL-85-13-500
Illustrate	RMD: Brand Name
	FL: Internal Rotation Frameless Torque Series
	85: Stator Diameter
	13: Stator height
	500: Rated Power

Series Name

RMD-FL

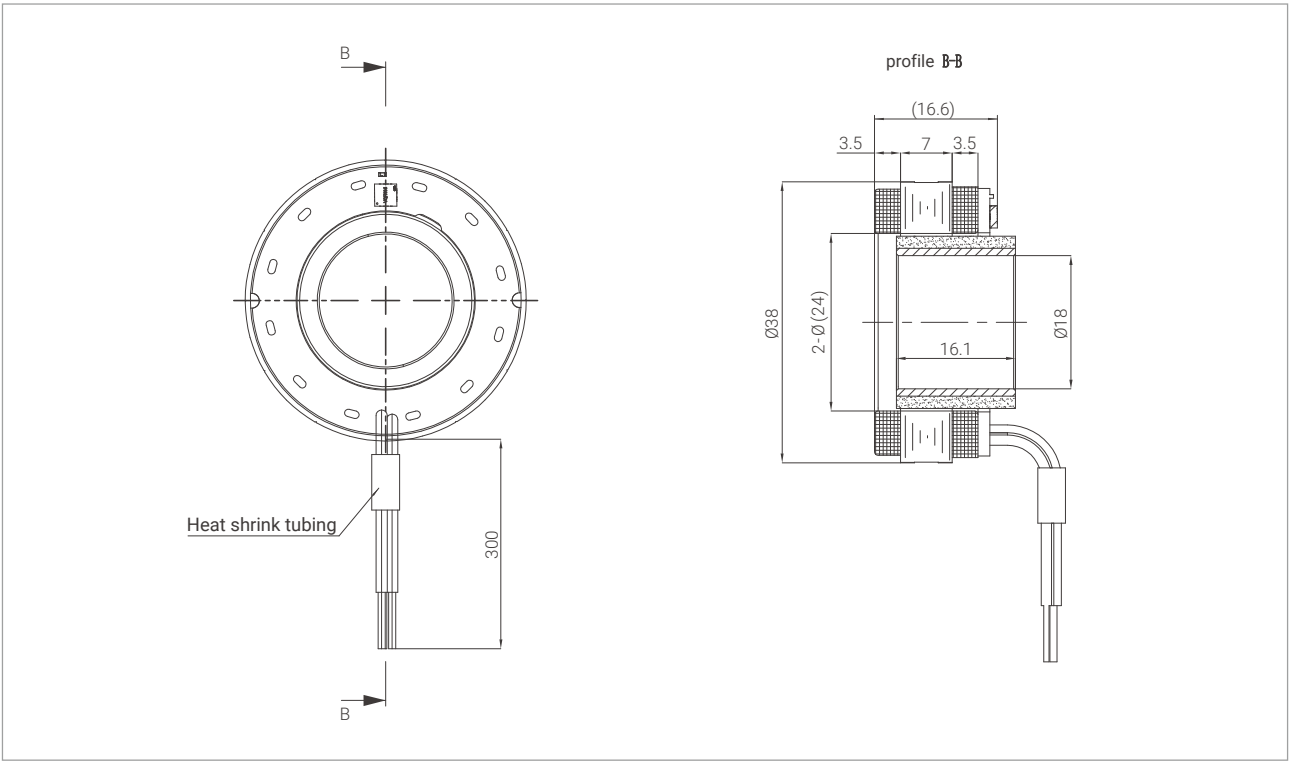
Motor Simplified Name

FL-38-08



Actuator Full Name	Encoder/Sensor
RMD-FL-38-08-100	Hall+INC

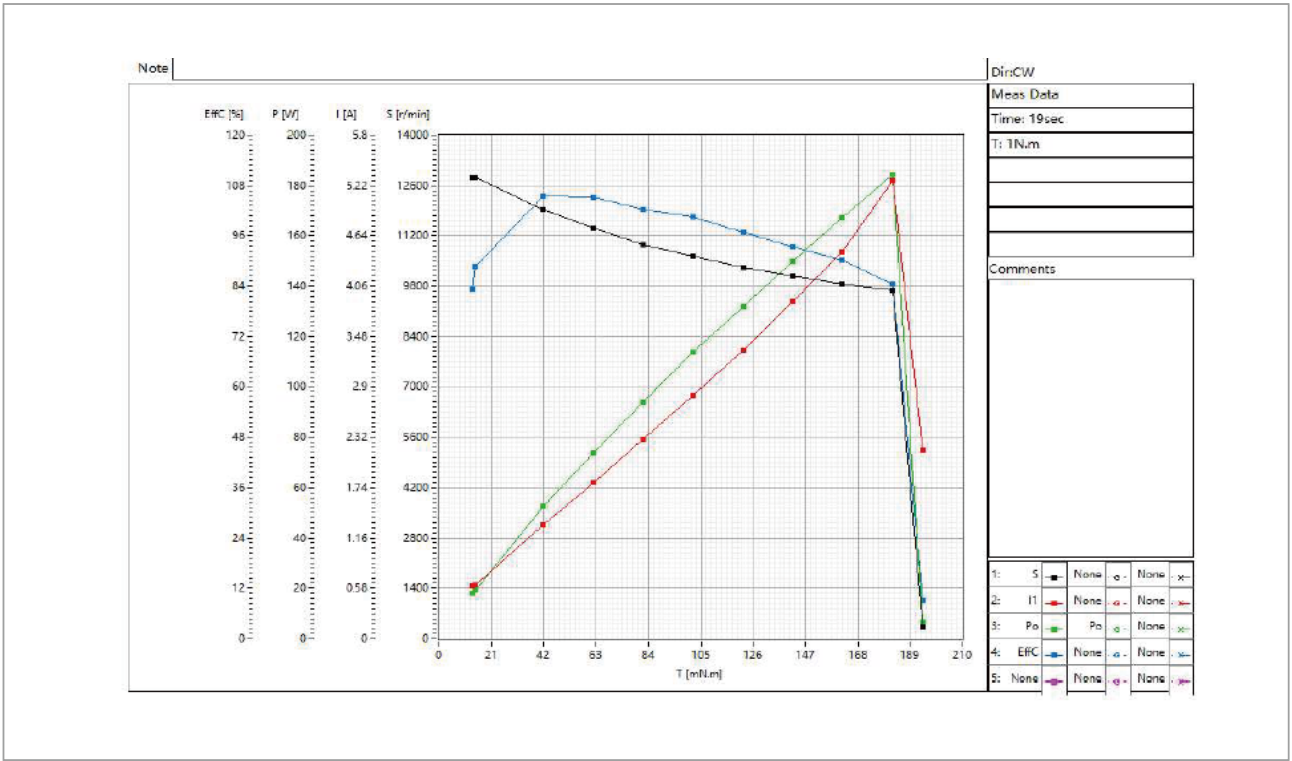
Installation Drawing



Product Parameters

Parameters	Unit	FL-38-08
Input Voltage	V	48
No Load Current	A	0.28
Rated Speed	RPM	9000
Rated Torque	N.m	0.1
Rated Power	W	94
Rated Current	A	2.75
Peak Torque	N.m	0.32
Peak Current	A	8.25
Efficiency	%	>80%
Motor Back-EMF Constant	Vdc/Krpm	2.54
Module Torque Constant	N.m/A	0.04
Motor Phase Resistance	Ω	1.07
Motor Phase Inductance	mH	0.89
Pole Pair	—	7
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.05
Insulation grade	—	F

Motor characteristic curve



Series Name

RMD-FL

Motor Simplified Name

FL-38-12



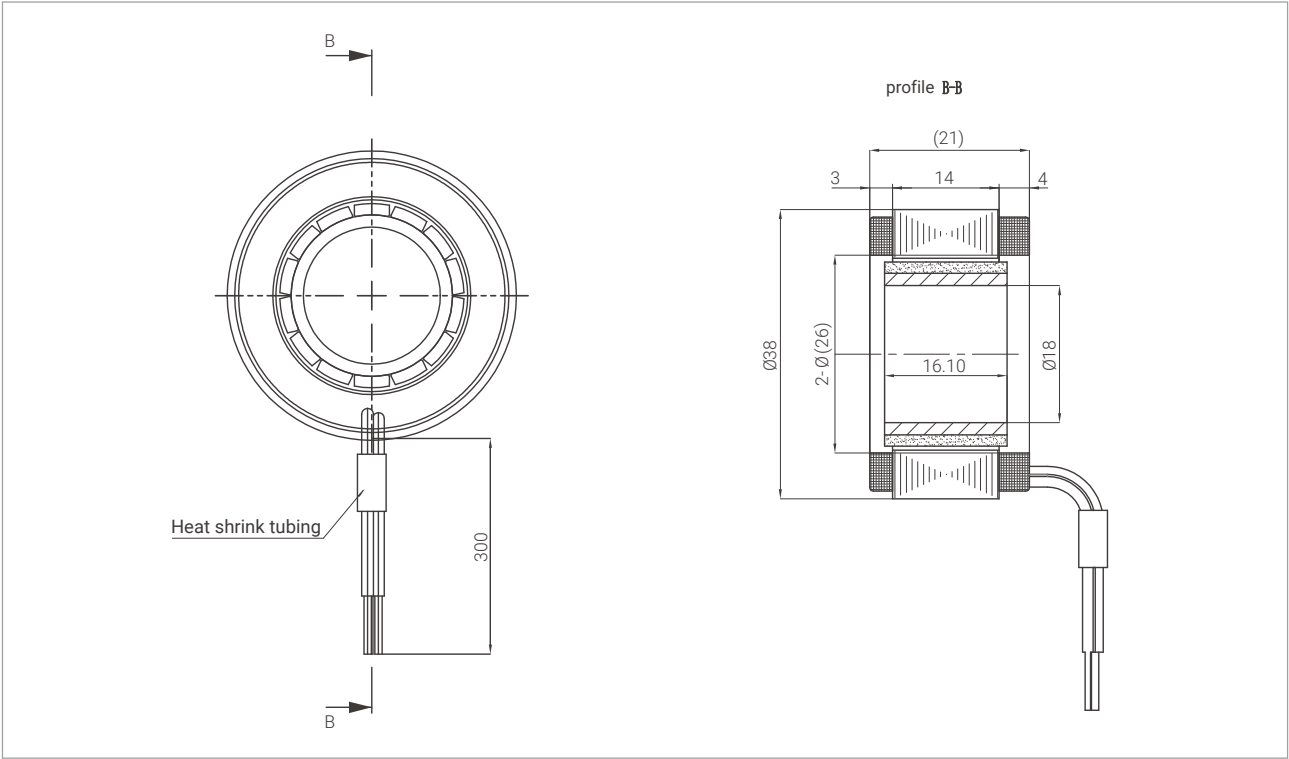
Actuator Full Name

RMD-FL-38-12-240

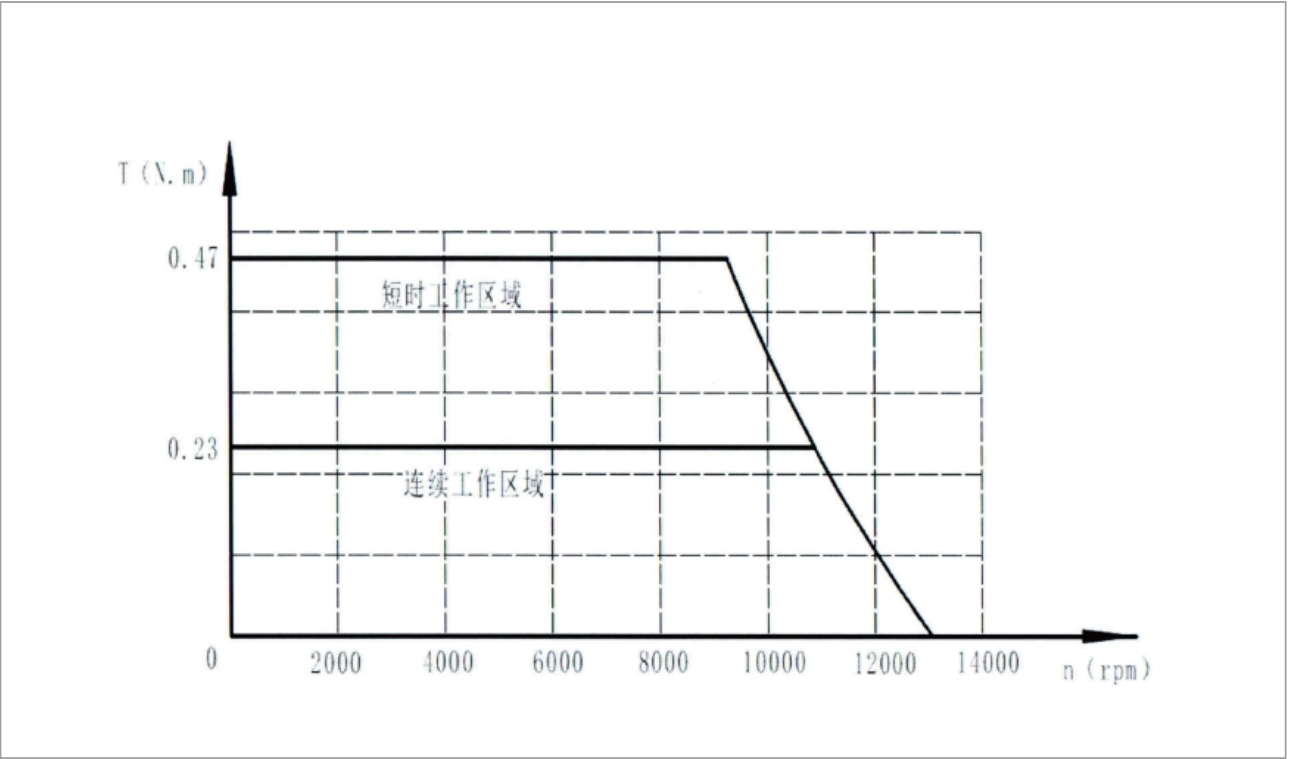
Product Parameters

Parameters	Unit	FL-38-12
Input Voltage	V	48
No Load Current	A	0.3
Rated Speed	RPM	9970
Rated Torque	N.m	0.23
Rated Power	W	240
Rated Current	A	6.1 (REF)
Peak Torque	N.m	0.47 (REF)
Peak Current	A	10.87
Efficiency	%	>80%
Motor Back-EMF Constant	Vdc/Krpm	2.4
Module Torque Constant	N.m/A	0.04
Motor Phase Resistance	Ω	0.5
Motor Phase Inductance	mH	0.27
Pole Pair	—	7
3 Phase Connection	—	Y
Encoder/Sensor Type	—	—
Weight	Kg	0.1
Insulation grade	—	F

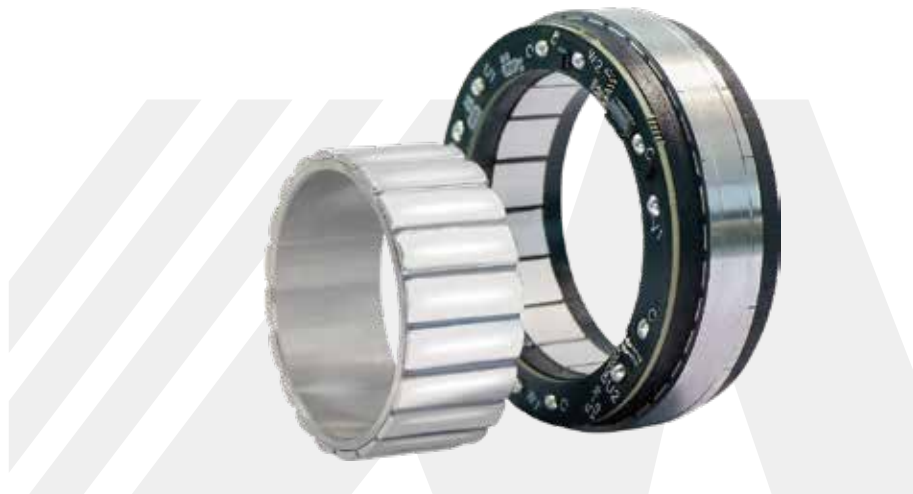
Installation Drawing



Motor characteristic curve

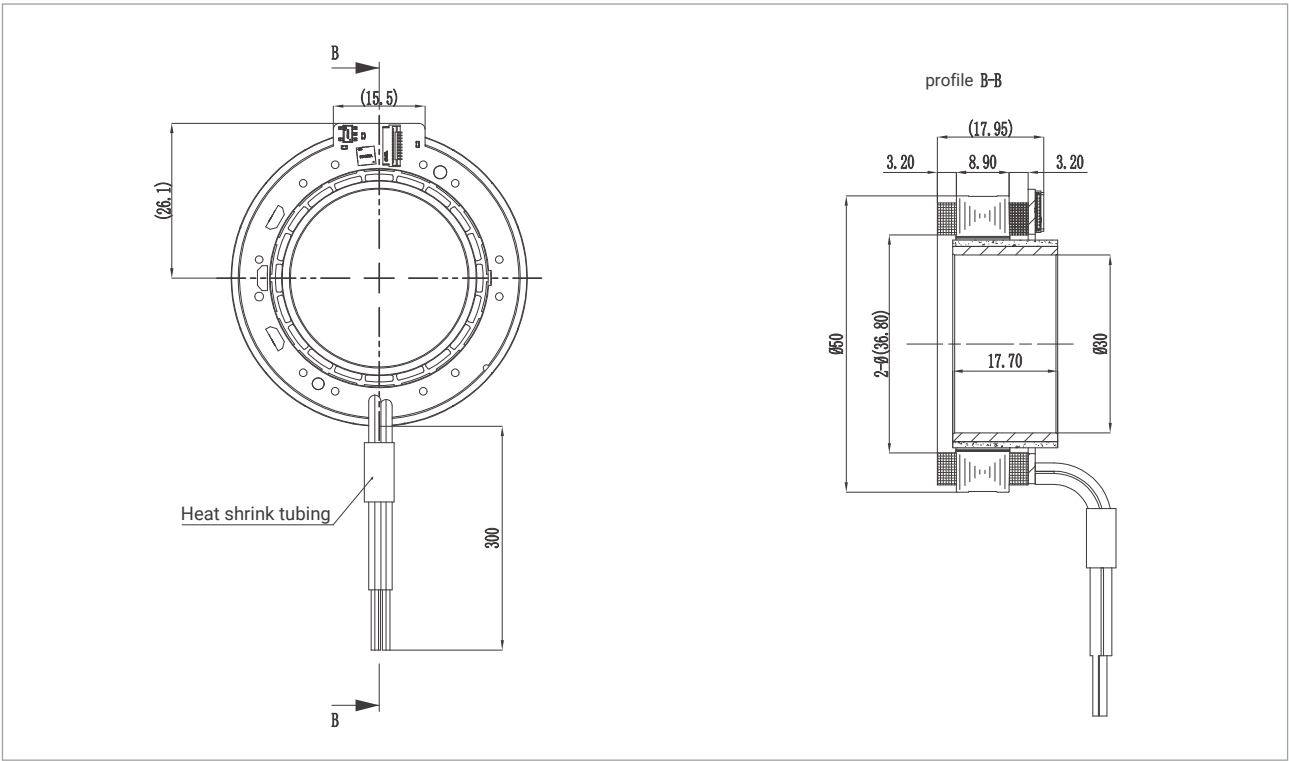


Series Name
RMD-FL
Motor Simplified Name
FL-50-08



Actuator Full Name	Encoder/Sensor
RMD-FL-50-08-180	Hall+INC

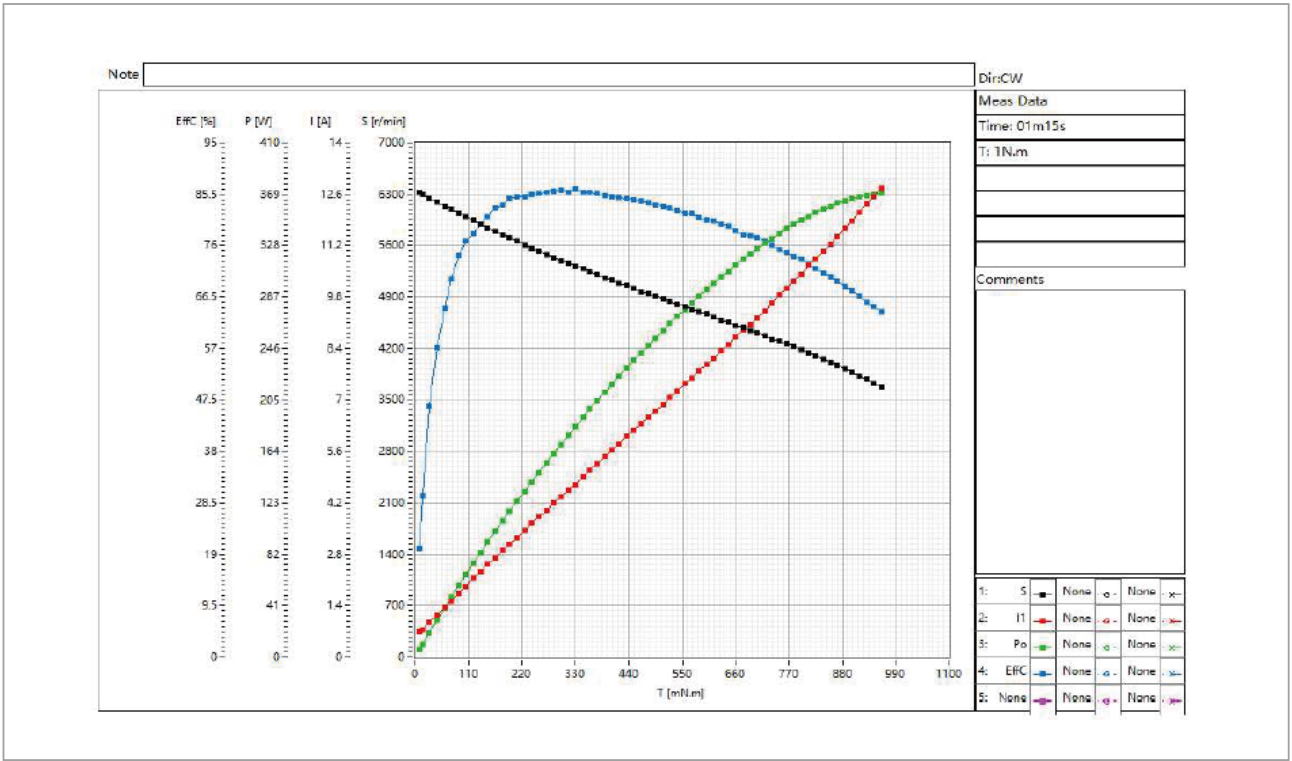
Installation Drawing



Product Parameters

Parameters	Unit	FL-50-08
Input Voltage	V	48
No Load Current	A	0.35
Rated Speed	RPM	5200
Rated Torque	N.m	0.3
Rated Power	W	163
Rated Current	A	3.9
Peak Torque	N.m	0.96
Peak Current	A	12.48
Efficiency	%	>83%
Motor Back-EMF Constant	Vdc/Krpm	5.37
Module Torque Constant	N.m/A	0.08
Motor Phase Resistance	Ω	0.62
Motor Phase Inductance	mH	0.43
Pole Pair	—	10
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.12
Insulation grade	—	F

Motor characteristic curve

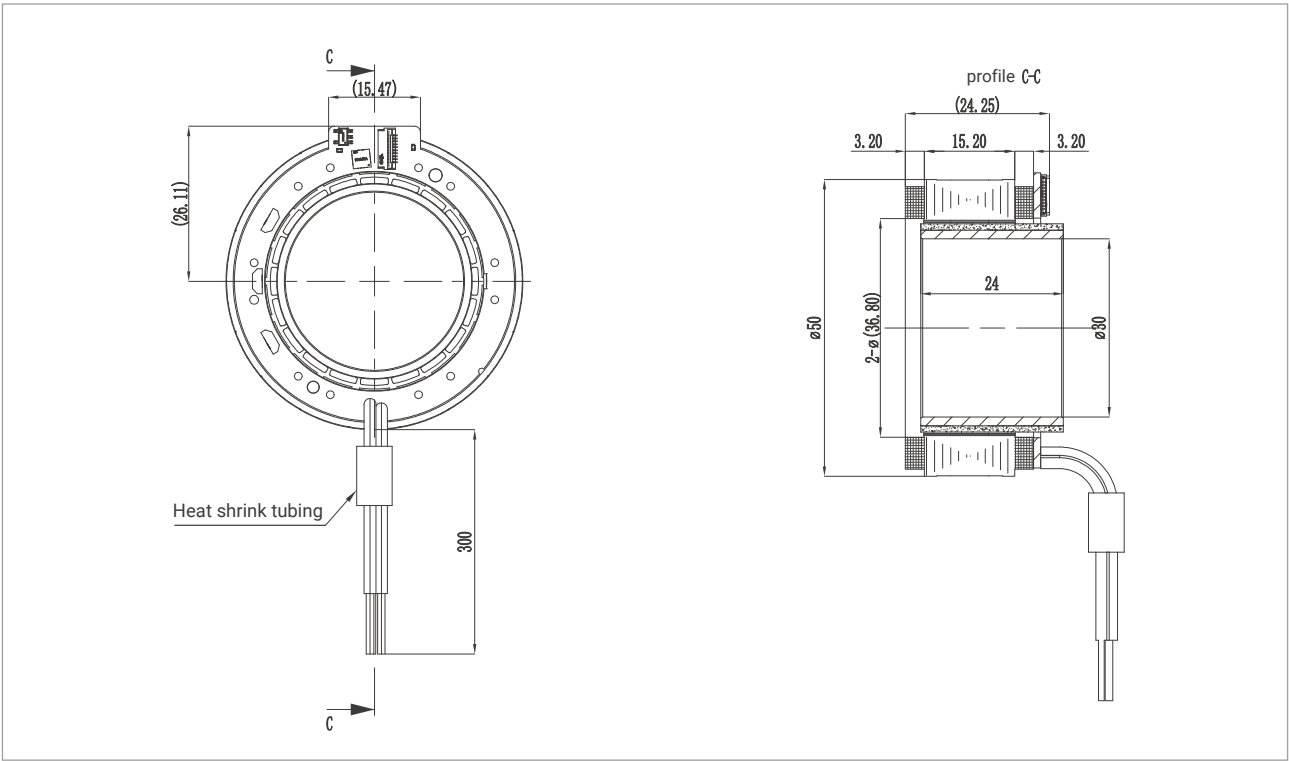


Series Name
RMD-FL
Motor Simplified Name
FL-50-15



Actuator Full Name	Encoder/Sensor
RMD-FL-50-15-200	Hall+INC

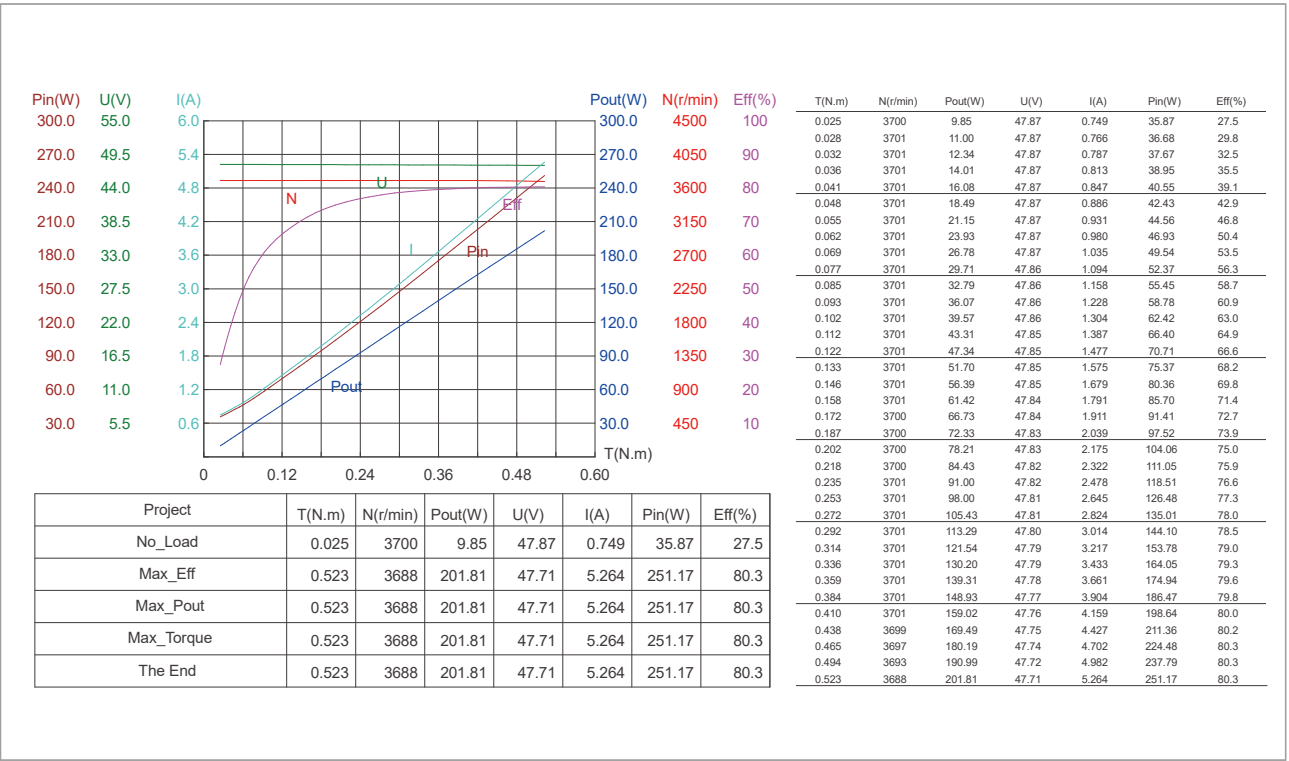
Installation Drawing



Product Parameters

Parameters	Unit	FL-50-15
Input Voltage	V	48
No Load Current	A	0.38
Rated Speed	RPM	3800
Rated Torque	N.m	0.51
Rated Power	W	203
Rated Current	A	4.3
Peak Torque	N.m	1.5
Peak Current	A	12.9
Efficiency	%	>81%
Motor Back-EMF Constant	Vdc/Krpm	8.12
Module Torque Constant	N.m/A	0.12
Motor Phase Resistance	Ω	0.7
Motor Phase Inductance	mH	0.47
Pole Pair	—	10
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.16
Insulation grade	—	F

Motor characteristic curve

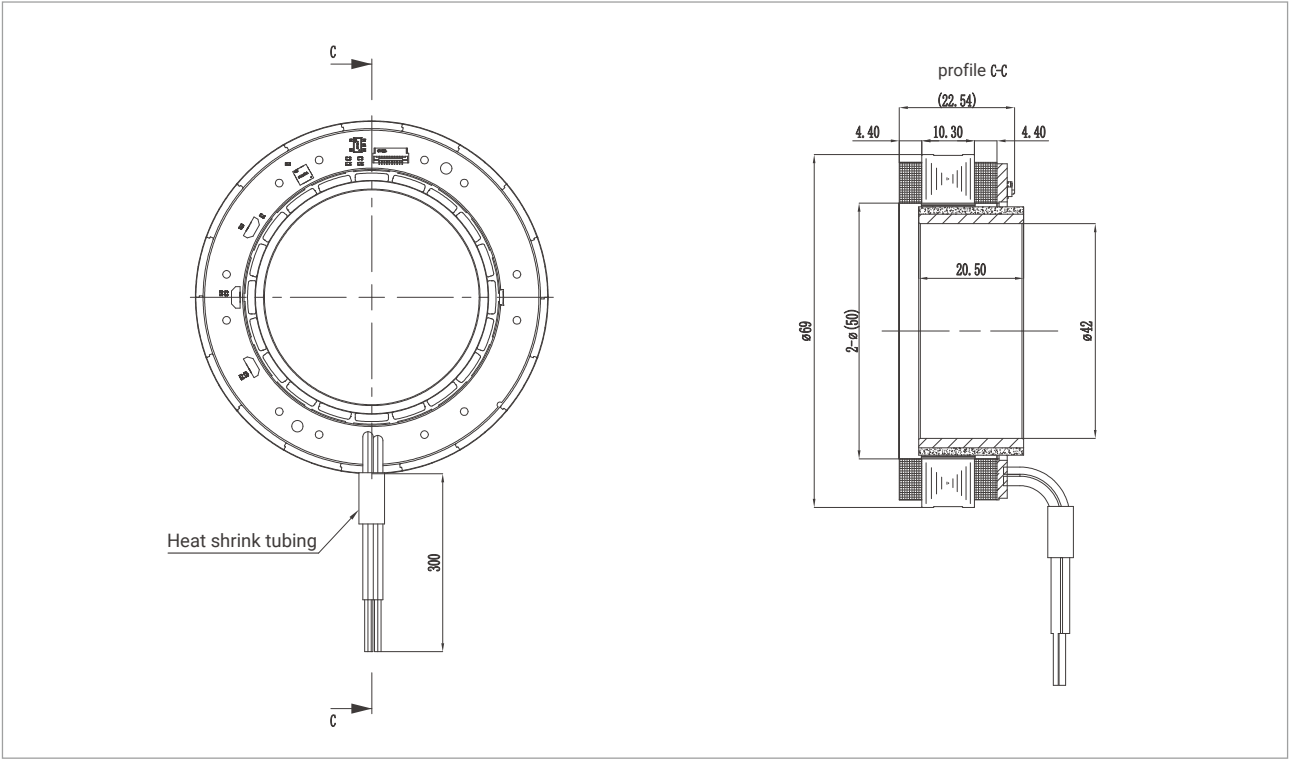


Series Name
RMD-FL
Motor Simplified Name
FL-70-10



Actuator Full Name	Encoder/Sensor
RMD-FL-70-10-250	Hall+INC

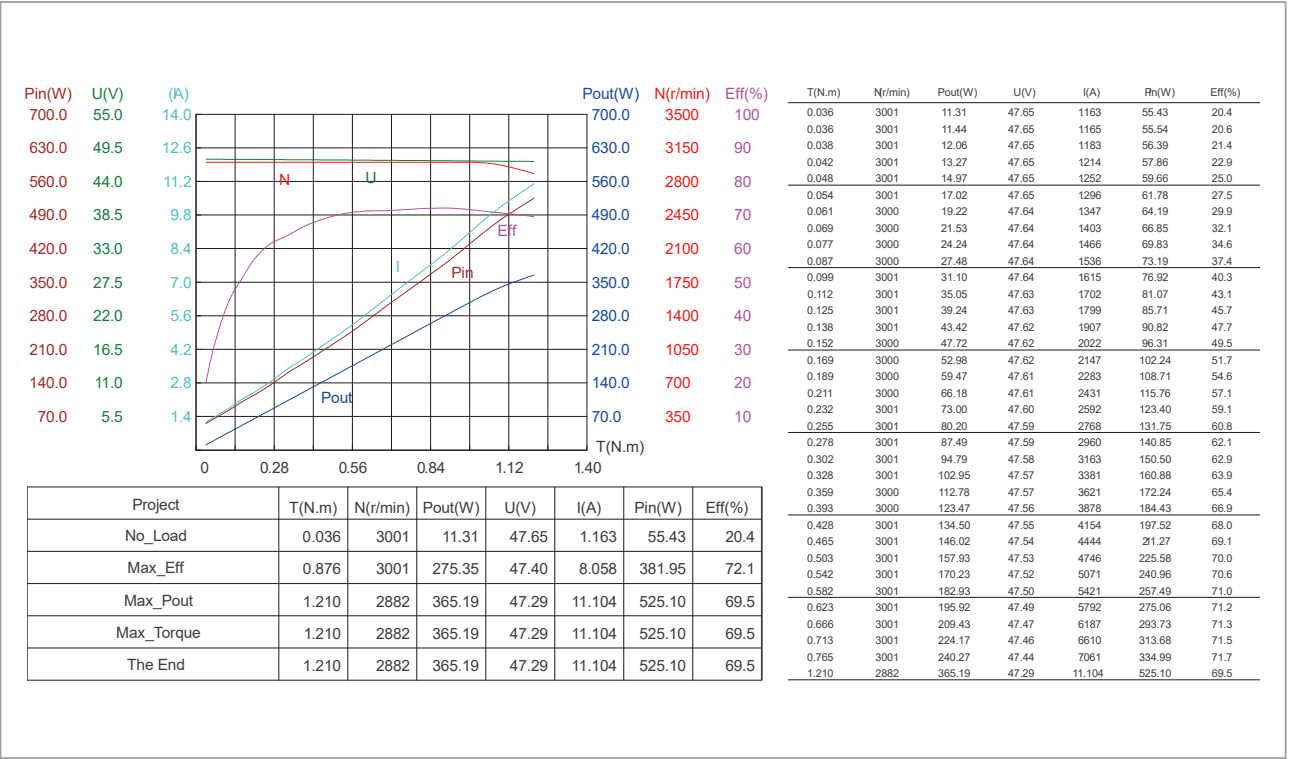
Installation Drawing



Product Parameters

Parameters	Unit	FL-70-10
Input Voltage	V	48
No Load Current	A	0.42
Rated Speed	RPM	3500
Rated Torque	N.m	0.55
Rated Power	W	201
Rated Current	A	4.4
Peak Torque	N.m	1.65
Peak Current	A	13.8
Efficiency	%	>82%
Motor Back-EMF Constant	Vdc/Krpm	8.34
Module Torque Constant	N.m/A	0.12
Motor Phase Resistance	Ω	0.36
Motor Phase Inductance	mH	0.51
Pole Pair	—	10
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.25
Insulation grade	—	F

Motor characteristic curve

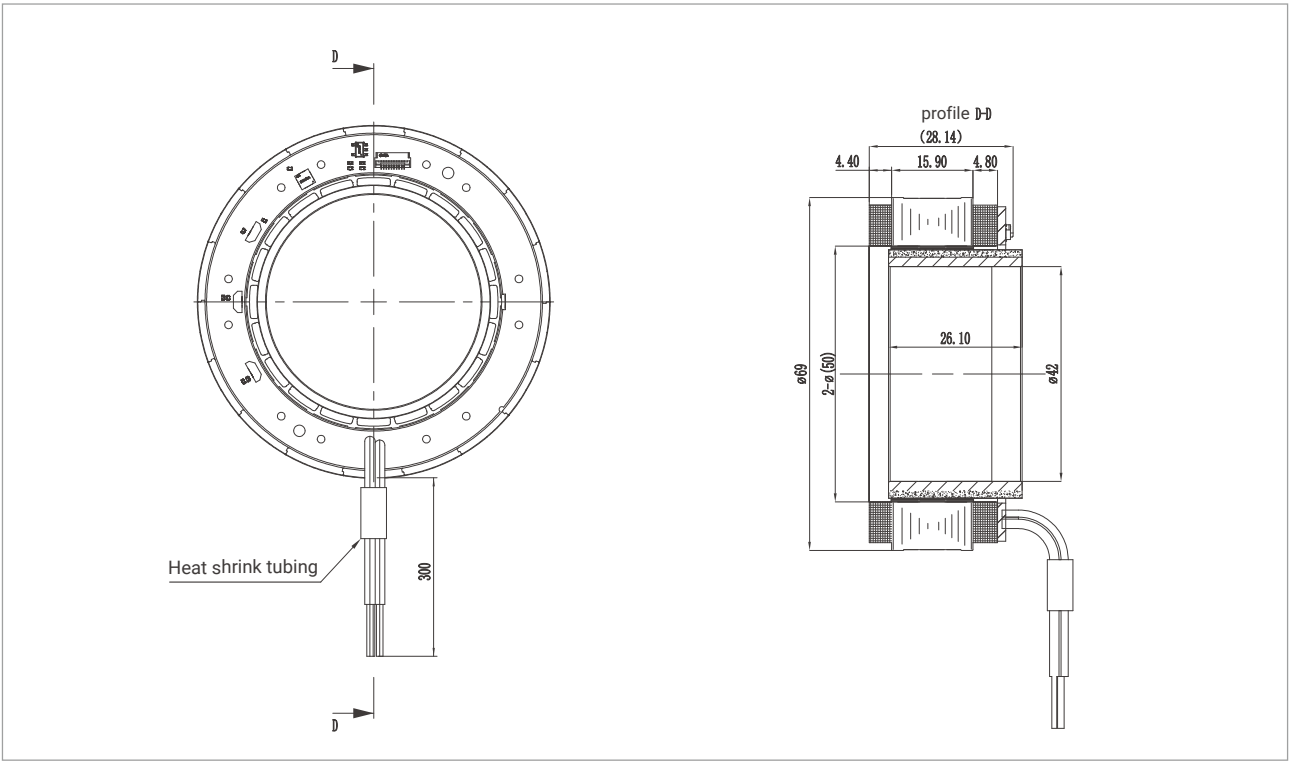


Series Name
RMD-FL
Motor Simplified Name
FL-70-16



Actuator Full Name	Encoder/Sensor
RMD-FL-70-16-360	Hall+INC

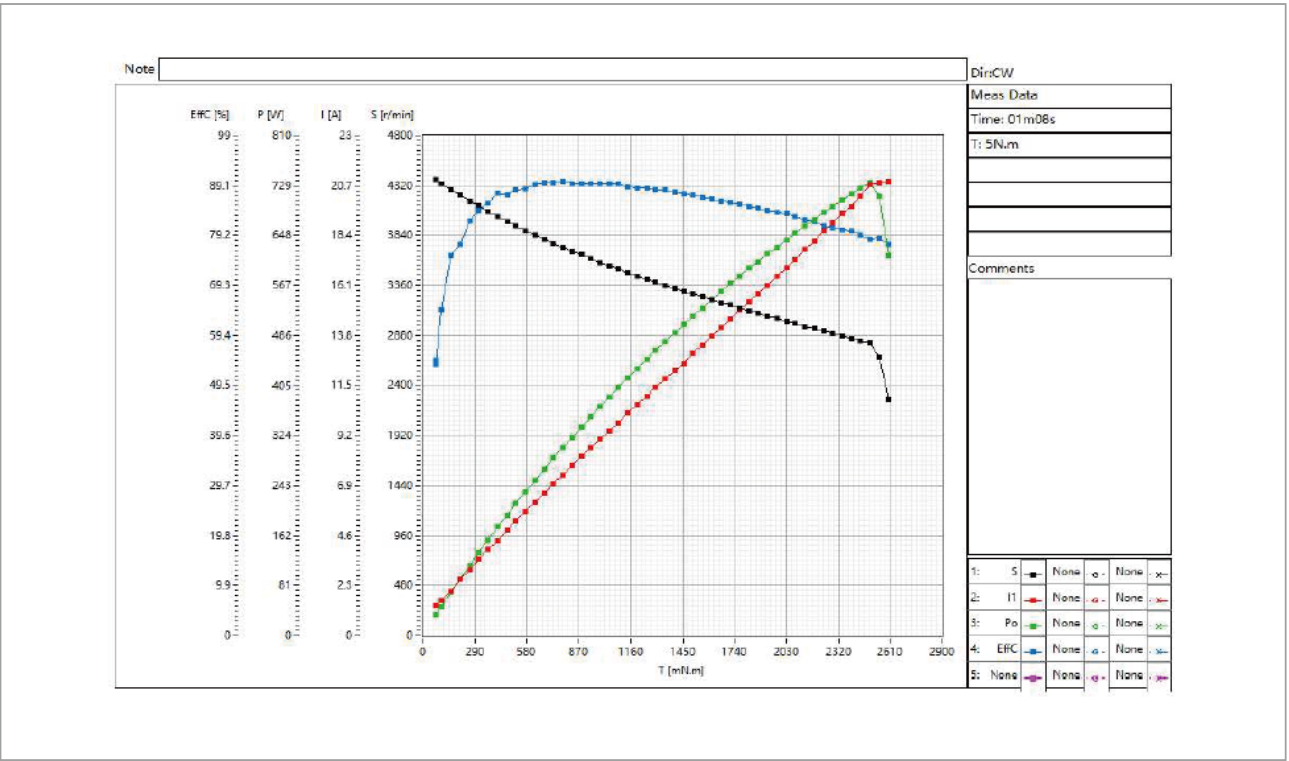
Installation Drawing



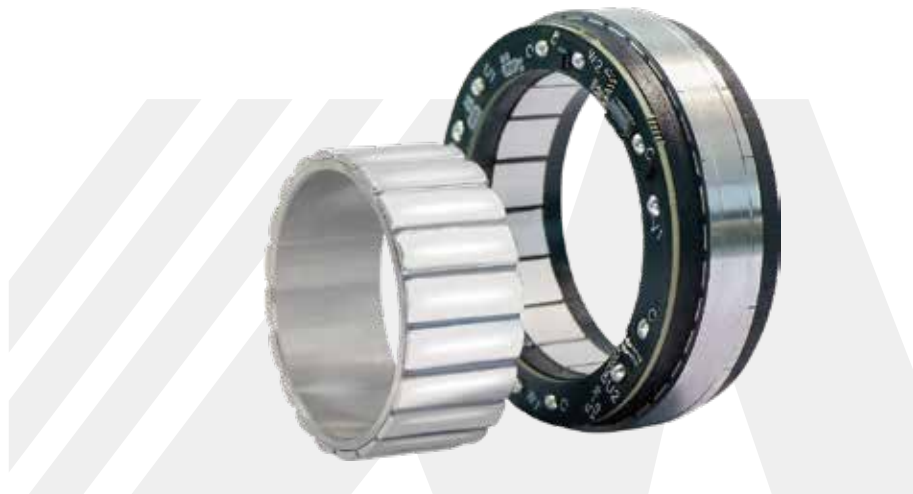
Product Parameters

Parameters	Unit	FL-70-16
Input Voltage	V	48
No Load Current	A	0.41
Rated Speed	RPM	3000
Rated Torque	N.m	1.3
Rated Power	W	408
Rated Current	A	9.2
Peak Torque	N.m	2.6
Peak Current	A	27.6
Efficiency	%	>83%
Motor Back-EMF Constant	Vdc/Krpm	9.86
Module Torque Constant	N.m/A	0.14
Motor Phase Resistance	Ω	0.25
Motor Phase Inductance	mH	0.38
Pole Pair	—	10
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.32
Insulation grade	—	F

Motor characteristic curve

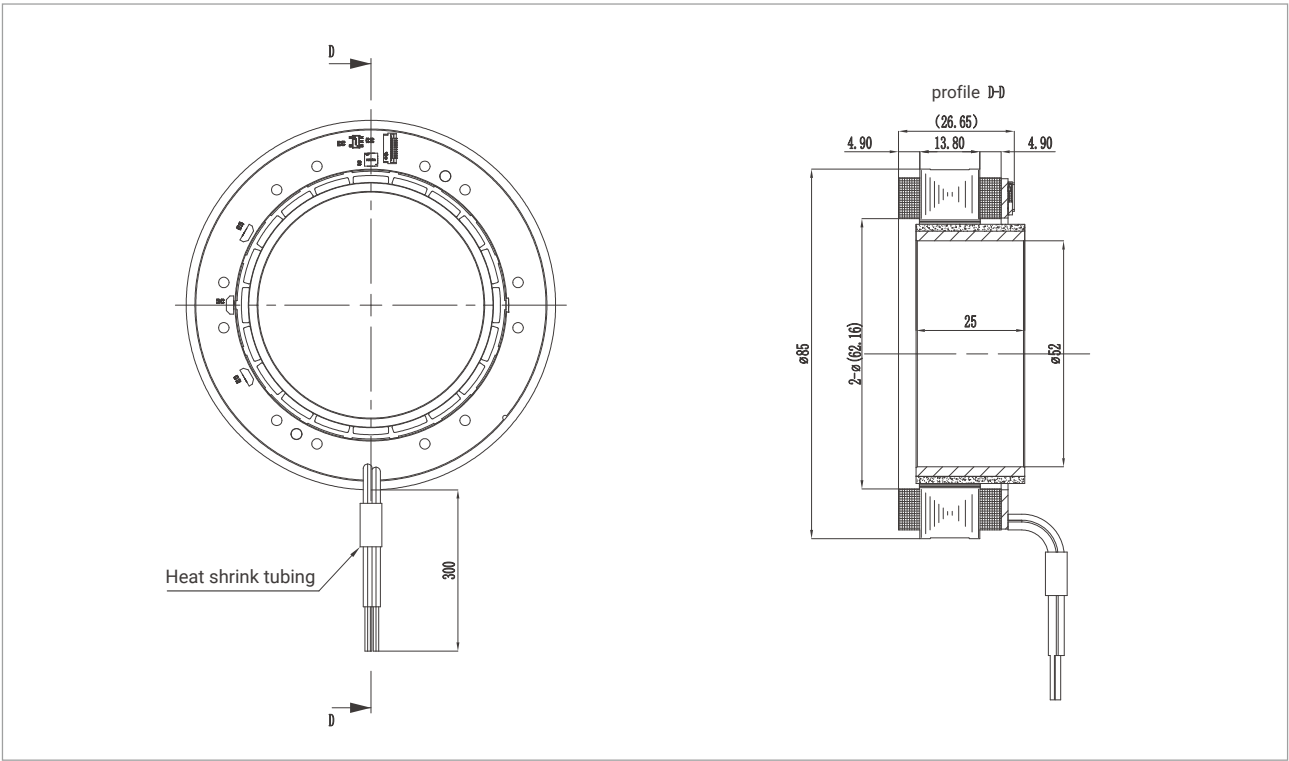


Series Name
RMD-FL
Motor Simplified Name
FL-85-13



Actuator Full Name	Encoder/Sensor
RMD-FL-85-13-500	Hall+INC

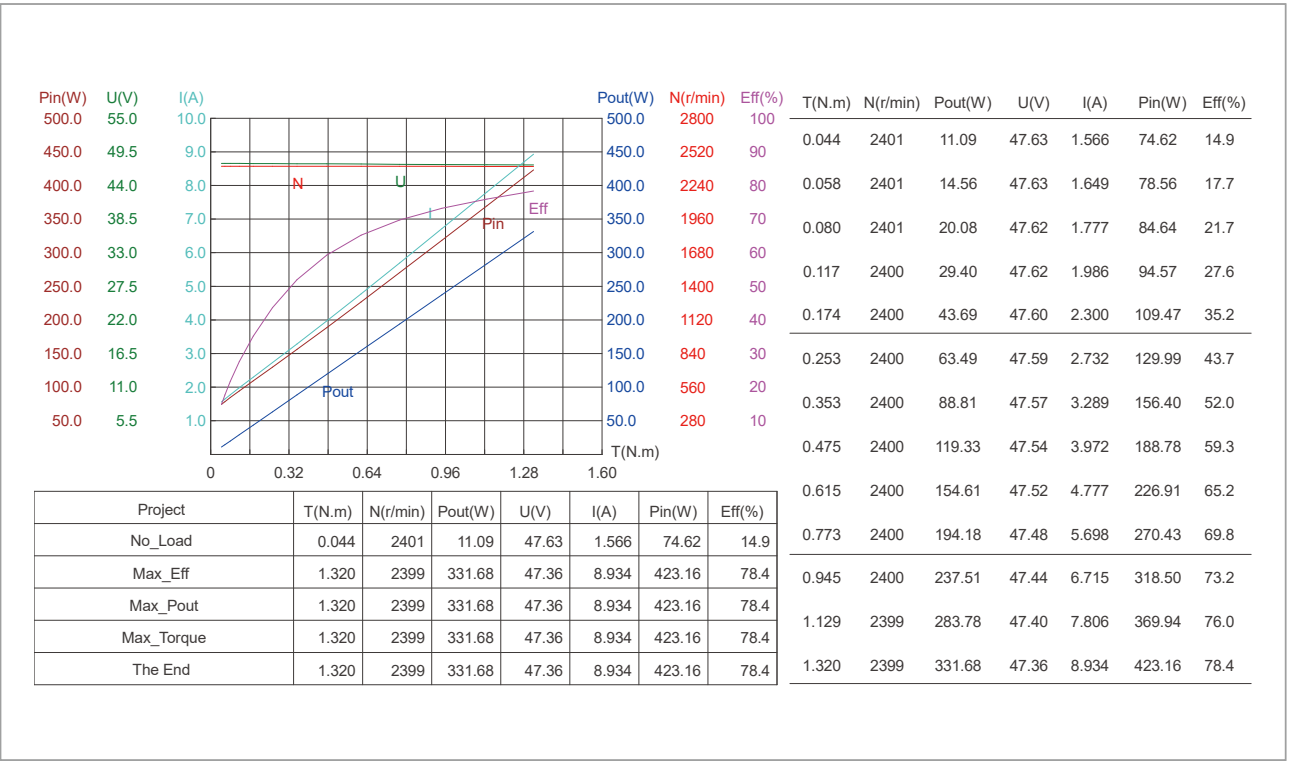
Installation Drawing



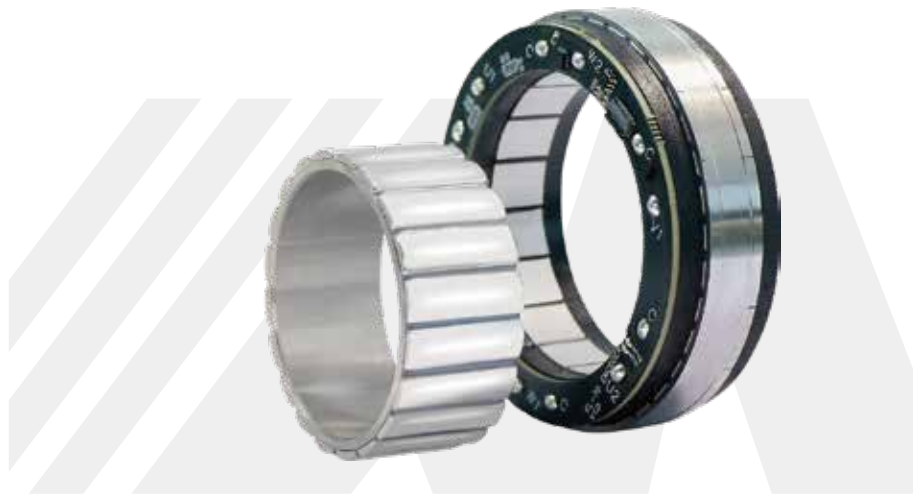
Product Parameters

Parameters	Unit	FL-85-13
Input Voltage	V	48
No Load Current	A	0.37
Rated Speed	RPM	2400
Rated Torque	N.m	1.3
Rated Power	W	408
Rated Current	A	9.4
Peak Torque	N.m	3.9
Peak Current	A	28.2
Efficiency	%	>82%
Motor Back-EMF Constant	Vdc/Krpm	9.65
Module Torque Constant	N.m/A	0.14
Motor Phase Resistance	Ω	0.16
Motor Phase Inductance	mH	0.33
Pole Pair	—	10
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.46
Insulation grade	—	F

Motor characteristic curve

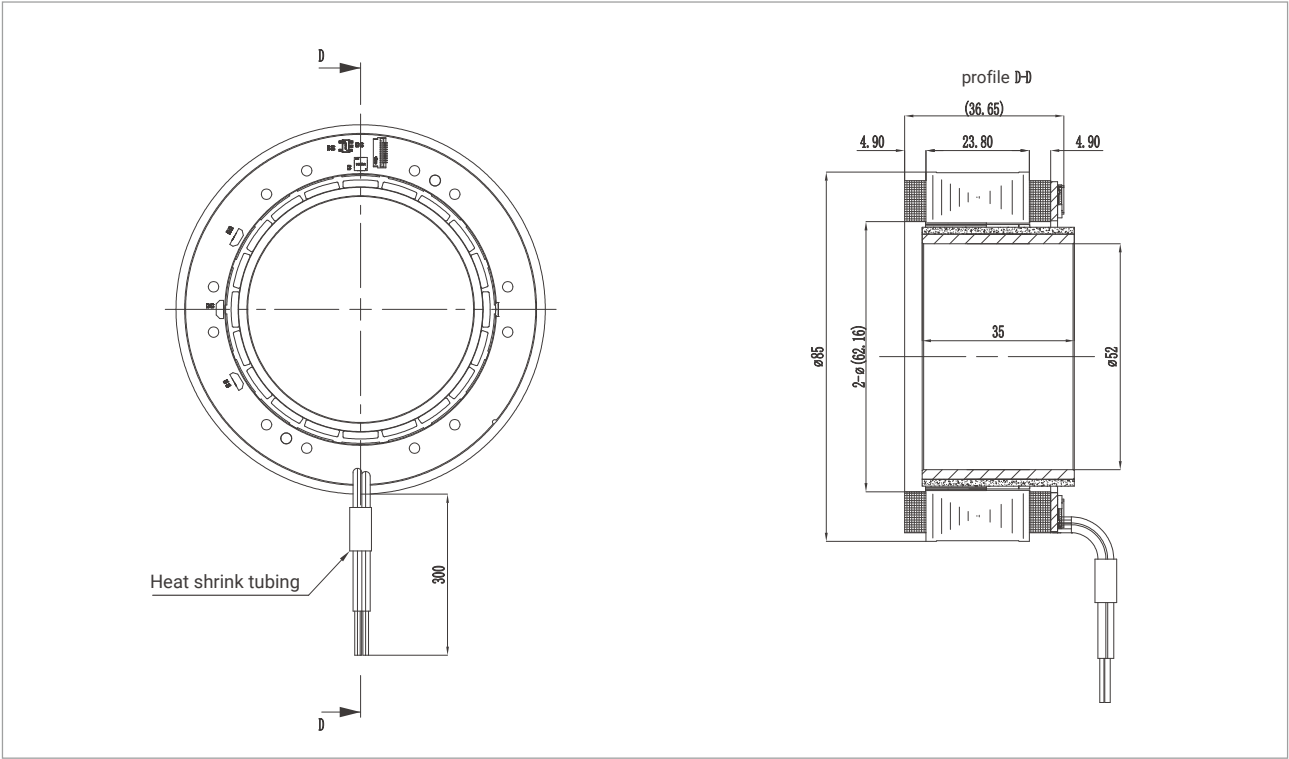


Series Name
RMD-FL
Motor Simplified Name
FL-85-23



Actuator Full Name	Encoder/Sensor
RMD-FL-85-23-750	Hall+INC

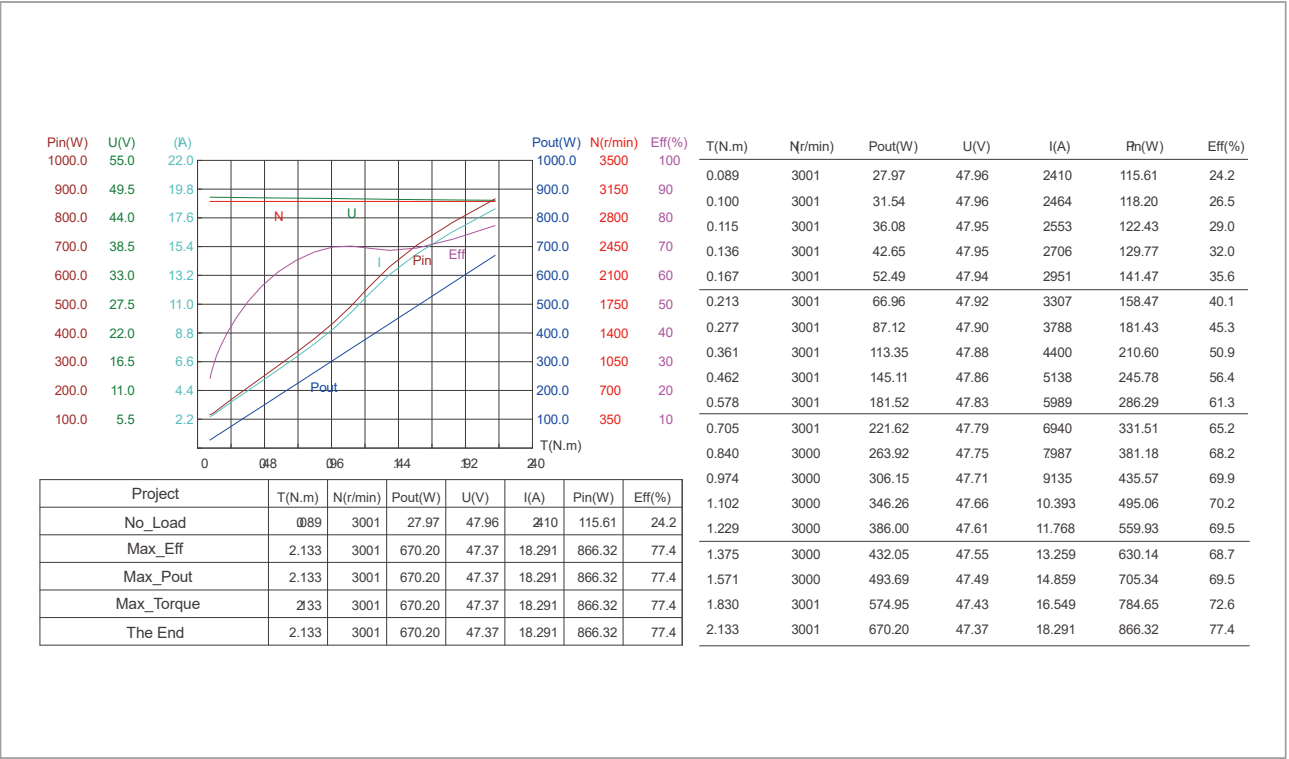
Installation Drawing



Product Parameters

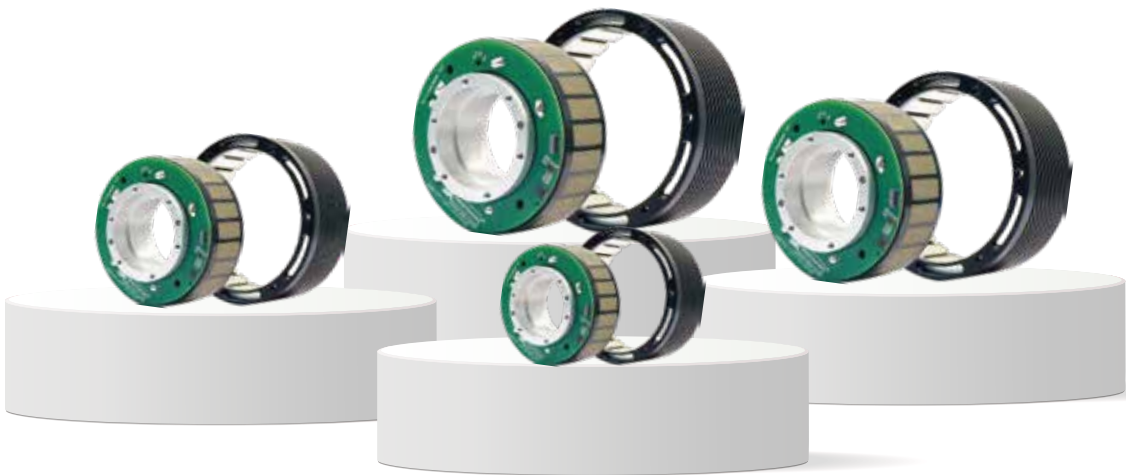
Parameters	Unit	FL-85-23
Input Voltage	V	48
No Load Current	A	0.38
Rated Speed	RPM	3500
Rated Torque	N.m	2
Rated Power	W	733
Rated Current	A	14.5
Peak Torque	N.m	6
Peak Current	A	43.5
Efficiency	%	>81%
Motor Back-EMF Constant	Vdc/Krpm	9.63
Module Torque Constant	N.m/A	0.14
Motor Phase Resistance	Ω	0.08
Motor Phase Inductance	mH	0.18
Pole Pair	—	10
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.61
Insulation grade	—	F

Motor characteristic curve



FLO Series Frameless Torque Motor

Outer Rotor



* The pictures of this series of products are subject to the actual products.

Product Features



Flexible integration



High Power Density



Good heat dissipation



High Precision

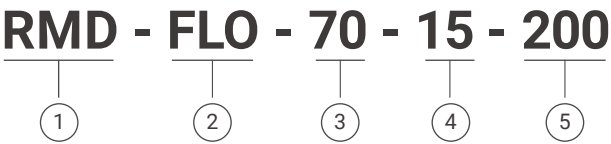


Light Weight



Hall+INC

RMD-FLO Series Naming Conventions



- 1

RMD:

Brand Name R-Reducer M-Motor D-Drive
- 2

FLO:

FLO Stands For The Series Name: Outer rotor frameless torque motor
- 3

70:

Rotor outer diameter 70mm
- 4

15:

Stator stack height 15mm
- 5

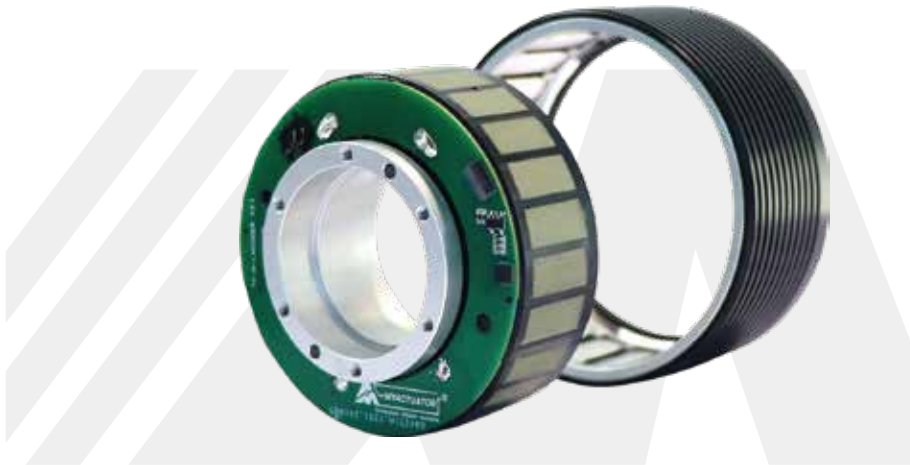
200:

Rated power 200W

For example

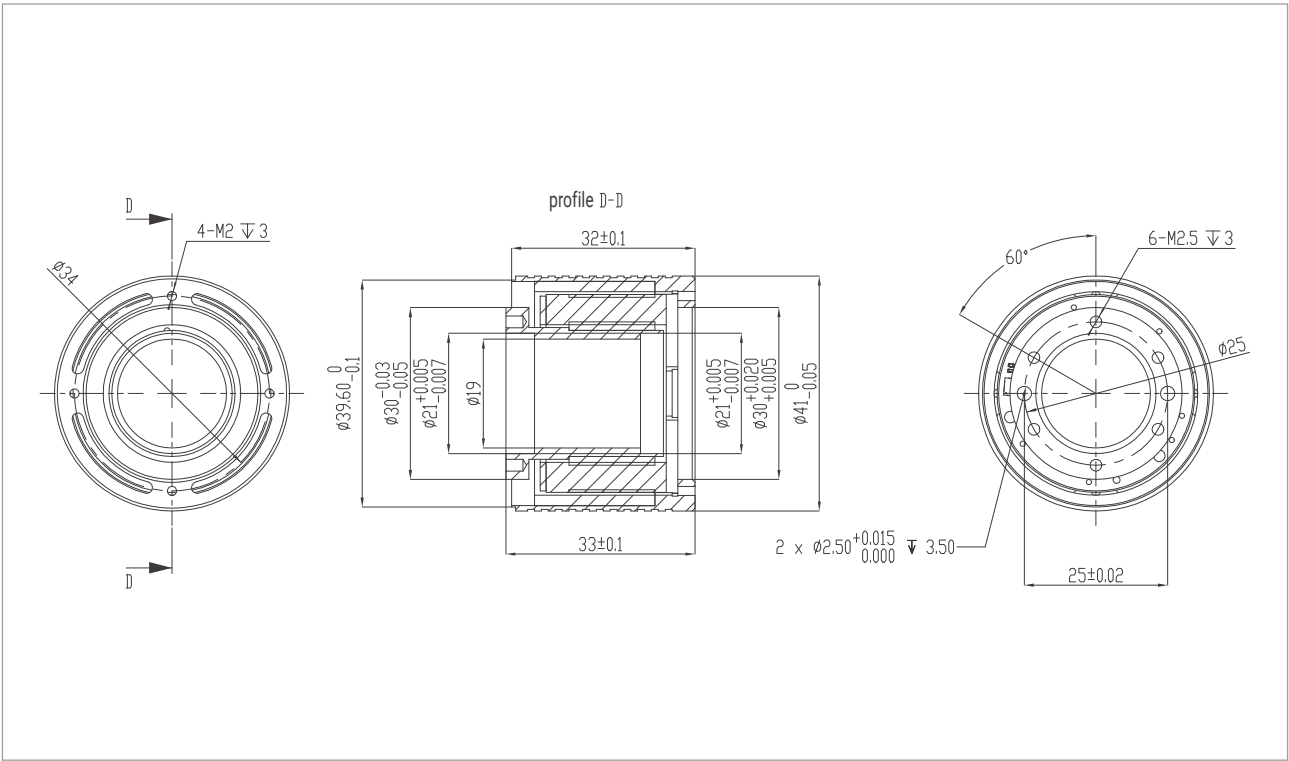
Model	RMD-FLO-90-15-400
Illustrate	RMD: Brand Name
	FLO: Outer rotor frameless torque motor
	90: Rotor outer diameter
	15: Stator stack height
	400: Rated power

Series Name
RMD-FLO
Motor Simplified Name
FLO-40-15



Actuator Full Name	Encoder/Sensor
RMD-FLO-40-15-50	Hall+INC

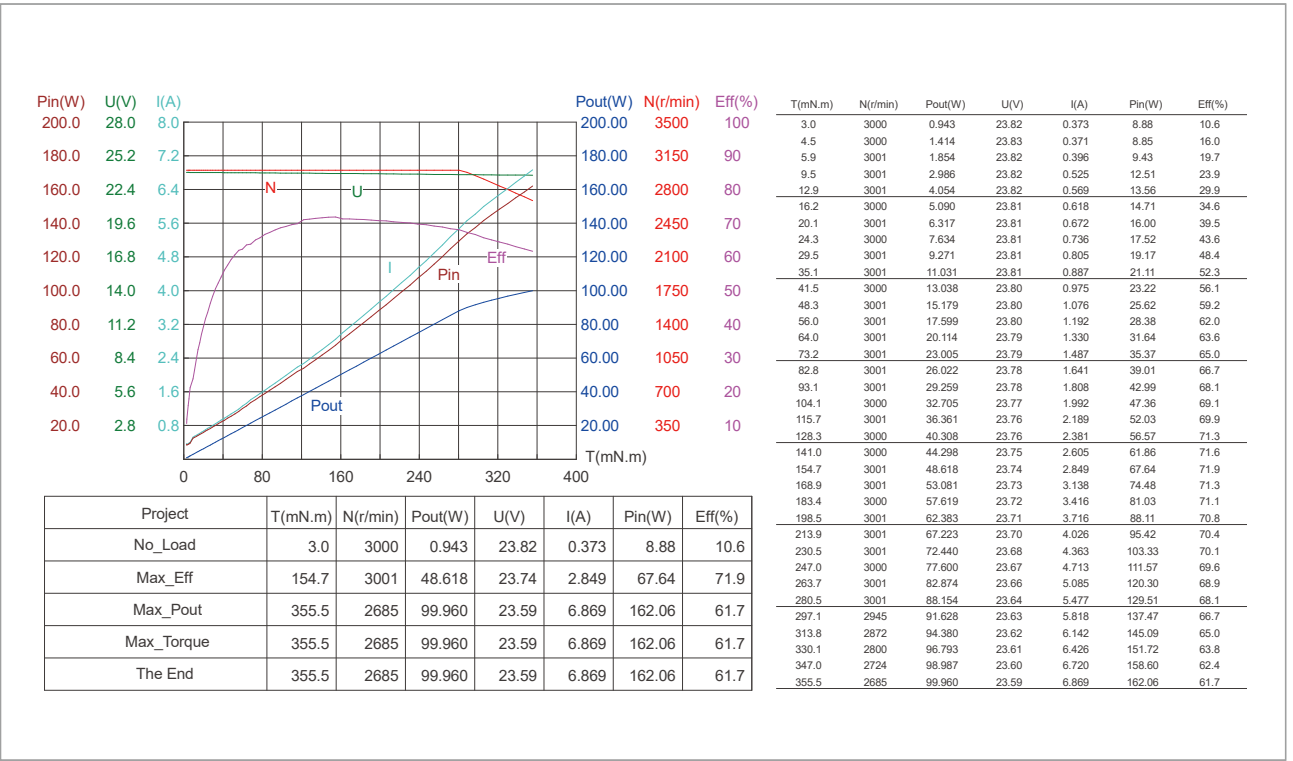
Installation Drawing



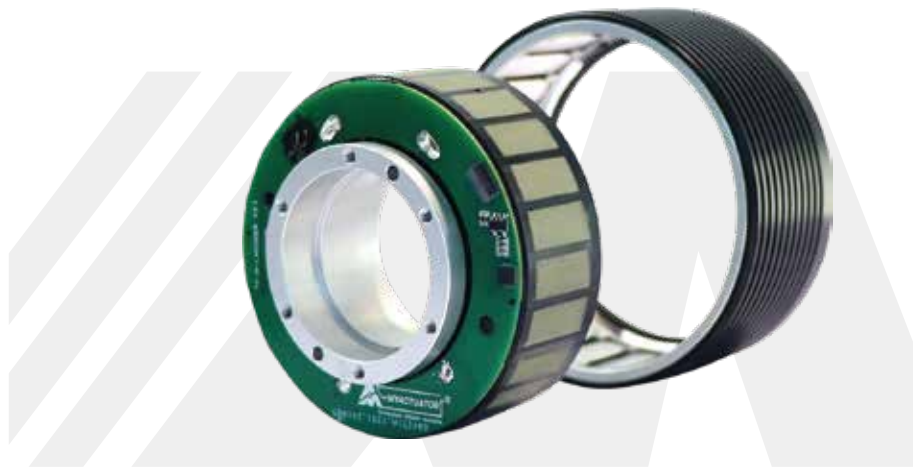
Product Parameters

Parameters	Unit	FLO-40-15
Input Voltage	V	24
No Load Current	A	0.44
Rated Speed	RPM	3000
Rated Torque	N.m	0.15
Rated Power	W	47
Rated Current	A	5.1
Peak Torque	N.m	0.3
Peak Current	A	12.8
Efficiency	%	>64%
Motor Back-EMF Constant	Vdc/Krpm	8
Module Torque Constant	N.m/A	0.03
Motor Phase Resistance	Ω	3.37
Motor Phase Inductance	mH	0.176
Pole Pair	—	13
Cogging torque	N.m	10
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	—
Insulation grade	—	F

Motor characteristic curve

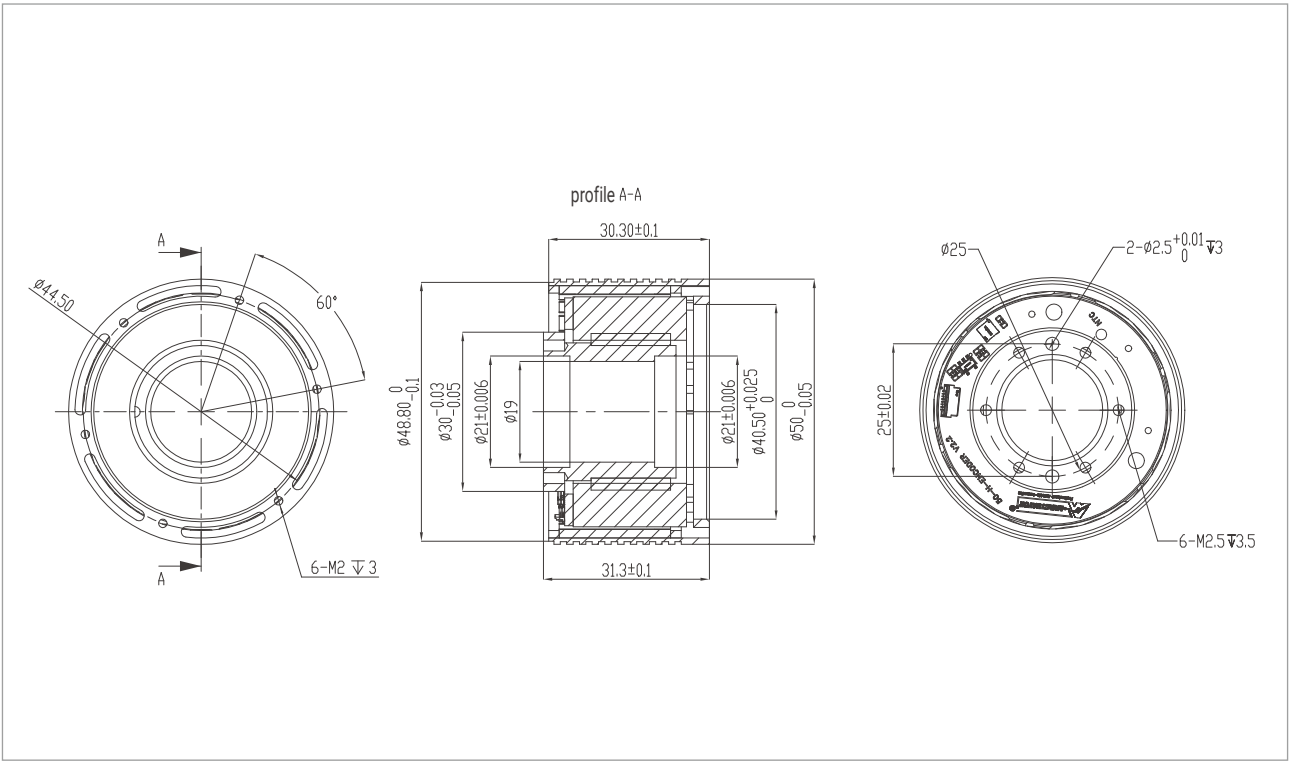


Series Name
RMD-FLO
Motor Simplified Name
FLO-50-15



Actuator Full Name	Encoder/Sensor
RMD-FLO-50-15-100	Hall+INC

Installation Drawing



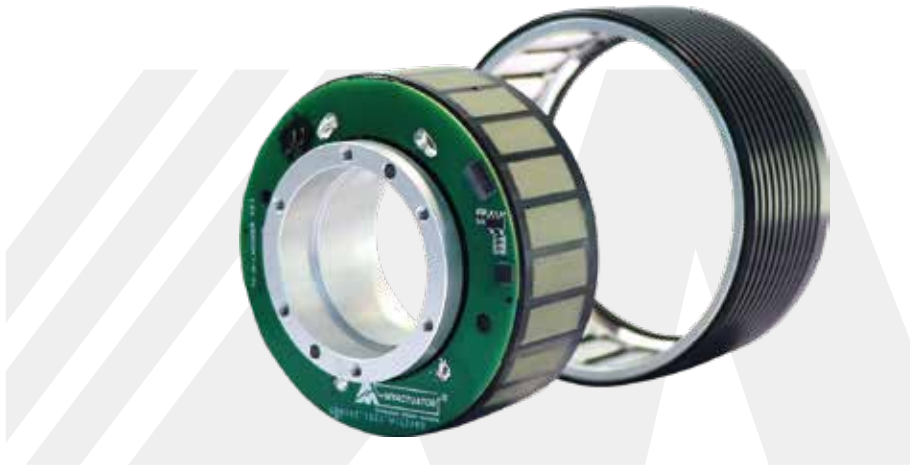
Product Parameters

Parameters	Unit	FLO-50-15
Input Voltage	V	24
No Load Current	A	0.32
Rated Speed	RPM	2500
Rated Torque	N.m	0.35
Rated Power	W	91.6
Rated Current	A	4.9
Peak Torque	N.m	0.87
Peak Current	A	14.7
Efficiency	%	>83%
Motor Back-EMF Constant	Vdc/Krpm	7.93
Module Torque Constant	N.m/A	0.07
Motor Phase Resistance	Ω	0.4
Motor Phase Inductance	mH	0.19
Pole Pair	—	13
Cogging torque	N.m	12
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.19
Insulation grade	—	F

Motor characteristic curve

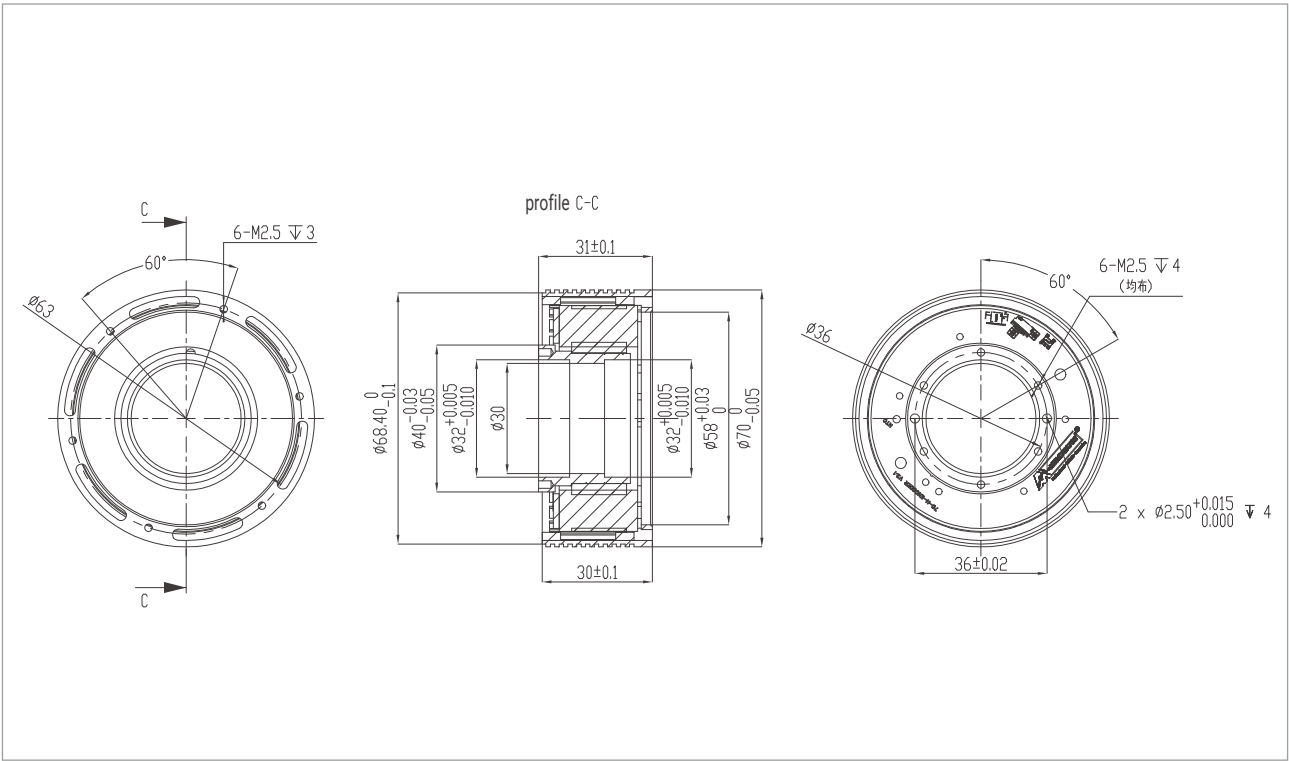


Series Name
RMD-FLO
Motor Simplified Name
FLO-70-15



Actuator Full Name	Encoder/Sensor
RMD-FLO-70-15-200	Hall+INC

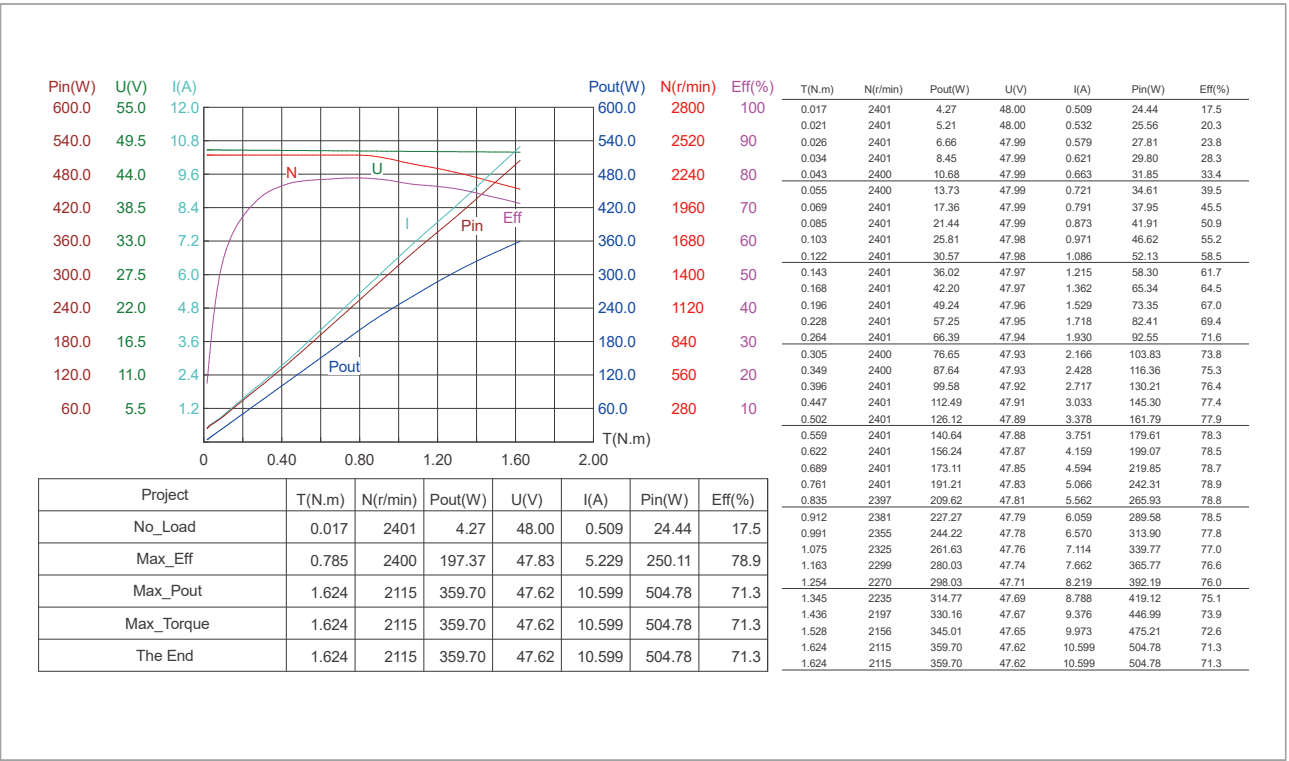
Installation Drawing



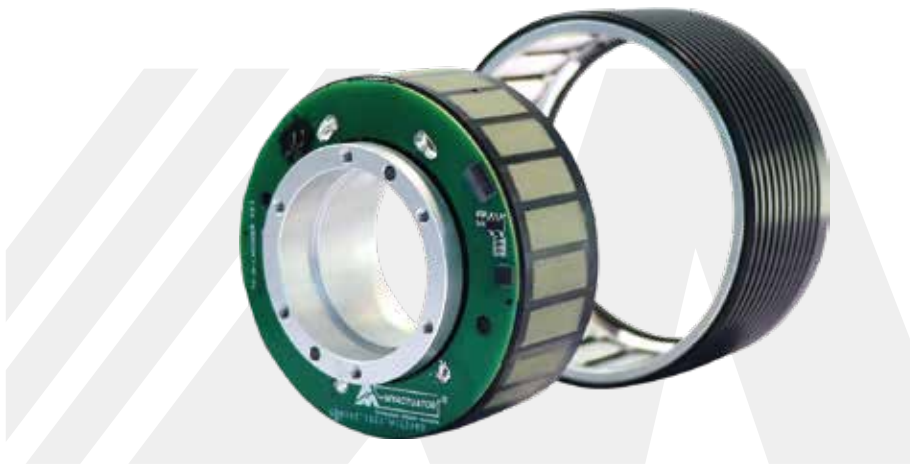
Product Parameters

Parameters	Unit	FLO-70-15
Input Voltage	V	48
No Load Current	A	0.21
Rated Speed	RPM	2400
Rated Torque	N.m	0.8
Rated Power	W	200
Rated Current	A	4.4
Peak Torque	N.m	2
Peak Current	A	13.2
Efficiency	%	>84%
Motor Back-EMF Constant	Vdc/Krpm	17
Module Torque Constant	N.m/A	0.18
Motor Phase Resistance	Ω	0.5
Motor Phase Inductance	mH	0.55
Pole Pair	—	13
Cogging torque	N.m	30.83
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.33
Insulation grade	—	F

Motor characteristic curve

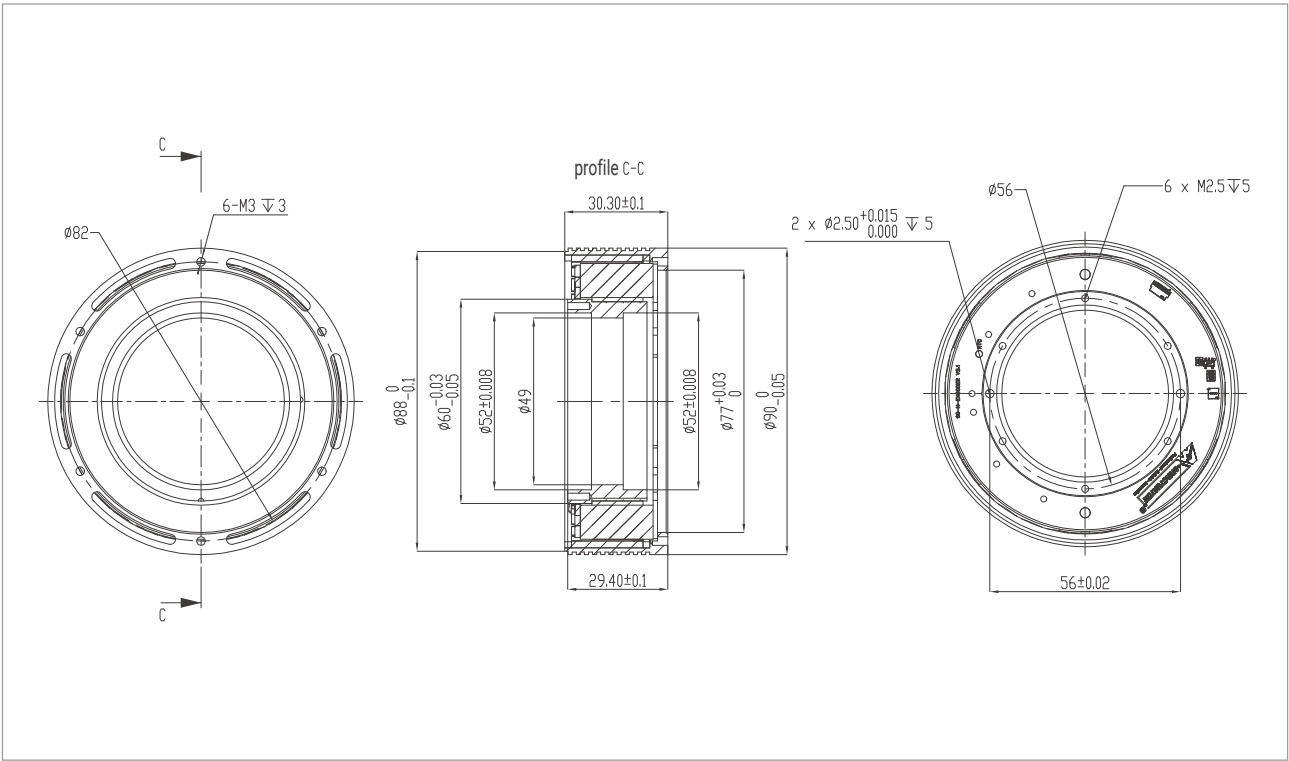


Series Name
RMD-FLO
Motor Simplified Name
FLO-90-15



Actuator Full Name	Encoder
RMD-FLO-90-15-400	Hall+INC

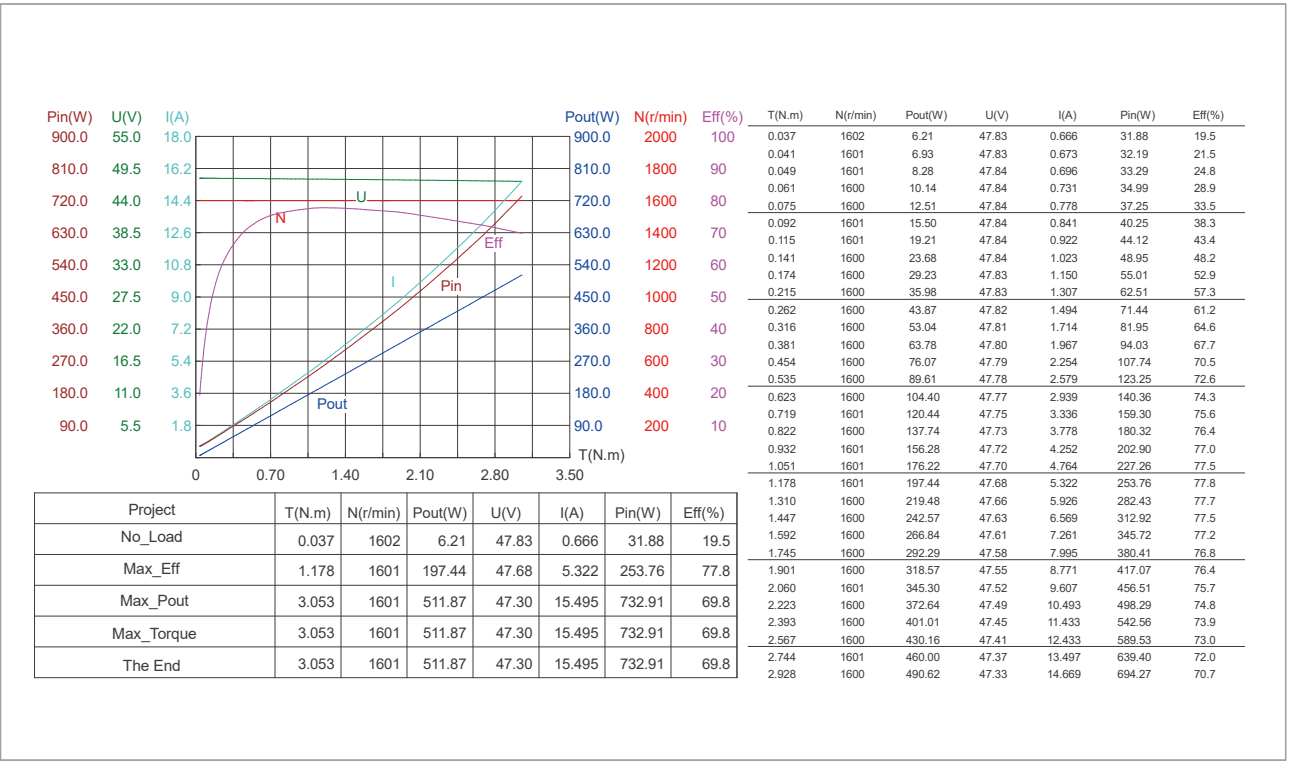
Installation Drawing



Product Parameters

Parameters	Unit	FLO-90-15
Input Voltage	V	48
No Load Current	A	0.35
Rated Speed	RPM	1600
Rated Torque	N.m	1.5
Rated Power	W	252
Rated Current	A	6.6
Peak Torque	N.m	3.75
Peak Current	A	19.8
Efficiency	%	>83%
Motor Back-EMF Constant	Vdc/Krpm	21.1
Module Torque Constant	N.m/A	0.23
Motor Phase Resistance	Ω	0.35
Motor Phase Inductance	mH	0.28
Pole Pair	—	17
Cogging torque	N.m	64
3 Phase Connection	—	Y
Encoder/Sensor Type	—	Hall+INC
Weight	Kg	0.46
Insulation grade	—	F

Motor characteristic curve





Core Components

- 01 Head

Direct Drive Actuator
L-5015*2
- 02 Upper arm

Harmonic Actuator
RH-17*6
- 03 Waist

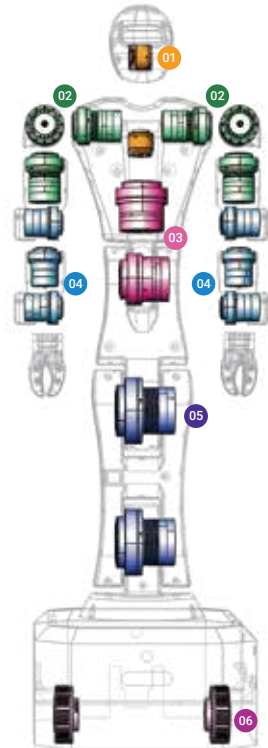
Harmonic Actuator
RH-25*2
- 04 Lower Arm

Harmonic Actuator
RH-14*6
- 05 Legs

Harmonic Actuator
RH-32*2
- 06 Bottom

Planetary Actuator
X10-40*2

Communication
CAN BUS



Component Parameters

Parameters		Unit	L-5015	X10-40	RH-14	RH-17	RH-25	RH-32
Gear ratio		—	-	7	100	100	100	100
Rated Voltage		V	24	48	48	48	48	48
Rated Power		W	100	265	28	91	282	282
Rated Torque		N.m	0.36	15	11	35	108	150
Rated Speed		RPM	350	165	25	25	25	18
Peak Torque		N.m	0.82	40	28	54	157	229
No Load Speed		RPM	680	-	30	30	30	20
Torque constant		N.m/A	0.23	-	4	7.4	10.1	6.9
Back-EMF Constant		Vdc/Krpm	-	-	19.2	19.2	19.2	26.7
Efficiency		%	-	1625	—	—	—	—
Control Accuracy		Degree	< 0.01	-	< 0.01	< 0.01	< 0.01	< 0.01
Inertia	N	kg.cm²	0.268	1625	0.29	0.52	1.94	6.86
	B	kg.cm²	-	-	—	0.56	2.32	8.32
Weight	N	Kg	0.174	1.15	0.78	1.11	2.42	4.32
	B	Kg	-	-	/	1.28	2.74	4.74
Communication		—	CAN BUS	CAN BUS	CAN BUS			
Encoder		—	14-BIT	-	Dual Encoder ABS-17BIT/ABS-17BIT			

1.3m

Half-scale Humanoid Power System Solution

Full body freedom

29

Machine quality

35 kg

Overall height

130 cm

Arm movement radius

420 mm

Wrist load

3 kg

Repeat positioning accuracy

0.1 mm

Core Components

- 01

Head

Planetary Actuator
X2-7*2
- 02

Wrist

Planetary Actuator
X4-10*4
- 03

The upper arm and elbow, forearm/ankle

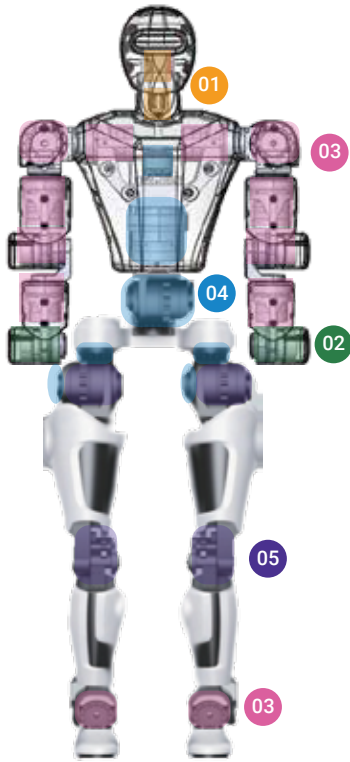
Planetary Actuator
X4-36*12
- 04

Hip side / Hip rotation / Waist

Planetary Actuator
X6-60*7
- 05

Front of the hip、Knee

Planetary Actuator
X8-120*4



Communication
CAN BUS

Component Parameters

Parameters	Unit	X2-7	X4-10	X4-36	X6-60	X8-120
Gear ratio	-	28.17	12.5	36	19.612	19.612
Rated Voltage	V	24	24	24	48	48
Rated Power	W	37	100	100	320	574
Rated Torque	N.m	2.5	4.0	10.5	20.0	43.0
Rated Speed	RPM	142	238	83	153	127
Peak Torque	N.m	7	10	34	60	120
No Load Speed	RPM	178	317	111	176	158
Torque constant	N.m/A	0.8	0.8	1.9	2.1	2.4
Back-EMF Constant	Vdc/Krpm	4.3	6.0	6.0	16.0	19.2
Efficiency	%	63	69.5	63.1	72.7	79.0
Control Accuracy	Degree	<0.01	<0.01	<0.01	<0.01	<0.01
Inertia	Kg.cm²	0.17	0.25	0.3	0.66	1.5
Weight	Kg	0.26	0.26	0.36	0.82	1.40
Communication	—	CAN BUS/EtherCAT			CAN BUS/EtherCAT	
Encoder	—	Dual Encoder ABS-17BIT/18BIT			Dual Encoder ABS-17BIT/17BIT	

0.8m

Educational Robot Power System Solutions

Full body freedom

24

Machine quality

20 kg

Overall height

80 cm

Arm movement speed

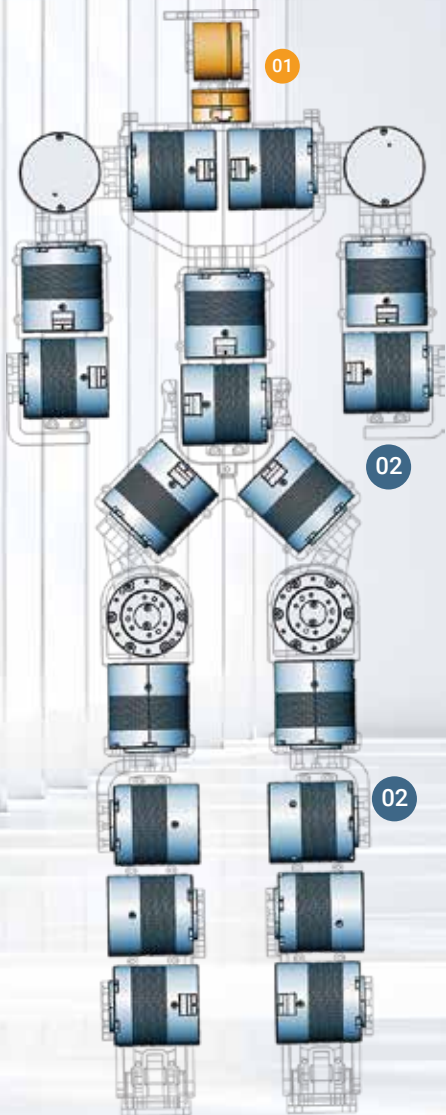
1m/s

Wrist load

2 kg

Rated Power

10~50 w



Core Components

01 Head

Direct Drive Actuator

L-4005*1



Direct Drive Actuator

L-4015*1



02 Upper body

Planetary Actuator

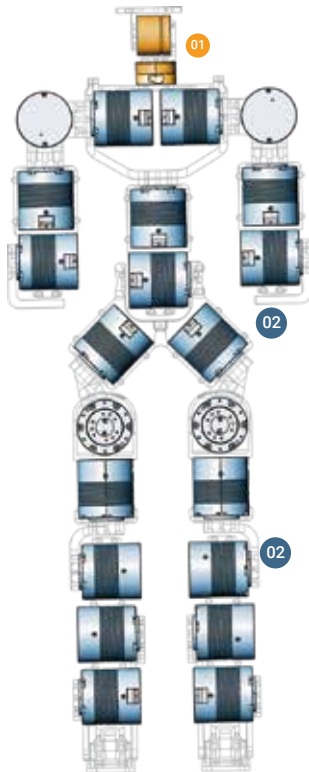
X4-36*12



02 Legs

Planetary Actuator

X4-36*10



Communication

CAN BUS

Component Parameters

Parameters	Unit	L-4005	L-4015	X4-36
Gear ratio	-	-	-	36
Rated Voltage	V	12	16	24
Rated Power	W	100	100	100
Rated Torque	N.m	0.07	0.22	10.5
Rated Speed	RPM	1600	520	83
Peak Torque	N.m	0.25	0.49	34
No Load Speed	RPM	2390	1010	111
Torque constant	N.m/A	0.05	0.12	1.9
Back-EMF Constant	Vdc/Krpm	-	-	6.0
Efficiency	%	-	-	63.1
Control Accuracy	Degree	< 0.01	< 0.01	<0.01
Inertia	Kg.cm ²	0.056	0.105	0.3
Weight	Kg	0.065	0.12	0.36
Communication	—	CAN BUS	CAN BUS	CAN BUS
Encoder	—	18-BIT	18-BIT	Dual Encoder ABS-17BIT/18BIT

Data Acquisition

Robotic arm

Power System Solutions

Degrees of Freedom

6 axis

Wingspan

420 mm

Deadweight

3 kg

Maximum load at the end

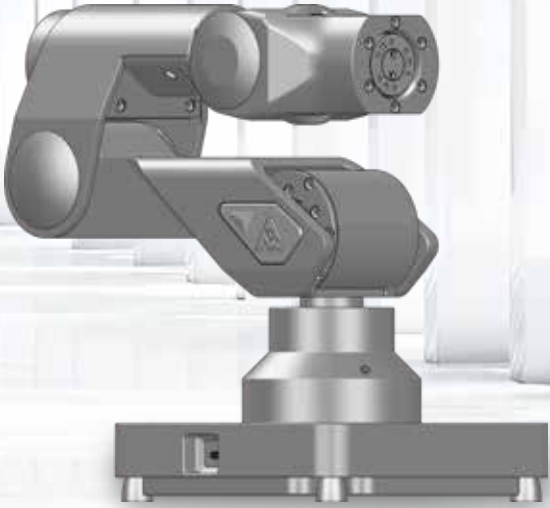
2 kg

Repeat positioning accuracy

0.1 mm

Maximum line speed

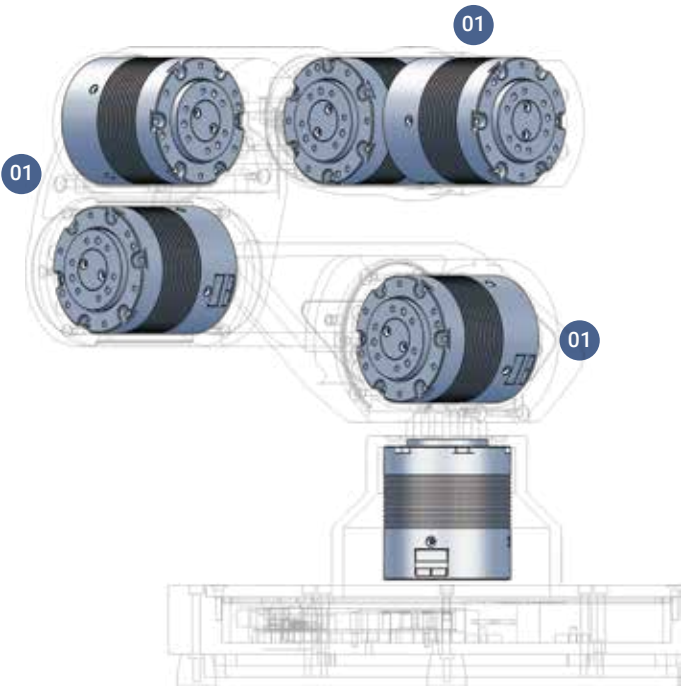
3 m/s



Core Components

01 Full axis

Planetary Actuator
X4-36*6



Communication

CAN BUS

Component Parameters

Parameters	Unit	X4-36
Gear ratio	-	36
Rated Voltage	V	24
Rated Power	W	100
Rated Torque	N.m	10.5
Rated Speed	RPM	83
Peak Torque	N.m	34
No Load Speed	RPM	111
Torque constant	N.m/A	1.9
Back-EMF Constant	Vdc/Krpm	6.0
Efficiency	%	63.1
Control Accuracy	Degree	<0.01
Inertia	Kg.cm ²	0.3
Weight	Kg	0.36
Communication	—	CAN BUS
Encoder	—	Dual Encoder ABS-17BIT/18BIT

Commercial-grade **Quadruped** Power System Solutions

Maximum joint torque
30 N.m

Rated voltage
24 v

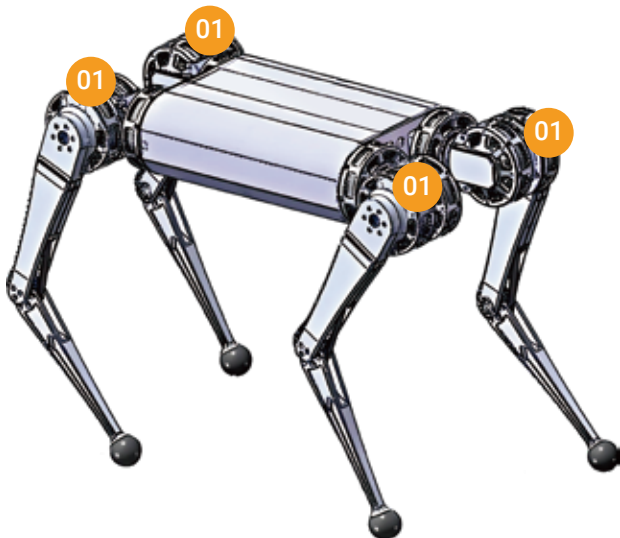
Maximum joint speed
224 RPM

Rated power
380 w



Core Components

01 **Four joints**
Planetary Actuator
X8-32*12



Communication
CAN BUS

Component Parameters

Parameters	Unit	X8-32
Gear ratio	-	9
Rated Voltage	V	24
Rated Power	W	380
Rated Torque	N.m	16
Rated Speed	RPM	224
Peak Torque	N.m	30
No Load Speed	RPM	300
Torque constant	N.m/A	1.3
Back-EMF Constant	Vdc/Krpm	11.9
Efficiency	%	82
Control Accuracy	Degree	<0.01
Inertia	Kg.cm²	1.43
Weight	Kg	0.55
Communication	—	CAN BUS
Encoder	—	Single Encoder ABS-18BIT

Industrial-grade **Quadruped** Power System Solutions

Maximum joint torque
320 N.m

Rated voltage
48 v

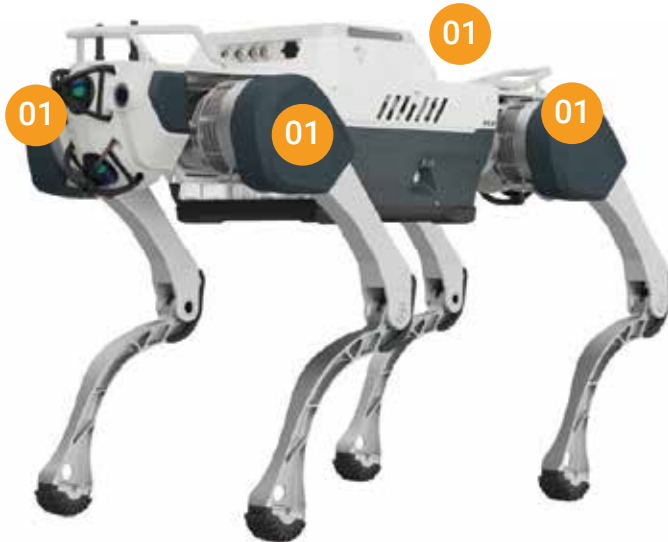
Maximum joint speed
100 RPM

Rated power
1000 w



Core Components

01 **Four joints**
Planetary Actuator
X12-320*12



Communication
CAN BUS

Component Parameters

Parameters	Unit	X12-320
Gear ratio	-	20
Rated Voltage	V	48
Rated Power	W	1000
Rated Torque	N.m	85.0
Rated Speed	RPM	100
Peak Torque	N.m	320
No Load Speed	RPM	125
Torque constant	N.m/A	3.3
Back-EMF Constant	Vdc/Krpm	17.9
Efficiency	%	75
Control Accuracy	Degree	<0.01
Inertia	Kg.cm²	12.9
Weight	Kg	2.37
Communication	—	CAN BUS
Encoder	—	Dual Encoder ABS-17BIT/17BIT

2-finger Gripper System Solution

Clamping force
20N

Deadweight
0.3kg

Rated voltage
24V

Rated current
1.44A

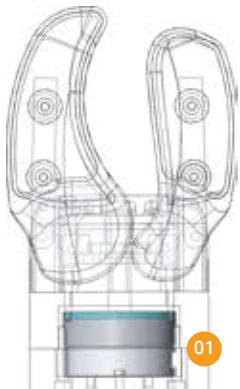
size
75(L) * 61(W) * 136(H) mm

Operating temperature
-20~55℃



Core Components

01 Joint
Direct Drive Actuator
L-4015*1



Communication
CAN BUS

Component Parameters

Parameters	Unit	L-4015
Gear ratio	-	-
Rated Voltage	V	16
Rated Power	W	100
Rated Torque	N.m	0.22
Rated Speed	RPM	520
Peak Torque	N.m	0.49
No Load Speed	RPM	1010
Torque constant	N.m/A	0.12
Back-EMF Constant	Vdc/Krpm	-
Efficiency	%	-
Control Accuracy	Degree	< 0.01
Inertia	Kg.cm²	0.105
Weight	Kg	0.12
Communication	—	CAN BUS
Encoder	—	18-BIT

3-finger Gripper System Solution

Clamping force
20_N

Deadweight
0.28_{kg}

Rated voltage
24_v

Rated current
1.44_A

size
85_(L) * 72_(W) * 130_(H) mm

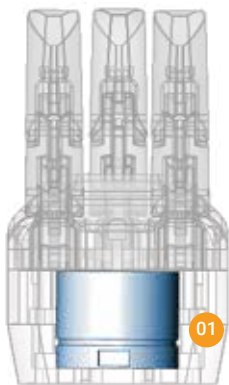
Operating temperature
-20~55℃



Core Components

01 Joint

Direct Drive Actuator
L-4015*1



Communication
CAN BUS

Component Parameters

Parameters	Unit	L-4015
Gear ratio	-	-
Rated Voltage	V	16
Rated Power	W	100
Rated Torque	N.m	0.22
Rated Speed	RPM	520
Peak Torque	N.m	0.49
No Load Speed	RPM	1010
Torque constant	N.m/A	0.12
Back-EMF Constant	Vdc/Krpm	-
Efficiency	%	-
Control Accuracy	Degree	< 0.01
Inertia	Kg.cm²	0.105
Weight	Kg	0.12
Communication	—	CAN BUS
Encoder	—	18-BIT

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